

Curriculum

School:	SET all Engineering programs(Except Data Science and BCA)
Program Code:	
Program Name:	B. Tech (Computer Science and Engineering)
Academic Year:	2021-22

GLOSSARY:

Programme Outcomes (POs): POs are statements that describe what the students graduating from any of the educational Programmes of the institution should be able to do on completion.

Programme Specific Outcomes (PSOs): PSOs are statements that describe what the graduates of a specific educational Programme should be able to do on completion.

Course Outcomes (COs): COs are statements that describe what students should be able to do on completion of the course.

NOTE:

Programme Outcomes (POs): Programme Outcomes (POs) are what knowledge, skills and attitudes a graduate should have at the time of graduation. While as at present, no agency has formally defined the POs of General Higher Education in a 3-year degree Programmes in India, POs of all professional Programmes in engineering and other areas are identified at the national level, by the concerned accrediting agency. POs are not specific to a discipline.

Course Outcomes (COs): COs are statements that describe what students should be able to do at the end of a course. They can be 6±2 for courses with 2 to 4 credits, and 8±2 for courses with 5 to 6 credits.

Vision of School:

Mission of School:

Programme Outcomes (POs):

PO1:	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2:	Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3:	Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4:	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5:	Modern Tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6:	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7:	Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8:	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9:	Individual and Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10:	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11:	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12:	Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcome (PSO):

Sample PSOs of Electrical and Electronics Engineering (Note: This is a sample PSOs, each program will have their own set of PSOs)

PSO1:	To prepare highly competent graduates with strong foundation in engineering and technology for successful career in industries, academics, and research organizations.
PSO2:	Succeed in the areas of Electrical and Electronics Engineering and other diverse fields by utilizing the fundamental knowledge of engineering, analytical and creative skills
PSO3:	To prepare graduates with holistic education approach that includes communication skills, ability to effectively coordinate and contribute as a member of multidisciplinary team, appreciates and respects diversity of opinion, considerate global, societal, environmental, and ethical issues.

Course Structure

L = Lecture	T = Tutorial	P = Practical	C = Credit
number of hours per week	number of hours per week	number of hours per week	Total Credits

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credits	19.5	19.0	20.0	23.0	23.0	21.5	23.0	15.0	164

Total Credits	160
Minor Specialization Credits	012
Total Credits Including Minor	172

CURRICULUM AND TEACHING SCHEME

Semester-I													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
CS168	Linear Algebra and Calculus	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
EE118	Basic Electrical and Electronics	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
EE119	Basic Electrical and Electronics Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS1006	Introduction to Web Design	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CS1007	Introduction to Web Design Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS1008	Introduction to Computer Programming	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS1009	Introduction to Computer programming laboratory	0	0	4	2	-NA-	-NA-	60	40	100	Emp	G	-
PH120	Engineering Physics	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
PH121	Engineering Physics Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-

ME111	Workshop Practices	0	0	3	1.5	-NA-	-NA-	60	40	100	Emp	G	-
	Total Credits				19.5								

CURRICULUM AND TEACHING SCHEME

Semester-II													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
CS193	Problem solving using Python	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CS194	Problem Solving Using Python Laboratory	0	0	4	2	-NA-	-NA-	60	40	100	Emp	G	-
CS143	Engineering Chemistry	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CS141	Engineering Chemistry Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS234	Digital Logic Design	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS154	Engineering Mathematics II	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
LC137	Communication Skills	0	0	2	1	-NA-	-NA-	60	40		SD	G	-
CE101	Applied Mechanics	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CE111	Applied Mechanics Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
ME113	Engineering Graphics	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-

	Total Credits				19								-
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CURRICULUM AND TEACHING SCHEME

Semester-III													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
CS157	Web Technology	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS158	Web Technology Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS231	Object Oriented Programming using Python	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS232	Object Oriented Programming using Python Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS233	Introduction to Microprocessor	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
EN102	Environmental Science					60	40	-NA-	-NA-	100	Emp	N, G	ES
CS239	Discrete Mathematics	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS229	Data Structures	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS230	Data Structures Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
	Interdisciplinary - 1	2	0	0	2	-NA-	-NA-	60	40	100	SD	L, R	-

	Total Credits				20								
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CURRICULUM AND TEACHING SCHEME

Semester-IV													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
CS330	Software Engineering	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS227	Database Management System	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS238	Database Management System Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS307	Computer Organization and Architecture	2	1	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS312	Design and analysis of algorithm	2	1	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS313	Design and analysis of algorithm Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS322	Information Security	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CS236	Theory of Computation	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
SS102	Humanities and Social Science	0	0	2	1	60	40	-NA-	-NA-	100	Emp	N, G	HV

	Critical thinking/Technical Reading	0	0	2	1	-NA-	-NA-	60	40	100	SD	G	-
	Interdisciplinary - 2	2	0	0	2	-NA-	-NA-	60	40	100	Emp	L, R	-
	Total Credits				23								

CURRICULUM AND TEACHING SCHEME

Semester-V													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
CS308	Computer Network	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS309	Computer Network Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
CS334	Mobile Application Development	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS344	Mobile Application Development Lab	0	0	4	2	-NA-	-NA-	60	40	100	Emp	G	-
CS404	Data Warehousing and Data Mining	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS240	Computer graphics	2	1	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS241	Computer graphics Laboratory	0	0	3	1.5	-NA-	-NA-	60	40	100	Emp	G	-

CS310	Operating Systems	2	1	0	3	60	40	-NA-	-NA-	100	Emp	G	-
	Operating Systems Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
PS314	Minor Project - I	1	0	3	2.5	-NA-	-NA-	60	40	100	Emp, SD	N	-
	Total Credits				23								

CURRICULUM AND TEACHING SCHEME

Semester-VI													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
	Big data analytics and Data Visualization	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
	Big data analytics and Data Visualization Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
PS415	KHOJ	3	0	0	3	60	40	-NA-	-NA-	100	SD	L, R	-
	Data Science and Machine Learning	3	0	0	3	60	40	-NA-	-NA-	100	Emp	R	-

	Data Science and Machine Learning Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
	Soft Skills	0	0	2	1	-NA-	-NA-	60	40	100	SD	G	-
CS324	Artificial Intelligence	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
CS325	Artificial Intelligence Laboratory	0	0	3	1.5	-NA-	-NA-	60	40	100	Emp	G	-
PS315	Minor Project - II	1	0	2	2	-NA-	-NA-	60	40	100	Emp	N	-
	Elective Theory – 1	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
	Elective Lab – 1	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
	Total Credits				21.5								

CURRICULUM AND TEACHING SCHEME

Semester-VII													
Teaching Scheme											Elements of Employability (Emp)/ Entrepreneurship (Ent)/ Skill Development (SD)	Relevance to Local (L)/ National (N)/ Regional(R)/ Global developmental needs (G)	Relation to Gender (G)/ Environment and Sustainability (ES)/ Human Values (HV)/ Professional Ethics (PE)
						Theory Course		Practical Course		Total marks			
Course Code	Course Name	L	T	P	C	Internal Examination (%)	End semester Examination (%)	Internal Examination (%)	End semester Examination (%)				
MG314	Entrepreneurship	3	0	0	3	60	40	-NA-	-NA-	100	Ent	G	-
CS419	Cloud Computing	3	0	0	3	60	40	-NA-	-NA-	100	Emp	G	-
	Cloud Computing Laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-

CS414	Major Project	1	0	4	3	60	40	-NA-	-NA-	100	Emp, SD	N	-
CS401	Research Methodology	2	0	0	2	60	40	-NA-	-NA-	100	SD	G	-
CS410	Cyber Security	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CS402	Compiler Design	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
CS403	Compiler Design laboratory	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
	Elective Theory – 2	2	0	0	2	60	40	-NA-	-NA-	100	Emp	G	-
	Elective Lab – 2	0	0	2	1	-NA-	-NA-	60	40	100	Emp	G	-
	Elective – 3	3	0	0	3	60	40	-NA-	-NA-	100	Ent	G	PE
	Total Credits				23								

CURRICULUM AND TEACHING SCHEME

[illegible]

Core Elective Subjects

Course code	Course Name
	Elective - 1
CS328	Numerical Computation using Sci_lab
CS352	Numerical Computation using Sci_lab Laboratory
CS349	Data Science using Python
CS351	Data Science using Python Laboratory
CS348	Core JAVA Programming
CS350	Core JAVA Programming Laboratory
	Elective - 2
CS423	Augmented Reality and Virtual Reality
CS435	Augmented Reality and Virtual Reality Laboratory
CS412	Computer Vision
CS413	Computer Vision Laboratory
CS434	Mathematical and Statistical Techniques
CS436	Mathematical and Statistical Techniques Laboratory
	Elective - 3
CS429	Management Information System
CS421	Green Computing
CS428	Enterprise Resource Planning

Minor Specialization Subjects

Course code	Course Name	Semester	L	T	P	C
	Fundamentals of Deep Learning	V	2	0	0	2
	Fundamentals of Deep Learning Laboratory	V	0	0	4	2
	Natural Language Processing	VI	2	0	0	2
	Natural Language Processing Laboratory	VI	0	0	4	2
	Robotics	VII	2	0	0	2
	Robotics Laboratory	VII	0	0	4	2
Total						12

Articulation Matrix COs and POs Mapping												
Semester-I	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Linear Algebra and Calculus	✓	✓		✓	✓	✓						
Basic Electrical and Electronics	✓	✓		✓	✓	✓						
Basic Electrical and Electronics Laboratory	✓	✓	✓	✓								
Introduction to Web Design	✓	✓	✓	✓								
Introduction to Web Design Laboratory	✓	✓			✓	✓						
Introduction to Computer Programming	✓	✓			✓	✓	✓					
Introduction to Computer programming laboratory	✓	✓			✓	✓	✓					
Engineering Physics	✓	✓	✓									✓
Engineering Physics Laboratory	✓	✓	✓									
Workshop Practices	✓							✓				✓
Semester-II	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Problem solving using Python	✓	✓	✓		✓			✓		✓		
Problem Solving Using Python Laboratory	✓	✓	✓		✓			✓		✓		
Engineering Chemistry	✓	✓		✓								
Engineering Chemistry	✓	✓		✓								
Digital Logic Design	✓	✓	✓	✓		✓						
Engineering Mathematics II	✓	✓		✓								
Communication Skills	✓				✓					✓		✓
Applied Mechanics	✓	✓		✓								
Applied Mechanics Laboratory	✓	✓		✓								
Engineering Graphics	✓	✓		✓								✓
Semester-III	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12

Web Technology	✓	✓	✓		✓	✓						
Web Technology Laboratory	✓	✓	✓		✓	✓						
Object Oriented Programming using Python	✓	✓	✓	✓	✓							
Object Oriented Programming using Python Laboratory	✓	✓	✓	✓	✓							
Introduction to Microprocessor	✓	✓		✓	✓		✓	✓				
Environmental Science	✓	✓	✓			✓	✓	✓			✓	✓
Discrete Mathematics	✓	✓	✓	✓	✓							
Data Structures	✓	✓	✓	✓	✓			✓	✓			
Data Structures Laboratory	✓	✓	✓	✓	✓			✓	✓			
Semester-IV	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Software Engineering	✓	✓	✓	✓	✓					✓	✓	
Database Management System	✓	✓	✓	✓	✓							
Database Management System Laboratory	✓	✓	✓	✓	✓							
Computer Organization and Architecture	✓	✓	✓	✓	✓		✓	✓				
Design and analysis of algorithms	✓	✓	✓	✓			✓	✓			✓	
Design and analysis of algorithms Laboratory	✓	✓	✓	✓			✓	✓			✓	
Information Security	✓	✓	✓	✓				✓				
Theory of Computation	✓	✓		✓		✓		✓				
Humanities and Social Science						✓			✓	✓		✓
Critical thinking/Technical Reading						✓			✓	✓		✓
Semester-V	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Computer Network	✓	✓	✓	✓	✓		✓		✓			
Computer Network Laboratory	✓	✓	✓	✓	✓		✓		✓			
Mobile Application Development	✓	✓	✓		✓			✓		✓	✓	
Mobile Application Development Lab	✓	✓	✓		✓			✓		✓	✓	

Data Warehousing and Data Mining	✓	✓	✓			✓	✓					
Computer graphics	✓	✓		✓	✓		✓	✓				
Computer graphics Laboratory	✓	✓		✓	✓		✓	✓				
Operating Systems	✓	✓		✓	✓				✓	✓		
Operating Systems Laboratory	✓	✓		✓	✓				✓	✓		
Minor Project - I	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓
Semester-VI	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Big data analytics and Data Visualization	✓	✓	✓	✓	✓			✓		✓		
Big data analytics and Data Visualization Laboratory	✓	✓		✓	✓			✓		✓	✓	
KHOJ	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Data Science and Machine Learning	✓	✓	✓	✓	✓			✓		✓		
Data Science and Machine Learning Lab	✓	✓		✓	✓			✓		✓	✓	
Soft Skills Laboratory						✓		✓	✓	✓		✓
Artificial Intelligence	✓	✓	✓	✓	✓			✓		✓		
Artificial Intelligence Laboratory	✓	✓	✓	✓	✓			✓	✓	✓		
Semester-VII	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Entrepreneurship						✓		✓	✓	✓		✓
Cloud Computing	✓	✓	✓	✓	✓		✓	✓				
Cloud Computing Laboratory	✓	✓	✓	✓	✓		✓	✓			✓	
Major Project	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓
Research Methodology	✓	✓	✓								✓	
Cyber Security	✓	✓	✓	✓	✓							
Semester-VIII	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Industry Internship	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓

PSO's - PO's Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PSO1	√		√		√	√					√	
PSO2	√	√	√	√	√	√	√				√	√
PSO3	√					√	√	√	√	√		
PSO4								√	√			
PSO5								√	√	√	√	√
PSO6	√	√	√	√	√						√	√

Syllabus of Courses

Bloom's Taxonomy Levels (BT Level):

1. Remember 2. Understand 3. Application 4. Analysis 5. Evaluation 6. Creation

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS168		Course Name	Linear Algebra and Calculus	
Year	1	Core Subject (Yes/No):		Lecture:	3
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-2020	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course covers topics from engineering mathematics. that serve as foundations for descriptions of electrical engineering devices and systems. It is the co-requisite mathematics course for Fundamentals of Computer Engineering				
Course Objective	The objective of this course is to disseminate the prospective engineers . with techniques in one variable & multivariable calculus, sequences & series and matrices. It aims to equip the students with concepts and tools at an intermediate to advanced level of mathematics and applications that they would find useful in their disciplines.				
Course Outcomes	Students will be able to: 1. Understand techniques of one variable. 2. Determine the sequence and series 3. Use multivariable calculus. 4. Find Areas and Volumes Multiple Integration 5. Compute and use concept of matrices.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	CALCULUS OF ONE VARIABLE Evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. Rolle's theorem, Mean value theorems, Taylor's and Maclaurin's theorems with remainders; Indeterminate forms and L's Hospital's rule; Maxima and minima. Newton's method, Taylor's theorem. General Leibniz rule, Logarithmic derivative Integral: Introduction, The Fundamental Theorem of Calculus, Common Integrals, The Sum and Multiple Rules, Integration by Substitution, Integration by Parts, Integration Using Partial Fractions.	8	1	CO1	PSO1

2	SEQUENCE AND SERIES Sequences: Introduction, Sequence explicit formula, Sequence recursive formula. Geometric sequence, using explicit formulas of geometric sequences, using recursive formulas of geometric sequences. Infinite sequences: Convergent and divergent sequences, limit of a sequence, Proof of sequence converges using the formal definition Series: Summation notation, Converting explicit series terms to summation notation, Arithmetic series formula, Finite geometric series, Geometric series with sigma notation, Finite geometric series formula justification. Partial sums: Introduction, formula for nth term from partial sum, term value from partial sum, infinite series as limit of partial sums Infinite geometric series: Infinite geometric series formula intuition, Proof of infinite geometric series as a limit. Infinite geometric series word problem - bouncing ball, repeating decimal Convergent & divergent geometric series (with manipulation) Power series, Taylor's series. Series for exponential, trigonometric and logarithmic functions.	7	2,3	CO2	PSO1
3	CALCULUS OF MULTIPLE VARIABLE Limit, continuity and differentiability: partial derivatives, directional derivatives, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers; Gradient, curl and divergence. Finding Areas and Volumes Multiple Integration: double and triple integrals (Cartesian and polar), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes by (double integration), Centre of mass and Gravity (constant and variable densities). Theorems of Green, Gauss and Stokes, orthogonal curvilinear coordinates, Simple applications involving cubes, sphere and rectangular parallelepipeds.	17	2,3	CO3	PSO2
4	MATRICES Linear Systems, Vectors and Spaces, Algebra of matrices, Inverse and rank of a matrix, rank-nullity theorem; System of linear equations; Symmetric, skew-symmetric and orthogonal matrices; Determinants; Eigen values and eigenvectors; Diagonalization of matrices; Cayley-Hamilton Theorem, Orthogonal transformation and quadratic to canonical forms. , L U Decomposition, Doolittle Algorithm : LU Decomposition, System of Linear Equations, Metric Space - Rk	12	2,3,4	CO4 CO5	PSO1 PSO2
5	Nil	Nil	Nil	Nil	Nil

Text books:

1. Erwin Kreyszig: Advanced Engineering Mathematics, 9th Ed. Wiley India Pvt. Ltd.
2. Joel Hass, George B. Thomas, Maurice D Weir, Thomas's Calculus, 12th Edition, Pearson.
3. James Stewart, Calculus, Brooks Cole; 7 edition (2012)

Reference books:

1. Stephen Timoshenko, J V Rao, D H Young, Engineering Mechanics 4th Edition, Tata McGraw-Hill Education, 2006.
2. T.M. Apostol, Calculus, vol. 1, 2nd Ed., Wiley India, 2006.
3. T.M. Apostol, Calculus, Vol. II, Wiley India, 2007.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	EE118		Course Name	Basic Electrical and Electronics	
Year	1	Core Subject (Yes/No):		Lecture:	3
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-2020	Prerequisites (If any)	Nil
Qualifies for	Project				
Course Description	Fundamentals AC & DC Electrical Circuits , Electrical Network Concept				
Course Objective	In a given circuit, identify polarities voltage and direction of current in loops and nodes.				
Course Outcomes	CO1: Alternating voltages and currents and their vector and time domain presentations. CO2: R -L, R-C and R-L-C series and parallel circuits, impedance and admittance, series and parallel resonance. CO3: star and delta connections and transformations CO4: Construction, Working and Principles of Different electrical machines. CO5: Forward and reverse biased characteristics of Diode CO6: Half-wave and Full wave diode rectifier characteristics CO7: Transistor Characteristic-NPN and PNP Configurations (CE, CB, CC). CO8: Introduction to Logic gates, Boolean Algebra, SOP & POS canonical form.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	AC Circuits Alternating voltages and currents and their vector and time domain presentations, average and rms values, form factor, phase difference, power and power factor, purely resistive inductive and capacitive circuits, R -L, R-C and R-L-C series and parallel circuits, impedance and admittance, series and parallel resonance, star and delta connections and transformations	8	2,3,4,5	CO1, CO2, CO3	PSO1 PSO3
2	ELECTRICAL MACHINES Construction and Working Principles of Single phase Transformer, DC Machine, Single phase and Three phase Induction motor.	8	1,2,4,5	CO4	PSO1 PSO4
3	ELECTRONIC DEVICES & CIRCUIT PN Junction, Forward and reverse biased characteristics of Diode, Halfwave and Full wave diode rectifier, Transistor Characteristic-NPN And PNP Configurations(CE,CB, CC)	10	1,2,3,4	CO5,CO6,CO7	PSO2,PSO3,PSO5
4	DIGITAL ELECTRONIC	10	2,3,4,5	CO8	PSO1,PSO2,PSO3

	Digital logic and Boolean algebra; Introduction of Logic Gates; truth table verification; Simplification of Boolean Function using Boolean theorems; Canonical and Standard Forms (SOP and POS) for Boolean Functions; implementation of simple SOP & POS logic circuits.				
5	Nil	9	Nil	Nil	Nil

Text books:

1. Hayt W. H., Kemmerly J. E, Durbin S. M., "Engineering Circuit Analysis", Tata McGraw Hill, 6th Edition, 2006.
2. Boylestad & Nashlesky, "Electronic Devices & Circuit Theory", PHI Publication, 2nd edition, 2000.

Reference books:

1. Edminister Joseph A., "Electrical circuits", Schaum's outline series, McGraw hill, 2nd edition, 1983.
2. G. K. Mithal , " Electronic Devices And Circuits" KHANNA PUBLISHERS. ISBN 9788174091777.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS144		Course Name	Introduction to Web Design	
Year	1	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2019-2020	Prerequisites (If any)	12th Science, Basic knowledge of Computer System
Qualifies for	Web Technology				
Course Description	This course is an introduction to web design and development of website in the information fields. It is designed to provide understanding of basic web development concepts and process, web design best practices, web page creation.				
Course Objective	1. To get familiar with basics of the Internet and Web. 2. Gain the knowledge about website hosting and Domain registration. 3. To acquire fundamental knowledge and skills for creation of web site. 4. To gain ability to create responsive website using Bootstrap. 5. To explore different online tools for website building.				
Course Outcomes	1. Learn the basic concepts of Internet and Web. 2. Use fundamental skills to maintain web server services required to host a website. 3. Select and apply mark-up languages for processing, identifying, and presenting of information in web pages. 4. Implement a responsive web site using Bootstrap. 5. Build Website using online website builder tools Wix, Weebly etc				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCITON TO WEB Introduction and Web Development Strategies, History of Web and Internet, Protocols governing Web, Connecting to Internet, to basic Internet services and tools, Websites: Static Websites and Dynamic Websites. c Types of Website: Internet Forums, Blog, Ecommerce, Anatomy of Web Page.	4	1,2	CO1	PSO1
2	DOMAIN REGISTRATION AND WEB HOSTING Domain Registration, Web hosting: Shared Hosting: Linux Hosting, Windows Hosting, Reseller Hosting: Linux Reseller Hosting, Windows Reseller Hosting. Specialized Hosting: WordPress Hosting, CMS Hosting, Ecommerce Hosting. Cloud: Cloud hosting, Cloud for business. Servers: KVM VPS Hosting, Linux Dedicated Servers,	5	1,2,3	CO2	PSO2,PSO3

	Windows Dedicated Servers Security: Digicerts & SSL, SiteLock, CodeGuard.				
3	WEB PAGE DESIGNING-HTML,CSS & XML HTML:HTML elements, Attributes, HTML Formatting list, table, images, frames, iframe ,forms, HTML Classes, ID, HTML Links, HTML Graphics, HTML Media. CSS: Types of CSS, CSS :Color, Background, Border, List, Tables, CSS Pseudo Classes, CSS Border Image, CSS Box Model, CSS Text Effects, CSS 2D Transforms, CSS 3D Transforms, CSS Animation, CSS User Interface. XML: Introduction to XML, XML Tree, XML DTD, XML Schema.	10	4,5	CO3	PSO3
4	BOOTSTRAP : Introduction to Bootstrap, Custom components: Album, Pricing, Checkout, Product, Cover, Carousel, Blog, Dashboard, Sign-in, Sticky footer. Framework: Starter template, Grid, Jumbotron. Navbars, floating labels, Offcanvas. BS Grid Basic, BS Typography, BS Tables, BS Images, BS List groups, BS Forms, BS Buttons, and BS button groups, Bootstrap Glyphicons Bootstarp Themes.	8	5	CO4	PSO4 PSO5
5	WEBSITE BUILDER : Working with DIY Website builder, Weebly, WIX.	3	4,5	CO5	PSO6

Text books:

- 1.Web Programming Step by Step by Marty Stepp, Jessica Miller, and Victoria Kirst
- 2.Web Technologies, Black Book, Dreamtech Press

Reference books:

- 1.HTML 5, Black Book, Dreamtech Press.
- 2.Web Design, Joel Sklar, Cengage Learning

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS142		Course Name	Introduction to Computer Programming	
Year	1	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018-2019	Prerequisites (If any)	12th Science, Basic knowledge of Computer System
Qualifies for	Object oriented programming and Operating System				
Course Description	This course covers the basics of programming in the "C" programming language and demonstrates fundamental programming techniques including the most common library functions and the usage of the pre-processor. It also introduces to object-based programming using C++. The problem solving begins with writing algorithm and drawing flowcharts which will be mapped on to a computer program. This course is designed for students having no or very little programming knowledge and experience.				
Course Objective	1. Understand the fundamentals of C programming. 2. Learn how to manage the Input and output Operations in C with the various control structures and looping statements and functions for transforming a problem solution into programs involving programming constructs. 3. Study the concept of array and OOP. 4. To get familiar with Class, Object and pointers using C++. 5. Understand how the files can be managed and dynamically how the memory can be allocated.				
Course Outcomes	1. Learn the basic concepts of algorithms, flowcharts, constants, variables, data types, operators, expressions and be able to design and solve complex problems. 2. To identify the loops, decision making statements and function to solve the problem. 3. Implement different Operations on arrays and understand basic concepts of OOP. 4. Understand the concept of Class, Object and pointers using C++. 5. Implement file Operations in C programming for a given application.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION TO COMPUTER PROGRAMMING IN C Algorithm, Flowchart, identifiers, basic data types and sizes, constants, variables, operators, expressions, type conversions, precedence and order of evaluation, input-output statements, statements and blocks.	6	1	CO1	PSO1
2	CONTROL STRUCTURES IN 'C'	12	1,2,3	CO1	PSO1

	Algorithm, Flowchart, identifiers, basic data types and sizes, constants, variables, operators, expressions, type conversions, precedence and order of evaluation, input-output statements, statements and blocks.			CO2	PSO2
3	ARRAY AND CONCEPTS OF OOP Arrays, two-dimensional and multi-dimensional arrays, structure and typedef, introduction, OOP paradigm, data encapsulation, inheritance, polymorphism, data abstraction, data hiding	9	3,4	CO3	PSO2
4	CONCEPTS OF OOP: Class declaration, object instantiation, pointers in C++ concepts, initialization of pointer variables, address arithmetic, pointers to object, new and delete operator.	9	4,5	CO4	PSO2
5	FILE MANAGEMENT: File Handling: Concept of files, File Opening Modes and closing of a file, Formatted reading from and writing to a file. Reading and writing processed arrays to files.	6	6	CO5	PSO6

Text books:

Text books:

1. B.A. Forouzan And R.F. Gilberg: Computer Science, A Structured Programming Approach Using C, Third Edition, Thomson.
2. E. Balaguruswamy, "Object Oriented Programming with C++", Fourth Edition, Tata Mcgraw Hill Publishing Pvt. Ltd

Reference books:

1. B .W. Kernighan, Dennis M. Ritchie: The C Programming Language, Phi/Pearson.
2. R. Lafore, "The Waite Group's Object Oriented Programming In C++", 3rd Edition, Galgotia Publications, 2001.
3. B. Stroustrup, "C++ Programming Language", 3rd Edition, Pearson Education, 1997

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	PH105		Course Name	Engineering Physics	
Year	1	Core Subject (Yes/No):		Lecture:	2
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2020-2021	Prerequisites (If any)	Basic understanding of Math's, Physics and chemistry
Qualifies for	Nil				
Course Description	This is an advanced course above college level physics meant for developing knowledge base for engineering. It mainly comprises of fundamentals of quantum mechanics, semiconductor physics, optics and materials. The course is designed to understand physics involved in these subjects. Exposure to various applications based on those concepts will be given. Semiconductors form the basis of existing technology. Hence understanding semiconductor physics is very essential. This requires some preliminary knowledge in quantum mechanics. The subject will then finally be developed in solid state electronics which is again at the heart of present technology. Optics has tremendous applications in engineering especially in non-destructive techniques and communication. This topic will give a wide understanding about optical phenomena, classical, modern as well as technology like optical fibres, lasers etc. Several problems will be solved which would give the students a better spirit of applying the newly learnt concepts and build their analytical powers.				
Course Objective	Physics is a science to fundamentally understand how things work in nature and around us. The importance of the subject is thus unparalleled for an engineering student of any branch. Some of the immediate and direct objectives of this course are listed below. 1. To develop clear understanding about concepts in physics which form basis of present day engineering practices and cutting edge technology. 2. To develop problem solving skills. 3. To have sufficient knowledge base for research and development work in the field of semiconductors, nanotechnology, materials, electromagnetic radiation, waves, optics and communication.				
Course Outcomes	CO1 To understand polarization property of light and understand their role in optical instrumentation. To understand two important optics phenomena- interference and diffraction and their role in achieving optical resolution of images. CO2 To learn basics of optical fibres and their applications in data communication. CO3 To understand of principle of lasers and information about various types of lasers with their applications CO4 To understand wave particle duality and the need to replace classical mechanical equations of motion meant for particles by wave				

	like equation of motion (Schrödinger equation). CO5 To understand energy levels of confined electrons in a quantum dot. To understand how energy bands are formed in solids based on the principles of quantum mechanics and draw/interpret energy band diagrams. CO6 To learn about Miller indices and how they can be used to describe crystal planes and directions. CO7 To learn basics of semiconductors that will help understand the physical concepts underlying the operation of semiconductor devices and predict the effect of device design variations on device performance. CO8 To understand physics of some important materials like superconductors, nanomaterials, dielectric and magnetic materials. These materials have tremendous engineering applications.
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Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	SPECIAL THEORY OF RELATIVITY: Michelson-Morley experiment, Galilean and Lorentz transformations, Time dilation, Length contraction, Energy mass relation	14	1,2	CO1	PSO1
2	QUANTUM MECHANICS Wave particle duality, Heisenberg uncertainty principle, Time dependent and time independent Schrödinger equation, Applications: particle in a 1-D box, harmonic oscillator, scattering from a step potential barrier, finite potential barrier, tunnelling through a barrier	14	3,4	CO3,CO4,CO5	POS1 PSO2
3	SOLID STATE PHYSICS: Types of crystal structures, Miller indices, Free electron theory of metals, Bloch's theorem, Band theory of solids, Kronig-Penny Model, Intrinsic and extrinsic, semiconductors, Hall effect, p-n junction diode	10	4,5	CO6,CO7	PSO3
4	INTRODUCTION TO MATERIAL SCIENCE Superconductors, Nanostructures, Polymers	3	1,2	CO7	PSO3
5	Soft matter, Smart materials	1	1,2	CO7,CO8	PSO2

Text books:

1. H.S. Mani and G.K. Mehta, Introduction to Modern Physics, 1st Edition, East-west Press Pvt. Ltd.-New Delhi, 2000
2. A. Beiser, Concepts of Modern Physics, 6th Edition, Tata McGraw Hill Education Pvt. Ltd., 2009
3. K.S. Krane, Modern Physics, 2nd Edition, Wiley India Pvt. Ltd., 2009

Reference books:

1. J. Bernstein, P.M. Fishbane and S. Gasiorowicz, Modern Physics, 1st Edition, Pearson Education, 2009
2. R. Resnick, Introduction to Special Relativity, 1st Edition, Wiley India Pvt.Ltd., 2007
3. R. Resnick and R. Eisberg: Quantum Physics, Of Atoms Molecules Solids Nuclei And Practicles, 2nd Edition, Wiley India Pvt Ltd., 2006
4. S. Gasiorowicz, Quantum Physics, 3rd edition, John Wiley & Sons, 2007

5. C Kittel, Introduction to Solid State Physics, 7th edition, John Wiley, 2009
6. M.S. Vijaya and G. Rangarajan, Materials Science, 1st edition, Tata McGraw Hill Publishing Company Ltd., 2004
7. William D. Callister, Jr., Materials Science and Engineering, 1st Edition, Wiley India Pvt. Ltd., 2010
8. Charles P. Poole, Jr., Frank J. Owens, Introduction to Nanotechnology, 1st Edition, Wiley India, 2007
9. Peter J. F. Harris, Carbon Nanotube Science, 1st edition, Cambridge University Press, 2009

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS193		Course Name	Problem solving using Python	
Year	1	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2019-2020	Prerequisites (If any)	Introduction to Programming
Qualifies for	Advance python programming				
Course Description	Introduction to Python Programming course is intended for students with little or no programming experience. It aims to provide students with an understanding of the role computation can play in solving problems and, regardless of their major, feel justifiably confident of their ability to write small programs that allow them to accomplish useful goals for an IOT application.				
Course Objective	1. To understand the nature of programming as human activity. 2. To learn and experience main components of programming process. 3. To understand main control structures of procedural programming languages. 4. To learn and being able to use major programming patterns				
Course Outcomes	Student will be able to: 1. Explain basic principles of Python programming language. 2. Make Simple application using python. 3. Understand any python code's structure and fundamental aspects. 4. Apply basic python for real time problem solutions.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	BASIS OF PYTHON PROGRAMMING Installation and Working with Python, Essentials of a Python program, Program Structure of Python, Basic Syntax, Variables and Identifiers, Built-In Data Types. Variable definition, Operators And Expressions, Constants And Literals, Basic Input/output Statement. Installation and Working with Python, Essentials of a Python program, Program Structure of Python, Basic Syntax, Variables and Identifiers, Built-In Data Types. Variable definition, Operators And Expressions, Constants And Literals, Basic Input/output Statement.	5	1,2	CO1	PSO1
2	LOOPING,CONDITIONAL & CONTROL STATEMENTS	7	PSO2	CO2	PSO2

	Decision Making - Conditions, Relational Operators, Logical Connectives, If-Else , Nested If-Else Statement, Replacement of switch case Loops: While loop, Do- while, For loop, Nested loops , Break, Continue and pass Statements. Arrays – One dimensional and multidimensional array, Array processing. String Manipulation - accessing Strings, Basic Operations , String slices and Function and Methods				
3	LIST,TUPLES & DICTIONARIES Lists: Introduction, Accessing list, Operations, Working with lists, Function and Methods. Tuple: Introduction, Accessing tuples, Operations, Working, Functions and Methods. Dictionaries: Introduction, Accessing values in dictionaries, Working with dictionaries, Properties, Functions.	6	2,3	CO3	PSO3
4	PYTHON UTILITY FUNCTION Dates and Time: Basic date and time classes, Different time formats, Converting between formats, Formatting dates and times, Parsing date/time information, Binary Data: What is Binary Data? , Binary vs. text, Using the Struct module	6	3,4	CO4	PSO4 PSO5
5	FUNCTION Defining a function, Python Built-in Functions , Calling a function, Types of functions, Function Arguments, Default Argument, Anonymous functions, Global and local variables, Custom Functions vs. Standard Functions, Refactoring, Making Functions Reusable, Functions as Data.	6	4,5	CO3	PSO2

Text books:

1. John V Guttag. "Introduction to Computation and Programming Using Python", Prentice Hall of India.
2. R. Nageswara Rao, "Core Python Programming", dreamtech.

Reference books:

1. Wesley J. Chun. "Core Python Programming - Second Edition", Prentice Hall.
2. Kenneth A. Lambert, "Fundamentals of Python – First Programs", CENGAGE Publication.
3. Luke Sneeringer, "Professional Python", Wrox.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS143		Course Name	Engineering Chemistry	
Year	1	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2019-2020	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	Thorough understanding regarding aromaticity, stereochemistry, spectroscopy as a tool for qualitative and quantitative analysis, catalytic processes, application of catalysts in various industries. The course also focuses on applications of phase diagram with respect to alloy formation. Contemporary subject areas like dye sensitized solar cells, corrosion and passivation, fuel cells, liquid crystals, polymers, composites, ceramics, refractory shall also be dealt with.				
Course Objective	This course introduces Organic, Inorganic and Analytical Chemistry. The student will be able to understand the concept in depth. He or she will be able to implement it in real time techniques.				
Course Outcomes	Student will be able to : 1. Understand chemical bonding. 2. Learn sophisticated analytical instrumentation techniques. 3. Understand water analysis and purification. 4. Implement aromatic compounds, liquid crystals. 5. Understand phase transitions, alloy formation, catalysis and their industrial applications				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INORGANIC AND ANALYTICAL CHEMISTRY Atomic Structure, Chemical Bonding, Coordination Chemistry, Basics of Analytical Chemistry, Spectroscopy, Chromatography, Water analysis, treatment and purification	9	1,2	CO1	PSO1
2	PHYSICAL AND ORGANIC CHEMISTRY Aromatic Compounds. Huckel's Rule. Aromatic Ions. Liquid crystals, classification, properties and applications. Phase equilibria: Phase rule, one component systems, two component system (simple eutectic system), Solid Solutions/Alloys, Hume-Rothery rules for Alloy formation. Catalysis and its Industrial applications.	9	2,3	CO2	PSO2
3	ENGINEERING CHEMISTRY Polymers/Macro-molecules, classification, thermal and mechanical properties. Green chemistry, Corrosion and passivation, Nano materials	9	3,4	CO3	PSO2
4	CERAMIC MATERIAL Ceramics & refractory materials, Polymer Matrix Composites, Fuel cells	9	3,4	CO4	PSO4, PSO5

5	DYE SENSITIVE SOLAR CELLS Dye-sensitized solar cells ,Portland Cement, Composition, Role of Gypsum and Plaster of Paris in Setting of Cement,Organic Light Emitting Diodes –OLEDs	9	5,6	CO5	PSO2
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Text books:

1. P. W. Atkins, Physical Chemistry, Oxford University Press, 9th Ed., 2011.
2. R. T. Morrison and R. N. Boyd, Organic Chemistry, Prentice Hall of India Pvt. Ltd., 5th Ed., 1990.
3. J. D. Lee, Concise Inorganic Chemistry, 4th Ed., ELBS, 1991.
4. D. K. Chakravarty, B. Vishwanathan, Heterogeneous Catalysis, New Age International, New Delhi, 2007.
5. Wiley Engineering Chemistry, 2nd Edition, Wiley India Pvt. Ltd., New Delhi, 2013.

Reference books:

1. G. M. Barrow, Physical Chemistry, 5th Ed., Tata McGraw – Hill, New Delhi, 1992.
2. D. A. McQuarrie & J. D. Simon: Physical Chemistry – A Molecular Approach, Viva Books, 1998.
3. L. G. Wade, Organic Chemistry, Pearson Education 6th Ed., 2006.
4. G. Solomons and C. Fryhle, Organic Chemistry, John Wiley & Sons (Asia) Pvt. Ltd.
5. M. J. Sienko and R. A. Plane: Chemical Principles and Applications, McGraw Hill, 1980.
6. Michael Smith, Jerry March, March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, 2007.
7. Pietro Tundo, Alvis Perosa, Fulvio Zecchini, Methods and Reagents for Green Chemistry: An Introduction, Wiley Interscience, 2007.
8. M.C. Gupta and A.P. Gupta, Polymer Composite, New Age International, New Delhi, 2007.
9. Charles P. Poole, & Frank J Owens Jr., Introduction to Nanotechnology, John Wiley and Sons, Inc., 2003.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS234		Course Name	Digital Logic Design	
Year	1	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018-2019	Prerequisites (If any)	Computer Workshop, Introduction to Computers
Qualifies for	Introduction to Microprocessor, Computer Organization and Architecture				
Course Description	The course introduces students to Foundation in design and analysis of the operation of digital gates. Design and implementation of combinational and sequential logic circuits. Concepts of Boolean algebra, Karnaugh maps, flip-flops, register, and counter along with various logic families and comparison of their behaviour and characteristics				
Course Objective	1. To understand number representation and conversion between different representation in digital electronic circuits. 2. To analyze logic processes using boolean algebra 3. Implement logical operations using combinational logic circuits. 4. To understand characteristics of memory and their classification. 5. To understand concepts of sequential circuits and to analyze sequential systems in terms of statemachines.				
Course Outcomes	1. Perform the conversion among different numbersystems; 2. Understand Boolean algebra and basic properties of Boolean algebra 3. Understand basic logic gates -- AND, OR & NOT, XOR, XNOR and Build simple logic circuits using basic logic gates 4. Design combinational logis circuit. 5. Design sequential logic circuits				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	NUMBER SYSTEM AND CODES Number systems, Complement Number Representation, Addition, Subtraction, Multiplication and Division using Different Number Systems; Representation of Binary Number in Sign-Magnitude, Sign 1's Complement and Sign 2's Complement Notation; Rules for Addition and Subtraction with Complement Representation;BCD,EBCDIC,ASCII,ExtendedASCII,Gray and other Codes.	8	1,2	CO1	PSO1
2	BOOLEAN FUNCTION AND ITS MINIMIZATION Simplification of Boolean Function using Boolean theorems; Canonical and Standard Forms (SOP and POS) for Boolean	10	1,2	CO2	PSO2

	Functions; Objectives of the Minimization Procedures; Karnaugh Map Method; Don't Care Conditions.				
3	COMBINATIONAL CIRCUIT USING DISCRETE LOGIC GATES Half Adder and Full Adder; Half Subtractor and Full Subtractor; Parity Generator and Checker; Code Converters; Carry look ahead generator; Binary Multiplier; Majority Circuits, Magnitude Comparator. COMBINATIONAL LOGIC USING MSI CIRCUIT Binary Parallel Adder; BCD Adder; Encoder, Priority Encoder, Decoder; Multiplexer and Demultiplexer Circuits; Programmable Logic Array (PLA) and Programmable Array Logic (PAL).	8	1,2,3	CO3	PSO3
4	FLIP FLOP CIRCUITS Basic Concepts of Sequential Circuits; Cross Coupled SR Flip-Flop Using NAND or NOR Gates; D-Type and Toggle Flip-Flops JK Flip-Flop & race Condition; Clocked Flip-Flops; Master Slave Configuration; Edge triggered D flip-flop; Elimination of Switch Bounce Using Flip-Flops; Flip-Flops With Preset and Clear	11	3,4	CO4	PSO4 PSO5
5	SEQUENTIAL LOGIC CIRCUIT DESIGN Sequential circuit; state table and state diagram; Design procedure; Basic Concepts of Counters and Registers; Shift Left and Shift Right Register; Registers With Parallel Load; Serial-In-Parallel-Out (SIPO) and Parallel-In-Serial-Out (PISO); Register Using Different Type of Flip-Flops; Ripple (asynchronous) counters; Up Down and Mod-N ripple counters; Design of Synchronous Counter Using State Diagrams and State Table; BCD Counters; Modulo-N Counter; Up Down Counter; Ring counter; Johnson Counter, Sequence Generators	8	5,6	CO5	PSO5,PSO6

Text books:

1. Morris Mano, "Digital Logic And Computer Design", Prentice Hall Of India, 2005.
2. A. Anand Kumar: "Fundamentals of Digital Circuits", Second Edition, Practice Using Integrated Circuits, TMH, 2013

Reference books:

1. Charles Kime: "Logic and Computer Design Fundamentals", Pearson Education, 2004.
2. Sandige Richard, "Modern Digital Design", McGraw-Hill, 1990.
3. Donald D. Givone, "Digital Principles and Design", Tata McGraw Hill, 2002.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS154		Course Name	Engineering Mathematics II	
Year	1	Core Subject (Yes/No):		Lecture:	3
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
		Year of Syllabus Revision:		Total Credit:	1
		Year of Introduction	2014-2015	Prerequisites (If any)	Nil
Qualifies for	Engineering Mathematics III				
Course Description	The course is aimed at developing the basic Mathematical skills for Engineering students that are imperative for effective understanding of Engineering subjects. The topics introduced will serve as basic tools for specialized studies in many Engineering fields				
Course Objective	This course is a fast-paced review of Algebra of one variable and an introduction to the first concepts of multivariable calculus. The course is intended to organize and review material through exercises.				
Course Outcomes	1. Know about necessary and sufficient condition for total differential equations. 2. Further students know about order differential equations and their transformations and solutions through methods. 3. Able to solve basic engineering models through partial differential equations such as wave equation, heat conduction equation etc. 4. Know about initial conditions and its applications to different engineering models. 5. Able to solve the problems of differentiation of functions.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	LINEAR ALGEBRA I Matrices and Determinants, linear transformations. Systems of linear equations. Gauss elimination, rank of a matrix, inverse of a matrix.	12	1,2	CO1	PSO1
2	LINEAR ALGEBRA II	12	2,3	CO2	PSO2
3	ORDINARY DIFFERENTIAL EQUATION I Ordinary Differential equations of the 1st order; exactness and integrating factors, Ordinary Differential equations of the 2nd order, variation of parameters, Picard's iteration method.	16	3,4	CO3	PSO3, PSO4
4	ORDINARY DIFFERENTIAL EQUATION II Ordinary linear differential equations of nth order. Solution of homogeneous and non homogeneous equations. Operator method,	10	4,5	CO4	PSO3, PSO4
5	ORDINARY DIFFERENTIAL EQUATION III Methods of undetermined coefficients and variation of parameters. Systems of differential equations.	6	5,6	CO5	PSO5

Text books:

1. E. Kreyszig, Advanced Engineering Mathematics, 9th Ed., Wiley India, 2011.

Reference books:

1. T.M. Apostol, Calculus, Vol. II, Wiley India, 2007.

2. W.E. Boyce and R.C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th ED., Wiley, International students' edition, 2009.

3. V. Krishnamurthy, V.P. Mainra and J.L. Arora, An Introduction to Linear Algebra, Affiliated East-West Press Pvt Ltd, New Delhi, 2007.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CE101		Course Name	Applied Mechanics	
Year	1	Core Subject (Yes/No):		Lecture:	2
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2014-2015	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	Applied mechanics bridges the gap between physical theory and its application to technology. It is the basis for most of engineering fields. This course will impart basic knowledge of engineering mechanics where in laws of physics are applied to solve engineering problems. This course covers rigid bodies and statics portion of mechanics. It provides a comprehensive knowledge and insight into the study of mechanics and its real life applications in the various fields of engineering.				
Course Objective	To familiarize students about the forces, moments, couples and its corresponding components. Ø To develop understanding of a rigid system and vector algebra. Ø To enable students to draw understand and analyze a free body diagram. Ø To enable students to obtain centroid and centre of gravity for two and three dimensional bodies. Ø To develop understanding on analysis of trusses, friction and hydrostatic forces.				
Course Outcomes	CO1 Understanding role and application of Mechanics in engineering discipline, Analysis of the rigid body, Resolution of forces, evaluating the resultant and determining the unknown forces by applying law of equilibrium and through knowledge of vector algebra. CO2 To be able to calculate moments in equivalent system of rigid bodies, Resolution of the moments and determination of couple of forces. CO3 Analyse trusses for determining the unknown support reactions and forces through the application of principle of equilibrium. CO4 Computation of concept of centre of gravity and centroid by different approaches. CO5 Understanding the laws of dry Friction, coefficients of friction, angles of friction and hydrostatics				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	STATICS OF PARTICLES Forces in a Plane: Force on a Particle. Resultant of Two Forces, Vectors, Addition of Vectors, Resultant of Several Concurrent Forces, Resolution of a Force into Components, Rectangular Components of a Force. Equilibrium of a Particle, Free-Body Diagrams. Forces in Space: Rectangular Components of a Force in Space, Equilibrium of a Particle in space.	6	1,2,3	CO1	PSO1,PSO3
2	RIGID BODIES: Equivalent system of forces; Principle of Transmissibility. Equivalent Forces, Vector Moment of a Force about a Point, Varignon's Theorem, Rectangular Components of	8	1,2,3	CO2	PSO1,PSO3

	the Moment of a Force, Moment of a Force about a Given Axis, Moment of a Couple, Equivalent Couples ,Addition of Couples, Resolution of a Given Force into a Force at O and a Couple.				
3	Equilibrium of Rigid Bodies: Free-Body Diagram, Equilibrium in Two and three Dimensions: Reactions at Supports and Connections for a Two-Dimensional and three dimensional Structure, Equilibrium of a Rigid Body in Two and three Dimensions.	6	2,3,4	CO3	PSO1,PSO2,PSO3
4	Distributed Forces: Centroids and center of Gravity: Areas and Lines: Center of Gravity of a Two-Dimensional Body, Centroids of Areas and Lines, First Moments of Areas and Lines, Composite Plates and Wires, Determination of Centroids by Integration	8	2,3,4	CO4	PSO1,PSO2,PSO3
5	Trusses, Friction and Hydrostatic forces: Definition of a Truss, Simple Trusses, Analysis of Trusses by the Method of Joints, Joints under Special Loading Conditions, Analysis of Trusses by the Method of Sections, Friction and Hydrostatic forces.	4	1,2,3	CO5	PSO1,PSO3

Text books:

1. F.P. Beer, E. R. Jhonston, D. F. Mazurek, P. Cornwell and E. Eisenberg: Vector Mechanics for Engineers: Statics and Dynamics, 9th Ed. Tata McGraw Hill, New Delhi, 2009.
2. J L Meriam and L. G. Kraige: Engineering Mechanics Statics, Volume 1, 6th Ed., Wiley India Pvt. Ltd, 2006.
3. J L Meriam and Kraige: Engineering Mechanics: Dynamics, 6th Ed., Wiley India Pvt. Ltd, 2010.

Reference books:

1. I H Shames: Engineering Mechanics: Statics and Dynamics, 4th Ed., Prentice – Hall of India Pvt. Ltd, New Delhi, 2004.
2. Stephen Timoshenko, J V Rao, D H Young, Engineering Mechanics 4th Edition, Tata McGraw-Hill Education, 2006.
3. R. C. Hibbeler, Ashok Gupta Engineering Mechanics - Statics and Dynamics 11th Edition, Pearson Education, 2009
4. Veenu Kumar, K. L. Kumar, Engineering Mechanics 4th Edition, Tata McGraw Hill Education, 2010.
5. Jacob P. DenHartog, Mechanics, Dover Publications, 1961.
6. M.S. Sivakumar, Stephen Crandall, Thomas Lardner, Norman Dahl, an Introduction to Mechanics of Solids, Tata McGraw Hill Education, 2012.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	ME113		Course Name	Engineering Graphics	
Year	1	Core Subject (Yes/No):		Lecture:	2
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	3
		Year of Syllabus Revision:		Total Credit:	3.5
	Practical	Year of Introduction	2012-2013	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	An Engineering Graphics is used to fully and clearly define requirements for engineered items. More than merely drawing of pictures, it is also a language - a graphical language that communicates ideas and information from one mind to another. The subject shall help students to understand the fundamentals and applications of Engineering Drawing, prepare 2D Drawings on drawing sheet as well as using Software (AutoCAD), Understand different types of drawings and its features, Use of drawing instruments. Introduced about standard methods of the drawing used worldwide. Also, to improve imagination power and creative				
Course Objective	1. To develop in students, graphic skills for communication of concepts, ideas and design of Engineering products. 2. To expose them to existing national standards related to technical drawings. 3, Importance of graphics in engineering applications – Use of drafting instruments – BIS conventions and specifications – Size, layout and folding of drawing sheets – Lettering and dimensioning.				
Course Outcomes	CO1 Understand standard concepts related drawings and tools used. CO2 Understand methods of representing standard objects in 2D. CO3 Understand the mathematics and methodologies behind designing of various curves CO4 Understand visualization and standard methods of representation. CO5 Reading and making of technical drawings CO6 Understand 2D CAD software such as AutoCAD and introduction to 2D Modeling in Creo.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to drawing instruments, lettering, lines and dimensioning, construction of simple geometrical figures.	4	1,2,3	CO1,CO2	PSO1
2	Simple orthographic projections, orthographic projections of points and lines. Orthographic projections of planes and solids, sections of solids. isometric views	10	1,2,3	CO2	PSO1
3	Development of surfaces, intersection of surfaces, isometric projections of combined solids, special curves	10	1,2,3,6	CO3,CO4	PSO1
4	Introduction to visualization and conceptual drawing concepts Introduction to different types of drawing used in industries. Different standards used in industries to make different drawings	6	1,2,3	CO5	PSO1,PSO3

	of a component/part or assembly. Reading of different drawings by providing examples from industry.				
5	Introduction to 2D CAD software, use of 2D CAD software for topics mentioned in Unit1 to 3	45	1,2,3,6	CO6	PSO1,PSO5

Text books:

Nil

Reference books:

1. N. D. Bhat and V. M. Panchal, Engineering Drawing , Charotar Publishing House, 2011
2. K. Venugopal, Engineering Drawing and Graphics + AutoCAD, New Age International, 2011
3. B. Agrawal and C. M. Agrawal, Engineering Drawing, Tata McGraw-Hill Publishing Company Limited, 2008.
4. K. R. Gopalakrishna, Engineering Drawings, Subhas Stores, Bangalore, 2001.
5. N. Sidheswar, P. Kanniah and V. V. S. Sastry, Machine Drawing, Tata-McGraw Hill, New Delhi, 1980.
6. T. E. French, C. J. Vireck and R. J. Foster, Engineering Drawing and Graphic Technology, 14th Ed, McGraw Hill, New York, 1993.
7. W. J. Luzadder and J. M. Duff, Fundamentals of Engineering Drawing, Prentice-Hall India, New Delhi, 2001.
8. Arunoday Kumar Engineering Graphics I and II, Tech–Max Publication, Pune.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS157		Course Name	Web Technology	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2017-2018	Prerequisites (If any)	Introduction to Programming, Web Technology
Qualifies for	Advanced Web Technology				
Course Description	This course is an introduction to web design and development of website in the information fields. It is designed to provide understanding of basic web development concepts and process, web design best practices, web page creation with X/HTML and CSS, web client-side scripting using JavaScript and server-side. The acquired knowledge will be used in and reinforced by developing a website that aligns to a set of user specified requirements.				
Course Objective	1.To get familiar with the fundamentals of Web and Internet technologies. 2.To design web pages using HTML, CSS, XML along with the scripting languages. 3.To understand the and implement server-side programming. 4.To learn and implement mysql database for website development. 5. To able to make use of JQuery and Ajax.				
Course Outcomes	1.To be able to learn basic concepts of Internet and Web Technologies. 2.To be able to design Website using HTML, CSS, XML and scripting languages. 3.To be able to expose to server-side programming languages. 4. Make web applications in the given scripting languages 5.To be able to understand and design mysql database using PHP.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION HTML: Background of HTML, CSS: Background of CSS, XML: Background of XML.	4	1,2	CO1 CO2	PSO2
2	JAVASCRIPT AND JSON JavaScript: JS Data types, Operators, Variables, Functions, Conditionals, Arrays, Objects, JS Events, and JS Forms. JSON: Introduction to JSON, JSON Syntax, JSON data types, JSON Objects, JSON Arrays, JSON Parse, JSON HTML.	5	1,2	CO1 CO2	PSO1
3	WORDPRESS Overview: Introduction of different Web Technology, What is WordPress, How WordPress Works, Setting up and Installing WordPress: Installation of server, Installation of WordPress, Installation of MY SQL, Website template customization, Website development.	5	6	CO4	PSO6
4	MySQL Introduction about Database, Data Types, DML, DDL, Aggregate functions, Data Time functions, Sub query and join, MySQL Introduction, MySQL Connect, MySQL Create, MySQL Insert,	5	1, 3	CO3 CO5	PSO1

	MySQL Select, MySQL Where, MySQL Order By, MySQL Update, MySQL Delete, phpMyAdmin.				
5	AJAX AND JQUERY Introduction to Ajax, setting up and implementing Ajax, PHP and AJAX, AJAX Database, AJAX XML, AJAX Live Search, AJAX RSS Reader, AJAX Poll. Introduction to jQuery, jQuery effects, jQuery HTML, jQuery AJAX, JQuery Widgets, JQuery Interactions.	11	2,1	CO2 CO3	PSO1, PSO6

Text books:

1. Web Programming, building internet applications, Chris Bates 2nd edition, Wiley Dremtech
2. Java Server Pages – Hans Bergsten, SPD O'Reilly
3. Java Script, D.Flanagan, O'Reilly, SPD

Reference books:

1. Beginning Web Programming-Jon Duckett WROX.
2. Programming world wide web, R.W. Sebesta. Fourth Edition, Pearson.
3. Internet and World Wide Web – How to program, Dietel and Nieto, Pearson.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS231		Course Name	Object Oriented Programming using Python	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2019-2020	Prerequisites (If any)	Programming in Python
Qualifies for	Advance Python Programming				
Course Description	This course in computer science is designed to introduce students to top-down structured problem-solving, algorithm development, procedural and data abstraction. Topics include selection, repetition, and sequence control structures; program design, development, test, and documentation; coding style, readability and maintainability; array, record and file manipulation. Students will examine elements of a basic python programming language.				
Course Objective	1. To teach basic object oriented programming concepts 2. To study Classes & Objects 3. To Study Input-Output, Modules & Packages 4. To Study OOP Building Blocks 5. Implement small projects using OOP				
Course Outcomes	Student will be able to 1. To implement basic object oriented programming concepts 2. To make programmes using Classes & Objects 3. To use Input-Output, Modules & Packages of python 4. Learn OOP Building Blocks 5. Implement projects using OOP paradigm				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION TO OOP What is object oriented programming? Why do we need object oriented. Programming characteristics of object oriented languages. Introduction to Python, Basics of a Typical Python Environment, Pre-processors Directives, Illustrative Simple Python Programs. Header Files and Namespaces, library files. Concept of objects, basic of object modeling, object classes, Specifying attributes and behaviors, Real time use of class	8	2	CO1	PSO6
2	CLASSES & OBJECTS Defining and using a class ,Instance attributes, Class attributes , Class decorators , Creating and initializing an object, New Style Classes, Inheritance and Mixins, Properties and Slots, Static and Class Methods , Instance Method, Abstract Base Classes, Method Overriding, Attributes and Functions , Decorators and Factories, Descriptors and Meta Classes	9	1	CO2	PSO1
3	INPUT-OUTPUT, MODULES & PACKAGES Files: Printing on screen, reading data from keyboard, Opening and closing file, Reading and writing files, File Position, File Attributes. Environment: sys	10	1,2	CO3	PSO2

	module, Command-line arguments, Standard data streams, Redirections, Exiting the program, os module, Environment variables, Working with directories, *args and **kwargs, Functions basics, The stack, Default parameters, globals() and locals() Decorators, Monkey patching, Lambdas, Generator functions and yield, Filter and map, is and id, Iterators, Garbage collector. Single/multi-dispatch, Modules: Importing module, Executing modules as scripts, Math module, Random module, Packages				
4	OOP BUILDING BLOCKS Introduction, Composition, Inheritance, More about inheritance, Avoiding inheritance, Composition, Encapsulation, Init Constructor, Polymorphism, Overriding Inheritance, Multiple Inheritance and the Lookup Tree, Exception, Exception Handling, Except clause, Try? finally clause, User Defined Exceptions	10	2	CO4	PSO1
5	EXTENDING AND EMBEDDING PYTHON Calling C/C++ from Python, Using ctypes, Extension Modules in C/C++, Raising Python Exceptions, Calling Python from C/C++, Embedding Python Interpreter, Importing Python Modules from C/C++, Converting Python Objects to C/C++, Invoking Python Functions from C/C++	8	3	CO5	PSO2

Text books:

1. John V Guttag. "Introduction to Computation and Programming Using Python", Prentice Hall of India
2. R. Nageswara Rao, "Core Python Programming", dreamtech
3. Wesley J. Chun. "Core Python Programming - Second Edition", Prentice Hall
4. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, "Data Structures and Algorithms in Python", Wiley

Reference books:

1. Kenneth A. Lambert, "Fundamentals of Python – First Programs", CENGAGE Publication
2. Luke Sneeringer, "Professional Python", Wrox
3. "Hacking Secret Ciphers with Python", Al Sweigart, URL <https://inventwithpython.com/hacking/chapters>

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS233		Course Name	Introduction to Microprocessor	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2016-2017	Prerequisites (If any)	Digital Logic Design
Qualifies for	Nil				
Course Description	The course introduces students to the architecture and instruction set of typical 8-bit microprocessor. It also deals with Assembly Language Programming using a macro-assembler. Input-output techniques and important programmable support chips used in microprocessor-based systems are discussed in detail.				
Course Objective	1.Introduce concepts of Microcomputer System 2.To teach basic microprocessor organization. 3.To teach microprocessor instruction set and assembly level programming. 4.Introduce higher generations of coprocessors. 5.Introduce higher generations of microprocessors				
Course Outcomes	1. To Understand concepts of Microcomputer System 2. To Understand the 8086 microprocessor and its functionality 3. To program the 8086 microprocessor instruction set and assembly level programming. 4. To provide knowledge about Coprocessor 5. To aware students about higher generations of microprocessors				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION OF MICROCOMPUTER SYSTEM CPU, I/O devices, clock, memory, bussed architecture, tristate logic, address bus, data bus and control bus. Semiconductor Memories: Development of semiconductor memory, internal structure and decoding, memory read and write timing diagrams, MROM, ROM, EPROM, EEPROM, DRAM Architecture of 8-bit Microprocessor: Intel 8085A microprocessor, Pin description and internal architecture	10	2	CO2	PSO2
2	EVOLUTION OF MICROPROCESSOR Internal microprocessor (8086 to Pentium) architecture of 8086; Programming Model, Real mode memory addressing, Introduction to protected mode memory addressing memory paging. Addressing modes: Data, program, Stack, memory-addressing modes	10	1	CO1	PSO6
3	INSTRUCTION SET OF 8086 Assembly language programming for 8086 microprocessor, Memory Segmentation. 16 and 32 – bit memory interfacing, various bus protocols like ISA, EISA, VESA, and PCI. Architecture Co-	10	1,2	CO3	PSO2

	processor (8087), programming with 8087, Multi-Processor System, Introduction to MMX technology.				
4	INTRODUCTION TO COPROCESSOR Architecture of Co- processor (8087), programming with 8087, Multi-Processor System, Introduction to MMX technology.	5	3	CO5	PSO6
5	INTRODUCTION TO PENTIUM AND ITS HIGHER GENERATIONS Pentium architecture, memory management. Assembler, debugger, Introduction to bit Slice processor, Signal processing processor and transputers, Introduction to development tools , MDS , logic analyzer , in-circuit emulator	10	3	CO4 CO5	PSO1

Text books:

1. Barry B. Brey, "The Intel Microprocessors: Architecture, Programming & Interfacing" PHI, 6th Edition, 2003.
2. D. V. Hall, "Microprocessor and Interfacing Programming & Hardware" TMH – 2nd Edition.

Reference books:

1. Advance Microprocessor, Deniel Tabak, TMH.
2. The Intel Microprocessors (Eight Edition): Barry B. Brey, Pub-Pearson (Prentice Hall).
3. The 8086 Microprocessor, Kenneth Ayala, Cengage Learning.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	EN102		Course Name	Environmental Science	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018 - 2019	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This is a fundamental Environmental Science course that covers topics pertaining to understanding the environment and relates and evaluates important factors influencing it. It also highlights the need for environmental protection and conservation on an utmost important basis. Additionally, this course emphasizes on the health concerns and aspects of sustainable development. The Credit Structure has its own importance in terms of studying the perturbation factors leading change in environmental conditions.				
Course Objective	1.To know the names of micro-organisms present in soil, air and identify them. 2. Should know the applications of biotechnology in their particular field. 3. Should know the basics part so that they know how they need to apply in industries in the near future				
Course Outcomes	1. Gaining in-depth knowledge on natural processes that sustain life and govern economy. 2. Predicting the consequences of human actions on the web of life, global economy and quality of human life. 3. Developing critical thinking for shaping strategies (scientific, social, economic and legal) for environmental protection and conservation of biodiversity, social equity and sustainable developmen				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	MULTIDISCIPLINARY NATURE OF ENVIRONMENTAL STUDIES Definition, scope and importance. Problems associated with environment. Understanding role of an individual in protecting and conserving nature. L	2	1	CO1	PSO4
2	Natural resources, Ecosystems, Biodiversity Renewable and non-renewable resources, Natural resources and associated problems. Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people. Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. Case studies. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Role of an	18	1	CO2	PSO4

	individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles. Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem, Ecological succession, Food chains, food webs and ecological pyramids, Introduction, types, characteristic features, structure and function of the forest ecosystem, Desert ecosystem, Grassland ecosystem and Aquatic ecosystems. Biodiversity and its conservation: Introduction – Definition: genetic, species and ecosystem diversity. Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values. In-situ and Ex-situ conservation methods of biodiversity.				
3	ENVIRONMENTAL POLLUTION Air pollution & health: Types of air pollutants, causes, Impact on human health, plants, materials, stratosphere, climate. Photochemical smog, global warming and green house effect, acid rain. Particulate matter, Occupational diseases, Indoor air pollution. Air quality guidelines in protecting public health and on climate change and its effects. Water pollution: Causes and effects of water pollution, physical, chemical and biological characteristics of water waste water treatment: primary, secondary and tertiary treatment methods. Control measures of water pollution.	15	2,4	CO2	PSO1
4	SOIL POLLUTION AND IT'S MANAGEMENT Soil pollution: causes, effects and control measures. Solid waste management: Sources and classification of solid waste, causes of solid waste pollution, Effects of solid waste pollution	8	1,2	CO3	PSO3
5	Control measures of urban and industrial wastes, examples of waste utilization	2	2,3	CO1	PSO4

Text books:

1. Air pollution: M. N. Rao & H. V. N. Rao
2. Water supply and sanitary Engineering: G. S. Birdi.
3. Waste Water supply engineering: S. K. Garg
4. Waste Water supply engineering: B. C. Punamia
5. Environmental Studies: Suresh K. Dhamejia

Reference books:

1. Bridge, J. & Demicco, R. 2008. Earth Surface Processes, Landforms and Sediment deposits. Cambridge University Press.
2. Duff, P. M. D. and Duff, D. (Eds.). 1993. Holmes' Principles of Physical Geology. Taylor & Francis.
3. Gupta, A. K., Anderson, D. M., & Overpeck, J. T. 2003. Abrupt changes in the Asian southwest monsoon during the Holocene and their links to the North Atlantic Ocean. Nature 421: 354-357

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS239		Course Name	Discrete Mathematics	
Year	2	Core Subject (Yes/No):		Lecture:	3
Semester	III	Elective Subject (Yes/No):		Tutorial	1
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	
	Tutorial	Year of Syllabus Revision:		Total Credit:	4
		Year of Introduction	2012-2013	Prerequisites (If any)	Mathematics-II
Qualifies for	Mathematics				
Course Description	In Mathematics III, topics covered are Partial differential equations, Functions of complex variables, Laplace transforms and Numerical methods. Classical methods involved in the formation of PDE's and the methods from finding their solutions are discussed. Models leading to II order PDE's and their solutions using the method of separation of variables are explained. Complex analysis is a branch of mathematics which is extensively used in electrical, nuclear, aerospace and mechanical engineering. Laplace transform is used in technology which transforms from time domain to frequency domain, transforms differential equations to algebraic equations and from convolution to multiplication. Numerical method is a course designed for engineers and scientists who look for approximate numerical solutions to the mathematically modeled phenomena instead of the analytical and graphical methods. The method of teaching includes lecturing followed by tutorials so that the students learn to apply the theoretical concepts of mathematics to the contextual problems. Students are taught to evaluate the iterations in Numerical Analysis manually to understand the convergence process of the approximate solution to the actual solution.				
Course Objective	1. To develop logical understanding of the subject. 2. To develop mathematical skill so that students are able to apply mathematical methods & principals in solving problem from Engineering fields. 3. To make aware students about the importance and symbiosis between Mathematics and Engineering.				
Course Outcomes	After completing the course, students should be able to: 1. Apply the fundamental concepts of Ordinary Differential Equations and Partial Differential Equations and the basic numerical methods for their resolution. 2. Solve the problems choosing the most suitable method. 3. Understand the difficulty of solving problems analytically and the need to use numerical approximations for their resolution. 4. Use computational tools to solve problems and applications of Ordinary Differential Equations and Partial Differential Equations.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	PARTIAL DIFFERENTIAL EQUATIONS: Formation of PDE, linear PDE of first order ($P \cdot p + Q \cdot q = R$), non linear PDE of first order, equations of the form: $f(p, q) = 0$, $z = px + qy + f(p, q)$, $f(z, p, q) = 0$, $f(x, p) = f(y, p)$, homogeneous linear PDE with constant coefficients, Second order partial differential equations, classification in to elliptic, parabolic and hyperbolic equations, models leading to 2nd order PDEs, boundary value problems involving 2nd order PDEs and their solution by methods of	14	1,5	CO1 CO3	PSO1

	separation of variables, heat, wave and Laplace equations and their solution by method of separation of variables and using Fourier series.				
2	LAPLACE TRANSFORMS: Definition, Linearity property, Laplace transforms of elementary functions, Shifting theorem Inverse Laplace transforms of derivatives and integrals, Convolution theorem, Application of Laplace transforms in solving ordinary differential equations, Laplace transforms of periodic, Unit step and Impulse functions.	14	1,5	CO1 CO2	PSO3
3	NUMERICAL METHODS: Finite difference, Newton's interpolation formulae, Lagrange's formula, numerical integration: trapezoidal rule, Simpson's rule. Solution of equations of one variable: Bisection method, Newton – Raphson method. Solution of system of linear equations: Gauss – Seidel and Jacobi methods. Numerical solution of differential equation: Euler's method, modified Euler method, Runge – Kutta method, finite difference methods for ordinary differential equations.	14	2,5	CO1	PSO1
4	FUNCTIONS OF COMPLEX VARIABLE: Limit, continuity and derivative of functions of a complex variable, analytic functions and their applications: Harmonic, conjugate harmonic and potential functions; Linear transformations of complex domains.	7	2,5	CO2	PSO1
5	TRANSFORMATION: Some special transformations, bilinear transformations, transformation by some standard functions, conformal mapping, applications of transformations and conformal mappings.	7	3,2	CO3	PSO3

Text books:

1. Dennis G. Zill., "Advanced Engineering Mathematics", IIIrd Ed., Narosa, 2009
2. E. Kreyszig., "Advanced Engineering Mathematics", 9th Ed., Wiley India, 2011
3. C. Henry Edwards, David E. Penny, Differential Equations and Boundary Value Problems Computing and Modeling, 3rd Ed., Pearson, 2013.

Reference books:

1. Grimaldi, R.P. "Discrete and Combinatorial Mathematics: An Applied Introduction", 4th Edition, Pearson Education Asia, Delhi, 2007.
2. Lipschutz, S. and Mark Lipson., "Discrete Mathematics", Schaum's Outlines, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 3rd Edition, 2010.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS229		Course Name	Data Structures	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2017-2018	Prerequisites (If any)	Introduction to Programming
Qualifies for	Design Analysis and Algorithm				
Course Description	The course introduces about how to design, write, and analyze the performance of C/C++ programs that handle structured data and perform more complex tasks, typical of similar to module of a larger software projects. The course aims at developing the skills in using generic principles for data representation & manipulation with a view for efficiency, maintainability and code reusability.				
Course Objective	1. The objective of the course is to teach 'c' programming (with an emphasis on problem solving) and introduce elementary data structures. 2. To introduce the fundamental concept of data structures and to emphasize the importance of data structures in developing and implementing efficient algorithms. 3. Topics include elementary data structures, (including arrays, stacks, queues, and lists), advanced data structures (including trees and graphs), the algorithms used to manipulate these structures, and their application to solving practical engineering problems. 4. Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data. 5. To explain the importance of the subject for various engineering fields and their uses over there.				
Course Outcomes	Students will be able to : 1. Formulate and apply 'C' programming, as a modern tool to solve engineering problems. 2. Demonstrate an understanding of basic data structures (such as an array-based list, linked-list, stack, queue, binary search tree) and algorithms. 3. Demonstrate the ability to analyze, design, apply and use data structures and algorithms to solve engineering problems and evaluate their solutions. 4. Describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms. 5. Discuss the efficiency of the principal algorithms for sorting, searching.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	REVIEW OF 'C' - ARRAYS, POINTERS Functions: Parameter passing call by value and call by reference, scope rules, functions and pointers, function returning pointer and pointer to function, String manipulations using arrays, pointer to pointer. Structure and Union: Passing and returning structure as parameter for function, structure and pointer. Recursion: Definition, writing recursive functions & how recursion works. File handling using C. Introduction to Data Structures: Concept of data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C'.	10	1,2	CO1 CO3 CO5	PSO1 PSO2

	Concept of Primitive and non primitive, linear and non-linear, static and dynamic, persistent and ephemeral data structures.				
2	LINEAR DATA STRUCTURES USING SEQUENTIAL ORGANIZATION Concept of sequential organization, Concept of Linear data structures, arrays as ADT, Storage representation of array – Row major and Column major & their address calculation, Multidimensional arrays, Concept of ordered list. Applications: Concept of Sparse Matrix, it's usage & representation using arrays, Algorithms for sparse matrix operations like addition, simple transpose, fast transpose & multiplication.	7	2,4	CO2 CO3 CO4	PSO1
3	LINEAR DATA STRUCTURES USING LINKED ORGANIZATION Limitations of static memory allocation. Dynamic memory allocation in C. Concept of linked organization, singly linked list, doubly linked list, circular linked list. Operations like insertion, deletion, traversal & other operations on these data structures. Applications: Representation & manipulation of polynomials using circular linked lists, Application of doubly linked list in dynamic storage management, garbage collection and compaction. Representation of polynomial using generalized linked list.	7	2,4	CO3 CO4	PSO3 PSO6
4	STACKS AND QUEUES Stacks: Concept of stack as ADT, Representation and implementation of stack using sequential & linked organization. Applications: Examples using implicit stack, Simulating recursion using explicit stack, Arithmetic expression conversion & evaluation, reversing a string, parsing: well- formed parenthesis checking, concept of multi-stack & it's representation. Queues: Concept of queue as ADT, Representation and implementation of linear queue & circular queue using sequential & linked organization. Applications: Josephus problem, Job scheduling, Queue simulation, Double ended queue	8	3,5	CO3 CO4	PSO6
5	SORTING AND SEARCHING TECHNIQUES & TREE AND GRAPH Need of sorting and searching, sorting order & stability in sorting. Sorting Techniques: Algorithms for Bubble sort, Selection sort, Insertion sort, Quick sort and Merge sort. Searching Techniques: Algorithms for Sequential search, Binary search, Fibonacci search & concept of Index Sequential search. Binary Trees: Definition and properties, Array representation of binary tree, Tree Traversals: Infix, Prefix and Postfix, Binary Search Tree, Priority Heap, applications of binary trees: Gaming, Expression and decision trees, Binary Search Tree, AVL Tree. Graph: Definition, Array representation of graph, Minimum spanning trees: Kruskal's and Prim's algorithms, shortest path algorithms, Max-flow min-cut theorem (transport network).	13	5,6	CO4 CO5	PSO1 PSO6

Text books:

1. Data Structures & Program Design In C, 2/E by Robert Kruse, C.L. Tondo, Bruce Leung, Shashi Mogalla.

Reference books:

1. An Introduction to Data Structures with Applications. by Jean-Paul Tremblay & Paul G. Sorenson Publisher-Tata McGraw Hill.
2. Data Structures using C & C++ -By Ten Baum Publisher – Prentice-Hall International.
3. Fundamentals of Computer Algorithms by Horowitz, Sahni, Galgotia Pub. 2001 ed.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS330		Course Name	Software Engineering	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018 - 2019	Prerequisites (If any)	Database Management Systems(CS237) Object Oriented Programming (CS 231) Web Technology (CS 227)
Qualifies for	Project				
Course Description	This course is aimed at helping students build up an understanding of how to develop a software system from scratch by guiding them thru the development process and giving them the fundamental principles of system development with object oriented technology using UML. The course will initiate students to the different software process models, project management, software requirements engineering process, systems analysis and design as a problem-solving activity, key elements of analysis and design, and the place of the analysis and design phases within the system development life cycle.				
Course Objective	1. Understand and choose appropriate process model. 2. Analyze the system requirement. 3. Understand design specification of software development. 4. Develop umbrella activities on system (RMMM, SQA, SCM) 5. Evaluate system design with test cases.				
Course Outcomes	Student will be able to 1. To write and survey solution for multidisciplinary case-study to choose appropriate Process model. 2. To design various system models with respect to different approaches. 3. To implicate various design specification of software. 4. To perform Cost/Effort estimation Planning. 5. To write different test cases according to system design.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION : Introduction to Software Engineering, Software Components, 8 Software Characteristics, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes, Software Quality Attributes. Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement	4	2,4	CO1	PSO1

	Models. Typical Application for each model, Agile methodology, Process and Project Metrics				
2	SYSTEM ANALYSIS : System Requirement, Stakeholders, Techniques for information gathering, Validating the requirements Feasibility Analysis, Tests for feasibility, Cost Benefit Analysis, Feasibility report. The system Proposal Data flow diagram and UML analysis diagrams	13	4	CO2	PSO2
3	SOFTWARE DESIGN : Software Design Specification– Abstraction, Modularity– Effective modular design, Cohesion and Coupling, Example of code for cohesion and coupling User Interface Design – Human Factors, Interface standards– User Interface Design Process Application Architecture for Information System, Deployment using UML diagrams, Component and deployment diagram for various architectures Design Patterns – Gang of four patterns	9	3,6	CO3	PSO3
4	SOFTWARE QUALITY ASSURANCE : Software estimation – Empirical estimation models – Cost/Effort estimation Planning – Work breakdown Structure, Gantt Chart. Evaluate schedule and cost variance Risk Management Risk Identification, Risk Assessment, Risk Projection, RMMM Software Configuration Management, SCM process, version and change control Quality metrics	10	4,5	CO4	PSO6
5	SOFTWARE TESTING : Software Testing Basic concept and terminology, Verification & validation, White Box Testing-, Black Box Testing, Integration, Validation and system testing, Object Oriented testing methods- Class Testing, Interclass testing, testing architecture, Behavioural testing	6	4,5	CO5	PSO6

Text books:

1. Roger Pressman, "Software Engineering: A Practitioners Approach", 6th Edition, McGraw Hill, 2010.
2. Ian Sommerville, "Software Engineering", 9th Edition, Addison Wesley, 2011
3. James F. Peters and Witold Pedrycz, "Software Engineering – An Engineering Approach", Wiley, 2000.

Reference books:

1. Roger S. Pressman, Software engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Ian Sommerville, Software engineering, Pearson education Asia
3. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
4. Software Engineering Fundamentals by Ali Behforoz & Frederick Hudson OXFORD

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS227		Course Name	Database Management System	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2017-2018	Prerequisites (If any)	Introduction to Computers, Introduction to programming, Data Structure
Qualifies for	Software Engineering				
Course Description	This course introduces students to basic database design and the use of databases in applications. Throughout the course students are introduced to Database and Database Management System, advantages and disadvantages of the different database models, comparison of relational model with the Structured Query Language (SQL); Understanding constraints and controversies associated with relational database model as well as the rules guiding transaction ACID. It includes the design using various diagrams like ER, Sequence, and Activity. The designed system will produce the 'reports' for specific queries.				
Course Objective	1.To introduce fundamental concepts of DBMS. 2.To design and create the Database model. 3.To understand the Normalization techniques to reduce the data Redundancy. 4.To Learn the query processing system, recovery techniques and management of transactions in database. 5. To empower and expose the students with database implementation using SQL and PL/SQL languages.				
Course Outcomes	1.To be able to apply analytical skills to map out the conceptual design for a real-life problem and explain the fundamental elements of Database management System. 2.To be able to draw the model of Relational Database using Entity Relationship (ER) model 3.To be able to Improve the database design by normalization. 4.To be able to evaluate the logical design to translate into a specific data model and physical design to meet system storage requirements. 5.To be able to evaluate Oracle features and Oracle related products for maintaining the integrity and performance of databases.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	DATA AND INFORMATION Data and Information: Limitations of Manual Data Processing, Advantages of DBMS over traditional computer file-based processing approach, Functions of DBA Elements of DBMS: DDL, DML, Entities, Sets and attributes. Data Base Tables: Keys - Primary, Secondary, Composite.	7	1	CO1	PSO1
2	RELATIONAL MODEL AND ENTITY-RELATIONSHIP MODEL Structure of relational databases, Domains, Relations, Relational algebra – fundamental operators and syntax, relational algebra	10	1,2	CO2	PSO2

	queries. Entity Relationship model: Basic concepts, Design process, constraints, Keys, Design issues, E-R diagrams, weak entity sets, extended E-R features – generalization, specialization, aggregation, reduction to E-R database schema.				
3	RELATIONAL DATABASE DESIGN Functional Dependency – definition, trivial and non-trivial FD, closure of FD set, closure of attributes, irreducible set of FD, Normalization – 1NF, 2NF, 3NF, Decomposition using FD-dependency preservation, BCNF, Multivalued dependency, 4NF, Join dependency and 5NF.	7	2,6	CO3	PSO2 PSO1
4	QUERY AND TRANSACTION MANAGEMENT Query Processing & Query Optimization: Overview, measures of query cost. Transaction Management: Transaction concepts, properties of transactions, serializability of transactions, testing for serializability, System recovery, Two Phase Commit protocol, Recovery and Atomicity, Log-based recovery, concurrent executions of transactions and related problems, Locking mechanism, solution to concurrency related problems, deadlock, two-phase locking protocol, Isolation, Intent locking	9	2,5	CO4	PSO6 PSO2
5	SQL AND PL/SQL CONCEPTS SQL Concepts: Basics of SQL, DDL, DML, DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator, aggregate functions, Built-in functions –numeric, date, string functions, set operations, sub-queries, correlated sub-queries, join, Exist, Any, All, view and its types., transaction control commands. PL/SQL Concepts: Cursors, Stored Procedures, Stored Function, Database Triggers.	12	1	CO5	PSO2 PSO3

Text books:

1. An introduction to Database Systems by C J Date -Wesley.
2. Understanding SQL by Martin Gruber, -BPB

Reference books:

1. An introduction to Database Systems, C J Date, Addison-Wesley.
2. Database System Concepts, Abraham Silberschatz, Henry F. Korth & S. Sudarshan, McGraw Hill.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS307		Course Name	Computer Organization and Architecture	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	IV	Elective Subject (Yes/No):		Tutorial	1
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
	Tutorial	Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018 - 2019	Prerequisites (If any)	Introduction to computers and programming (CS 139)
Qualifies for	Embedded Systems				
Course Description	This course teaches the Von-Neumann Architecture of computer system in detail. Students learn about internal components and peripherals of a computer system. Advanced topics such as parallel processing, multicore computers and graphic processing units are taught in this course.				
Course Objective	1. To teach basic organization of computer system . 2. To understand central processing unit. 3. Study of pipelining and vector processing. 3. Study of parallel processing . 4. Study of Control unit operation .				
Course Outcomes	1. Comprehend organization of computer system and assembling of a computer 2. To understand the Central Processing Unit. 3. . Identify performance issues and Design issues in parallel processing 4. Understand when to use a GPU co-processor				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION TO COMPUTER SYSTEM: Computer organization and architecture, evolution, brief history, embedded systems, ARM architecture, Performance issues, Ahmdahl's Law, benchmarks and specification, computer components, functions, interconnection structures, Bus interconnection, PCI, Computer memory, cache memory, Internal memory, external memory, I/O modules, DMA, DCA, operating systems, Operating system support, Number systems, Computer Arithmetic, Positional Arithmetic, Binary system, Hexadecimal system, Digital Logic: Combinational circuits, gates, sequential circuits.	15	1	CO1	PSO1
2	THE CENTRAL PROCESSING UNIT : ,Instruction sets: Characteristics and functions, types of operands, types of operations, intel x86. Instruction sets: Addressing modes and formats, Addressing mode, assembly language. Processor structure and function: processor organization, register organization, ARM processor	7	1,2	CO2	PSO6

3	PIPELINE AND VECTOR PROCESSING: Reduced instruction set computers: RISC pipelining, MIPS, SPARC, RISC v/s CISC, compiler based register organization. Instruction Level Parallelism and Superscalar processors: Design Issues, intel core microarchitecture, ARM Cortex. Vector Processing, Array Processors	8	2,3	CO3	PSO2
4	PARALLEL ORGANIZATION : Parallel processing: Multiple processor organization, symmetric multiprocessor, Multithreading and chip microprocessors, clusters, non-uniform memory access, cloud computing Multicore computers: Hardware and software performance issues, multicore organization, heterogeneous multicore organization General purpose graphic processing unit: GPU v/s CPU, CUDA basics, intel generation 8 GPU.	7	3,4	CO4	PSO2
5	THE CONTROL UNIT : Control Unit operation: Micro operations, hardwired implementation Microprogrammed control: Microinstruction sequencing, microinstruction execution	8	3,4	CO2	PSO2 PSO6

Text books:

1. Computer Organization and Architecture: Designing for Performance (10th Edition), William Stallings, Prentice Hall Pearson.

Reference books:

1. M. Morris Mano, Computer System Architecture, Pearson
2. Andrew S. Tanenbaum and Todd Austin, Structured Computer Organization, Sixth Edition, PHI
3. M. Murdocca & V. Heuring, Computer Architecture & Organization, WILEY
4. John Hayes, Computer Architecture and Organization, McGrawHill

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS312		Course Name	Design analysis and algorithm	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	IV	Elective Subject (Yes/No):		Tutorial	1
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
	Tutorial	Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018 – 2019	Prerequisites (If any)	Introduction to computers and programming (CS 139), Data Structures
Qualifies for	Major Project				
Course Description	This course enables to develop problem solving by analysing algorithms and their running time. It also gives insight into analysing space complexity in order to write efficient programs.				
Course Objective	1. To teach theoretical concepts underlying computer algorithm development. 2. Teach problem solving by different methods. 3. Synthesize efficient algorithms in common engineering design situations. 4. To teach analysing algorithms for time and space complexity through different methods such as statistical methods and dynamic programming. 5. Ability to understand how the choice of data structures and the algorithm design methods impact the performance of programs.				
Course Outcomes	Student will be able to: 1. Choose methods for problem solving. 2. Analyze the performance of algorithms and analyse algorithms to find mean running time. 3. Understand different algorithmic design strategies. 4. To understand how the choice of data structures and the algorithm design methods impact the performance of programs. 5. Apply the algorithms and design techniques to solve problems;				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION TO ALGORITHMS AND ANALYSIS What is an algorithm? Role of Algorithm in computing, Sets, Functions and Relations, Vectors and Matrices, Linear Inequalities and Linear Equations, Functions and Sets, Growth Function, Probabilistic Analysis and Randomized Algorithms, Notations for algorithm analysis: Asymptotic Notations	8	1,4	CO1 CO4	PSO2
2	SORTING AND ORDER STATISTICS Heapsort, Quicksort, Sorting in Linear Time: Bucket sort, Radix sort and Counting sort, Medians and Order Statistics, Divide and Conquer method Introduction, Recurrence and different methods to solve recurrence, Multiplying large Integers Problem, Problem Solving using divide and conquer algorithm - Binary Search, MaxMin problem, Sorting (Merge Sort, Quick Sort), Matrix Multiplication, Exponential.	12	2,5	CO2	PSO1

3	ELEMENTARY DATA STRUCTURES Elementary Data Structures, Hash Tables, Binary Search Trees, Red-Black Trees, Augmenting Data Structures B-Trees, Fibonacci Heap, Van Emde Boas Trees, Data Structures for Disjoint Sets.	8	2,4	CO5	PSO1
4	ADVANCED DATA STRUCTURES Graph Algorithms: Elementary Graph Algorithms, Minimum Spanning Trees, Single Source Shortest Paths, All-Pairs Shortest Paths, Maximum Flow.	8	2,4,6	CO4 CO5	PSO2 PSO6
5	ADVANCED DESIGN AND ANALYSIS TECHNIQUES Dynamic Programming, Greedy Algorithms, Amortized Analysis, Multithreaded Algorithms, Matrix Operations, Linear Programming, Polynomials and the FFT, Number-Theoretic Algorithms, String Matching, Computational Geometry, NP-Completeness, Approximation Algorithms	9	2,3,5	CO3	PSO2

Text books:

1. Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles, Narasimha Karumanchi (Textbook)
2. Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest, "Introduction to Algorithms", Prentice Hall of India. (Textbook)

Reference books:

1. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, PHI.
2. Fundamentals of Algorithms by Gills Brassard, Paul Bratley, PHI.
3. Introduction to Design and Analysis of Algorithms, Anany Levitin, Pearson.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS322		Course Name	Information Security	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	2
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2019-2020	Prerequisites (If any)	Computer Networking (CS 308), Operating System (CS 310)
Qualifies for	Cyber Security				
Course Description	This course focuses on the fundamentals of information security that are used in protecting both the information present in computer storage as well as information travelling over computer networks. Interest in information security has been spurred by the pervasive use of computer-based applications such as information systems, databases, and the Internet. Information security has also emerged as a national goal in the United States and in other countries with national defense and homeland security implications. Information security is enabled through securing data, computers, and networks. In this course, we will look into such topics as fundamentals of information security, computer security technology and principles, access control mechanisms, cryptography algorithms, software security, physical security, and security management and risk assessment. By the end of this course, you will be able to describe major information security issues and trends, and advise an individual seeking to protect his or her data.				
Course Objective	1. Identify cryptographic algorithm to secure information. 2. Interpret concepts of security, authentication and authorization. 3. Illustrate the software security practices. 4. Categorize the attacks in each layer of OSI model. 5. Explain the different ways to secure a web.				
Course Outcomes	Student will be able to 1. Choose appropriate cryptographic algorithm to specified problem definition 2. Explain various concepts of Authentication and Authorization 3. Classify different kind of attacks. 4. Elucidate plethora of ways for web security.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	CRYPTOGRAPHIC TECHNIQUES Security Goals, Threats and Attack on Information, Classic Cryptography, Symmetric Key Cryptography-DES, AES, Public and Private Key Cryptography – RSA, Diffie-Hellman, Hash Function – MD5, SHA-1, Digital Signature.	6	1	CO1	PSO6
2	AUTHENTICATION Authentication Methods and Protocols, Password based authentication, Token Based Authentication, Biometric Authentication. Digital Certificates, X.509 Directory Services, PKI, Needham Schroeder, Authentication Protocol, Single sign on,	5	2,3	CO2	PSO1

	Kerberos, Authentication Protocol, Federated Identity Management.				
3	ACCESS CONTROL Access control Policies: DAC, MAC, RBAC, Access control Matrix, ACLs and Capability Lists, Multiple level security models: Biba and Bell La Padula Models, Multilateral security, Covert channel, CAPTCHA	5	1,2	CO3	PSO1
4	SOFTWARE SECURITY Software Flaws, Buffer Overflow, Incomplete Mediation, Race conditions. Malware: Viruses, Worms, Trojans, Logic Bomb, Bots, Rootkits. Miscellaneous Software Attacks: Salami attack, Linearization Attacks, Trusted Computing: Software reverse engineering, Digital Rights management.	4	1,4	CO4	PSO2
5	NETWORK SECURITY Network security basics, TCP/IP vulnerabilities Layer wise. Packet Sniffing, ARP spoofing, port scanning, IP spoofing, TCP syn flood, DNS Spoofing, Internet Security Protocols: SSL, TLS, IPSEC, Secure Email and S/MIME, Denial of Service: Classic DOS attacks, Source Address spoofing, ICMP flood, SYN flood, UDP flood, Distributed Denial of Service, Defenses against Denial of Service Attacks. Firewalls, Intrusion Detection Systems: Host Based and Network Based IDS, Honey pots. Web Security User Authentication and session management, Cookies, Secure HTTP, SQL Injection Techniques, Cross Site Scripting. Cross-Site Request Forgery, Session Hijacking and Management, Phishing Techniques, Web Services Security.	10	2,3	CO4	PSO2

Text books:

1. Ranjan Bose, "Information Theory, Coding and Cryptography", 2 nd Edition, Tata McGraw Hill, 2008.
2. Trappe and Washington, "Introduction to Cryptography with Coding theory", Pearson, 2007.

Reference books:

1. Cryptography And Network Security, Principles And Practice Sixth Edition, William Stallings, Pearson
2. Information Security Principles and Practice By Mark Stamp, Wiley India Edition
3. Cryptography & Network Security, Forouzan, Mukhopadhyay, McGrawHill

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS236		Course Name	Theory of Computation	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2016-2017	Prerequisites (If any)	Discrete Mathematics
Qualifies for	Compiler Design				
Course Description	This course introduces formal models of computation and the problems that they can solve. It presents Turing machines and equivalent models of computation. It also discusses the fundamental limitations of what can be computed. It covers finite state machines, regular expressions and regular grammars as well as context-free languages and grammars and non-context free grammars. It includes algorithms and decision procedures for regular and context-free languages, Turing machine, decidability and complexity analysis				
Course Objective	1. Construct finite state machines and the equivalent regular expressions. 2. Prove the equivalence of languages described by finite state machines and regular expressions. 3. Construct pushdown automata and the equivalent context free grammars. 4. Prove the equivalence of languages described by pushdown automata and context free grammars. 5. Construct Turing machines and Post machines. 6. Prove the equivalence of languages described by Turing machines				
Course Outcomes	Students will be able to: 1. Understand the importance of automata as a modelling tool of computational problems. 2. Understand the role of context-free languages and their limitations. 3. Understand the basis of theory of computation, in particular the role of key problems in defining classes of equivalent problems from a computational perspective. 4. Understand the limitations of computational procedures.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	AUTOMATA Strings, Alphabet, Language, Operations, Finite State Machine, definitions, finite automation model, acceptance of strings and languages, on deterministic finite automation, deterministic finite automation, Equivalence between NFA and DFA, Conversion of NFA into DFA, minimization of FSM, equivalence between two FSM's, Moore and Mealy machines.	9	2,5	CO1 CO2	PSO3
2	REGULAR EXPRESSIONS Regular sets, regular expressions, identity rules, manipulation of regular expressions, equivalence between RE and FA, inter conversion, Pumping lemma, Closure properties of regular sets(proofs not required), regular grammars, right linear and left linear grammars equivalence between regular linear grammar and FA, inter conversion between RE and RG.	9	2,5	CO1 CO2	PSO6

3	CONTEXT FREE GRAMMARS Context free Grammars, Derivation trees, Left Most Derivations, Right Most Derivations, and Ambiguity in Context-Free Grammars, Specifications of Context Free Grammars, Normal Forms, Chomsky Normal Form (CNF), and Greibach Normal Form (GNF).	9	2,5,6	CO3 CO4	PSO3
4	TURING MACHINE Turing machine, definition, model, design of TM, Computable Functions, recursive enumerable language, Church's Hypothesis, Counter machine, types of TM's (Proofs not required).	9	2,5,6	CO2	PSO2
5	CLASSES OF PROBLEMS Chomsky hierarchy of languages, linear bounded automats and context sensitive language, Introduction to DCFL and DPDA, LR(O) Grammar, decidability of problems, Universal Turing Machine, undecidability of post's correspondence problem. Turing reducibility, definition of P and NP problems, NP complete and NP hard Problem	9	1,3	CO3	PSO2

Text books:

1. Hopcroft, John E.; Motwani, Rajeev; Ullman, Jeffrey D. (2013). Introduction to Automata Theory, Languages, and Computation (3rd ed.). Pearson. ISBN 1292039051
2. Peter Linz, "An Introduction to Formal Language and Automata", Third Edition, Narosa Publishers, New Delhi, 2002

Reference books:

1. An introduction to automata theory and formal languages By Adesh K. Pandey, Publisher: S.K. Kataria & Sons
2. Introduction to computer theory By Deniel I. Cohen, Joh Wiley & Sons, Inc

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	SS102		Course Name	Humanities and Social Science	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-2020	Prerequisites (If any)	Basic of Science
Qualifies for	Nil				
Course Description	Humanities and Social Science aims to develop communication competence in prospective engineers so that they are able to communicate information as well as their thoughts and ideas with clarity and precision. This course will also equip them with the basic skills required for a variety of practical applications of communication such as applying for a job, writing reports and proposals, facing an interview and participating in a group discussion. Further, it will make them aware of the new developments in technical communication that have become part of business organizations today. First three units emphasize on communication skills and next two units focus on human ethics, values and professional behavior.				
Course Objective	1. Understand the role of communication in personal & professional success. 2. Develop awareness of appropriate communication strategies. 3. Prepare and present messages with a specific intent. 4. Analyze a variety of communication acts. 5. Ethically use, document and integrate sources.				
Course Outcomes	Student will be able to 1. Analyze interpersonal communication skills. 2. Display competence in oral, written, and visual communication. 3. Students will prepare and write messages with the intent of explaining a specific strategy for communicating with cultural awareness. 4. Explain how to improve communication in your social relationships, especially family and intimate relationships. 5. Students will ethically use, document, and integrate sources within their preparation outlines and presentations				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Business communication covering, Role of communication in information age; concept and meaning of communication; skills necessary for technical communication; Communications in a technical organization; Barriers to the process of communication and sola; Style and organization in technical communication covering, Listening, speaking, reading and writing as skills; Objectivity, clarity, precision as defining features of technical communication; Various types of business writing: Letters, reports, notes, memos; Language and format of various types of business	7	1,3	CO1 CO2	PSO5

	letters; Language and style of reports; Report writing strategies; Analysis of a sample report;				
2	Communication and personality development covering, Psychological aspects of communication, cognition as a part of communication; Emotional Intelligence; Politeness and Etiquette in communication; Cultural factors that influence communication; Mannerisms to be avoided in communication; Language and persuasion; Language and conflict resolution; Language Laboratory emphasizing Listening and comprehension skills; Reading Skills; Sound Structure of English and intonation patterns; Oral Presentation and professional speaking covering, Basics of English pronunciation; Elements of effective presentation; Body Language and use of voice during presentation; Connecting with the audience during presentation; Projecting a positive image while speaking; Planning and preparing a model presentation; Organizing the presentation to suit the audience and context; Basics of public speaking; Preparing for a speech.	10	1,3	CO1 CO2	PSO5
3	Career Oriented Communication covering, Resume and Biodata: Design & style; Applying for a job: Language and format of job application. Job Interviews: purpose and process; How to prepare for interviews; Language and style to be used in interview; Types of interview questions and how to answer them; Group Discussion: structure and dynamics; Techniques of effective participation in group discussion; Preparing for group discussion; Advanced Techniques in Technical Communication covering, Interview through telephone/video-conferencing; Power-point presentation: structure and format; Using e-mail for business communication; Standard e-mail practices; Language in e-mail; Using internet for collecting information; Referencing while using internet materials for project reports; Writing for the media; Language laboratory training in speaking skills covering oral presentations, mock interviews and model group discussions through the choice of appropriate programmes. Career Oriented Communication covering, Resume and Biodata: Design & style; Applying for a job: Language and format of job application. Job Interviews: purpose and process; How to prepare for interviews; Language and style to be used in interview; Types of interview questions and how to answer them; Group Discussion: structure and dynamics; Techniques of effective participation in group discussion; Preparing for group discussion; Advanced Techniques in Technical Communication covering, Interview through telephone/video-conferencing; Power-point presentation: structure and format; Using e-mail for business communication; Standard e-mail practices; Language in e-mail; Using internet for collecting information; Referencing while using internet materials for project reports; Writing for the media; Language laboratory training in speaking skills covering oral presentations, mock interviews and model group discussions through the choice of appropriate programmes. Career Oriented Communication covering, Resume and Biodata: Design & style; Applying for a job: Language and format of job application. Job Interviews: purpose and process;	11	1,3	CO3	PSO3 PSO5

	How to prepare for interviews; Language and style to be used in interview; Types of interview questions and how to answer them; Group Discussion: structure and dynamics; Techniques of effective participation in group discussion; Preparing for group discussion; Advanced Techniques in Technical Communication covering, Interview through telephone/video-conferencing; Power-point presentation: structure and format; Using e-mail for business communication; Standard e-mail practices; Language in e-mail; Using internet for collecting information; Referencing while using internet materials for project reports; Writing for the media; Language laboratory training in speaking skills covering oral presentations, mock interviews and model group discussions through the choice of appropriate programmes.				
4	Understanding Harmony in the Human Being - Harmony in Self: Understanding human being as a co-existence of the sentient 'I' and the needs of Self ('I') and 'Body' - happiness and physical facility, Health; Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship; Understanding values in human-human relationship; Meaning of Justice (nine universal values in relationships) and program for its fulfillment to ensure mutual happiness; Trust and Respect as the foundational values of relationship	5	2	CO4	PSO3 PSO5
5	Professional Ethics: Natural acceptance of human values; Definitiveness of Ethical Human Conduct; Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order; Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco -friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems; Case studies of typical holistic technologies, management models and production systems; Strategy for transition from the present state to Universal Human Order: a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers b. At the level of society: as mutually enriching institutions and organizations.	9	2,4	CO5	PSO3 PSO5

Text books:

1. Human Values and Professional Ethics by R R Gaur, R Sangal, G P Bagaria, Excel Books, New Delhi, 2010.
2. Organizational Behavior, Stephen P. Robbins, Timothy A. Judge, Neharika Vohra, Pearson; Sixteenth edition, 2016.

Reference books:

1. A New Look into Social Sciences, Sheikh Sabir, A.M. Shiekh and Jaya Dwadshiwar, Sage Publication New Delhi.
2. Social Problems in India, Ram Ahuja, Rawat Publication New Delhi.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS308		Course Name	Computer Network	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	V	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018-2019	Prerequisites (If any)	Introduction to computers and programming (CS 139), Web Technology (CS 227)
Qualifies for	Distributed Operating System, Sensors and Cloud computing				
Course Description	This course delves into the details of internet and communication between computer systems. It helps students understand networking concepts using the OSI reference model. OSI, abbreviation for Open Systems Interconnection is a conceptual model for studying networks, its components and protocols. It partitions the communication system into 7 abstract layers and studies importance of each layer in detail				
Course Objective	1. To teach basic organization of network and internet through OSI model. 2. To introduce different hardware components used for networking computers. 3. Study of protocols used in networking. 4. Study of networking architectures 5. Studying physical aspects of networking such as bandwidth allocation, hardware and tools				
Course Outcomes	Student will be able to 1. Choose organization of networking elements 2. Understand working of different protocols for layers of OSI models 3. Apply concepts learned in domains like cloud computing and distributed systems. 4. Apply the networking concepts 5. Undertsand the security aspects				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to Networking and network models : Computer networks, Internet, History of Internet, Network models, Client Server model, Physical organization of network: LAN, MAN, WAN, Network topologies, OSI model protocols in network: Layered Protocol Principle, Logical Connection, TCP/IP Protocol Suite	5	1,2	CO1	PSO1
2	Physical Layer : Data and Signals: Analog & Digital Signals, Transmission Media: Guided Media, Unguided Media, Transmission Loss: Noise, Attenuation and Distortion, Performance Metrics: Bandwidth, Throughput, Delay, Bandwidth Delay Product, Jitter, Digital	6	1,2	CO2	PSO1

	Transmission: Analog to Digital Conversion, Digital to Digital Conversion, Transmission modes: Parallel and Serial Transmission, Analog Transmission: Shift Keying Techniques and Modulation Techniques, Multiplexing: FDM, WDM, TDM, Switching: Circuit Switched Networks, Packet Switched Networks				
3	Data Link Layer : Introduction: Nodes and Links, Services, Link Layer Addressing: Address Resolution Protocol, Error Detection and Correction: Block Coding, Cyclic codes, Checksum, Forward Error Correction. Data Link Control: Framing, DLL Protocols: Simple, Stop and Wait, Piggybacking, HDLC, Point to Point Protocol, Media Access Control Protocols: ALOHA, CSMA/CD, CSMA/ CA, Controlled Access: Reservation, Polling, Token Passing, FDMA, TDMA. Wired LANs: Ethernet and Ethernet Protocols, SONET and ATM. Wireless LANs: 802.11, Wimax, Connecting Devices, Virtual Lans	8	1,2	CO3	PSO5 ,PSO6
4	Network Layer : Packetizing, Packet Switching and Forwarding, Internet Protocol (IP): IPV4, IPV6, Difference between both, Mobile IP. Routing: Unicast, Multicast and Broadcast. Routing Protocols: Distance Vector Routing, Link State Routing, Routing Information Protocol, OSPF, BGP, IGMP	8	1, 2	CO4	PSO1, PSO3
5	Transport Layer and Application Layer : Transport Layer Protocols: Transport Layer Services, Port Numbers, Transport Layer Protocols: Simple Protocol, Stop & Wait Protocol, Go Back N protocol, Selective Repeat, UDP, TCP, TCP flow, error and congestion control, SCTP .Providing Services, Application Layer Paradigms, Client Server Programming, Application Programming Interface. Protocols: HTTP, FTP, TELNET, SSH, DNS. Network Management and network Security	15	1	CO5	PSO1, PSO5

Text books:

1. Computer Networks (4th edition), Andrew Tanenbaum, Prentice Hall Data Computer Networking- A Top-Down approach, 5th edition, Kurose and Ross, Pearson.
2. Computer Networking and the Internet (5th edition), Fred Halsall, Addison Wesley

Reference books:

Data Communications and Networking (5th edition), Behrouz Forouzan, McGraw Hill

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS334		Course Name	Mobile Application Development	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	V	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-2020	Prerequisites (If any)	Introduction to Programming, Web Technology
Qualifies for	Mini Project, Minor Project				
Course Description	This course was designed to Mobile computing devices have become ubiquitous in our communities. In this course, we focus on the creation of mobile solutions for various modern platforms, including major mobile operating systems. Topics include mobile device architecture, programming languages, software engineering, user interface design, and app distribution.				
Course Objective	1. Design mobile based applications. 2. Able hybrid application development platform. 3. Develop mobile based applications. 4. Analyze and troubleshoot the problems while developing application. 5. Deploy application on available online app stores.				
Course Outcomes	1. Learn the basic concepts of mobile application development. 2. To understand the android development. 3. Design the Graphic user interface for android application. 4. Implement the android mobile application and make use of database. 5. Understand the iOS application development and Xcode IDE				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to Mobile Application Development Introduction, Importance of Mobile Applications, Types of Mobile Applications: Native, Hybrid and Web Applications, advantages and disadvantages of Different types of Applications, Application Development Environment, Smartphone Application Platforms: Android, iOS and Windows, Similarities and differences between platforms, Mobile Application development process – Mobile Software Engineering, Frameworks and Tools	6	1,2	CO1	PSO1
2	Introduction to Android Introduction to Android Platform, History of Android, Virtual device, development tools, Java packages, emulators, services, Structure and lifecycle of an application for Android system, Android Layouts, Deployment of applications: localization of applications, application signing, version management, licenses	8	1,2	CO2	PSO1

3	Android XML and GUI based Design Graphical User Interface: preparing containers and components, management of component layout, event handling, XML functions and keywords, Responsive designs.	8	1,2,3,6	CO3	PSO4 ,PSO5
4	Content Providers and Database Processing of application resources, content providers, file system, Data persistence: backups, databases, Application security and permissions: security architecture, application signing, user identification, file access, declaration and verification of permissions, Network communication and internet applications, Wi-Fi connections, Geographical location: use of GPS data, Bluetooth communication: basics, permissions, Bluetooth device discovery, device connectivity as a client, server creation, connection management, Push notification, Database design and implementation: Interfacing with SQLite, use of preferences and content providers	16	1,2,3,6	CO4	PSO1, PSO3
5	Introduction to iOS and Xcode (Case Study) IOS Architecture, Frameworks, MVC Design Pattern, Application Life Cycle, Features, Introduction to Objective-C, General Concepts of Objective-C, Introduction to Swift, General Concepts of Swift, Introduction to Xcode, Deployment in App store.	17	1,2,3,6	CO5	PSO1, PSO5

Text books:

1. J2ME: The Complete Reference, James Keogh, Tata McGrawHill
2. Android application development for java programmers. By James C. Sheusi. Publisher: Cengage Learning, 2013.

Reference books:

1. Lauren Darcey and Shane Conder, —Android Wireless Application Development||, Pearson Education, 2nd ed. (2011)
2. Learning iOS Programming: From Xcode to App Store by Alasdair Allan published by O'Reilly Media; 3rd edition, 2013.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS404		Course Name	Data Warehousing and Data Mining	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	V	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-2020	Prerequisites (If any)	Data Modelling, Databases and Database Management Systems, or equivalent.
Qualifies for	Big data, Data Analytics, Data Science				
Course Description	Data has relevance for managerial decisions and is accumulating at an incredible rate due to technological advances. A data warehouse is a database offloaded from the operational systems and used for reporting. The main source of the data is cleaned, transformed, catalogued and made available for use by managers and other business professionals for data mining, online analytical processing, market research and decision support. However, the means to retrieve and analyze data, to extract, transform and load data, and to manage the data dictionary are also considered essential components of a data warehousing system. Data mining is seen as an increasingly important tool by modern business to transform data into business intelligence giving an informational advantage. It is currently used in a wide range of profiling practices, such as marketing, surveillance, fraud detection, and scientific discovery. Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in databases, perform prediction and forecasting, and generally improve their performance through interaction with data. It is currently regarded as the key element of a more general process called Knowledge Discovery that deals with extracting useful knowledge from raw data.				
Course Objective	1. Be familiar with mathematical foundations of data mining tools. 2. Understand and implement classical models and algorithms in data warehouses and data mining. 3. Characterize the kinds of patterns that can be discovered by association rule mining, classification and clustering. 4. Master data mining techniques in various applications like social, scientific and environmental context. 5. Develop skill in selecting the appropriate data mining algorithm for solving practical problems.				
Course Outcomes	Students will be able to: 1. Understand the functionality of the various data mining and data warehousing component. 2. Appreciate the strengths and limitations of various data mining and data warehousing models. 3. Explain the analyzing techniques of various data. 4. Describe different methodologies used in data mining and data ware housing 5. Compare different approaches of data warehousing and data mining with various technologies				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Data Warehousing: Data warehousing Components –Building a Data warehouse — Mapping the Data Warehouse to a Multiprocessor Architecture – DBMS Schemas for Decision Support – Data Extraction, Cleanup, and Transformation Tools –Metadata.	7	1,2	CO1,CO2	PSO2
2	Business Analysis: Reporting and Query tools and Applications – Tool Categories – The Need for Applications – Cognos Impromptu – Online Analytical Processing (OLAP) – Need – Multidimensional Data Model – OLAP Guidelines – Multidimensional versus Multi-relational OLAP – Categories of Tools – OLAP Tools and the Internet	5	1,2	CO2	PSO1,PSO2
3	Data Mining: Introduction – Data – Types of Data – Data Mining Functionalities – Interestingness of Patterns – Classification of Data Mining Systems – Data Mining Task Primitives – Integration of a Data Mining System with a Data Warehouse – Issues –Data Pre-processing.	8	1,2	CO2,CO4	PSO3 PSO5
4	Association Rule Mining And Classification: Mining Frequent Patterns, Associations and Correlations – Mining Methods – Mining various Kinds of Association Rules – Correlation Analysis – Constraint Based Association Mining – Classification and Prediction – Basic Concepts – Decision Tree Induction – Bayesian Classification – Rule Based Classification – Classification by Back propagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods – Prediction	13	1,2	CO3	PSO2,PSO3
5	Clustering And Trends In Data Mining: Cluster Analysis – Types of Data – Categorization of Major Clustering Methods – K-means– Partitioning Methods – Hierarchical Methods – Density-Based Methods –Grid Based Methods – Model-Based Clustering Methods – Clustering High Dimensional Data – Constraint – Based Cluster Analysis – Outlier Analysis – Data Mining Applications	12	1,2	CO4	PSO3PSO5

Text books:

1. Alex Berson and Stephen J.Smith, “Data Warehousing, Data Mining and OLAP”, Tata McGraw – Hill Edition, Thirteenth Reprint 2008.

Reference books:

1. Pang-Ning Tan, Michael Steinbach and Vipin Kumar, “Introduction to Data Mining”, Person Education, 2007.
2. K.P. Soman, Shyam Diwakar and V. Aja, “Insight into Data Mining Theory and Practice”, Eastern Economy Edition, Prentice Hall of India, 2006

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS240		Course Name	Computer graphics	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	V	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2016-2017	Prerequisites (If any)	Nil
Qualifies for	NIL				
Course Description	The purpose of this course is to study the process of creating graphics using computer programs				
Course Objective	1. To teach creation of graphics using computers 2. Teach different algorithms used in the process 3. To design basic animations using latest graphics package software. 4. Students will be able to apply the knowledge, techniques, skills and modern tools to become successful professionals in graphics industries				
Course Outcomes	1. Draw basic lines and 2 dimensional figures using computer programs				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to computer graphics, lines, line segments, vectors, pixels and frame buffers, vector generation, DDA and Bresenham's line and circle drawing algorithms, anti-aliasing, thick lines, character generation: Stroke Principle, Starburst Principle, Bit map method, display of frame buffer. Graphics Primitives: Display devices, Interactive devices: Tablets, touch panels, mouse, joysticks, track balls, light pen etc., Data generating devices: Scanners and digitizers, primitive operations, display file structure, algorithms and display file interpreter, Text and line styles. Introduction, representation, entering Polygons, Polygon filling: Seed fill, Edge fill, scan conversion algorithm, filling with patterns.	8	1,2,3,6	CO1	PSO2
2	2D Transformations: Introduction, matrices, Scaling, Rotation, homogeneous coordinates, Translation, Co-ordinate transformation, rotation about an arbitrary point, inverse transforms and shear transforms Introduction, segment table, segment creation, deletion, renaming. Image transformations, raster techniques Windowing and Clipping: Introduction, viewing transforms, 2D clipping, Cohen-Sutherland outcode algorithm, Polygon Clipping, Sutherland-Hodgeman algorithm, Generalized clipping. 3-D Transformations: Introduction, 3-D geometry, primitives, transformations, Rotation about an arbitrary axis, Concept of	12	1,2,3,6	CO3	PSO2,PSO3

	parallel and perspective projections, Viewing parameters, 3D clipping, 3D viewing transformations				
3	Introduction, Back-face removal algorithm, Z buffers, scan-line, Painters algorithm, Warnock algorithm, hidden line methods, binary space partition. Light, Color and Shading: Introduction, Diffused illumination, point source illumination, shading algorithm, reflections, shadows, ray tracing, Colour models and tables, shading algorithm, transparency	8	1,2,3,6	CO3,CO5	PSO3,PSO5
4	Curves and Fractals: Introduction, Curve generation, Interpolation, interpolating algorithms, interpolating polygons, B-Splins and corners, Bezier curves, Fractals, fractal lines and surfaces (With complete mathematical treatment of this unit)	12	1,2,3,6	CO2,CO3	PSO2,PSO3
5	Development of 3D structures and their projection in various angles. Generation of isometric views from given information of front view, top view and side view and vice versa	5	1,2,3,6	CO4	PSO3PSO5

Text books:

2. Jiawei Han and Micheline Kamber, "Data Mining Concepts and Techniques", Third Edition, Elsevier, 2012.

Reference books:

1. D. Rogers, "Procedural Elements for Computer Graphics", 2nd Edition. Tata McGraw- Hill Publication, 2001, ISBN 0 - 07 - 047371 - 4.
2. J. Foley, V. Dam, S. Feiner, J. Hughes, "Computer Graphics Principles and Practice", 2nd Edition, Pearson Education, 2003, ISBN 81 - 7808 - 038 - 9.
3. F. Hill, "Computer Graphics: Using OpenGL", 2nd Edition, Pearson Education, 2003 ISBN 81 - 297 -0181 - 2.
4. D. Hearn, M. Baker, "Computer Graphics - C Version", 2nd Edition, Pearson Education, 2002, ISBN 81 - 7808 - 794 – 4

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS310		Course Name	Operating Systems	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	V	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018-2019	Prerequisites (If any)	Introduction to computers and programming
Qualifies for	Distributed Operating System, Sensors and Cloud computing				
Course Description	This course studies computer from its system software i.e. the operating system perspective. The operating system (OS) allows users to perform the basic functions of a computer. Operating system acts as intermediary between hardware and the user programs. The OS manages all software and peripheral hardware, and accesses the central processing unit (CPU) for memory or storage purposes. The purpose of this course is to provide an overview of computer operating systems.				
Course Objective	1. To learn the fundamentals of Operating Systems. 2. To learn the mechanisms of OS to handle processes and threads and their communication. 3. To learn the mechanisms involved in memory management in contemporary OS. 4. Provide comprehensive introduction to understand the underlying principles, techniques and approaches which constitute a coherent body of knowledge in operating systems. 5. To understand the basic components of a computer operating system including Scheduling, Memory Management, I/O Management, and the interactions among the various components.				
Course Outcomes	1. To learn the fundamentals of Operating Systems. 2. To learn the mechanisms of OS to handle processes and threads and their communication. 3. To learn the mechanisms involved in memory management in contemporary OS. 4. Provide comprehensive introduction to understand the underlying principles, techniques and approaches which constitute a coherent body of knowledge in operating systems. 5. To understand the basic components of a computer operating system including Scheduling, Memory Management, I/O Management, and the interactions among the various components.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Operating System Overview: Introduction to Operating Systems, Operating System Objectives and Functions.	4	1,2	CO1,CO2	PSO1
2	Processes and Process Management, Threads and Concurrency, Threads Case Study, Thread Implementation Considerations, Thread Performance Considerations.	7	1,2	CO2	PSO2,PSO3
3	Scheduling, Memory Management, Inter-Process Communication	6	1,2	CO2,CO3	PSO2,PSO3,PSO5
4	Synchronization Constructs, I/O Management, Resource Virtualization. Remote Services	7	1,2	CO3,CO4	PSO4,PSO5
5	Distributed File Systems, Distributed Shared Memory, Data Centre Technologies.	6	1,2	CO5	PS2,PS3

Text books:

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts. Sixth edition. Addison-Wesley (2003).

Reference books:

1. Andrew Tanenbaum, Modern Operating Systems, Prentice Hall.
2. William Stallings, Operating Systems, Prentice Hall.
3. Harvey M. Deitel, An introduction to operating systems. Addison-Wesley.
4. Andrew Tanenbaum & Albert Woodhull, Operating Systems: Design and Implementation. Prentice-Hall.
5. Douglas Comer, Operating System Design - The XINU Approach. Prentice-Hall.
6. A.M. Lister, Fundamentals of Operating Systems. Macmillan (1979).

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	PS314		Course Name	Minor Project	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	1
Semester	VI	Elective Subject (Yes/No):		Tutorial	0
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	3
		Year of Syllabus Revision:	2021	Total Credit:	2.5
	Practical	Year of Introduction	2019-20	Prerequisites (If any)	Minor Project
Qualifies for					
Course Description	Minor Project is a significant component of higher education degrees which helps to consolidate students' learning by providing an opportunity for applying a broad range of knowledge to a real life problem while developing other useful skills. Students work in close proximity with their assigned guides who guide them on a learning path during their project work. Research claims that the quality of overall outcome is dependent on the supervisor/student relationship.				
Course Objective	Students will try to: 1. Apply fundamental and disciplinary concepts and methods in ways appropriate to their principal areas of study. 2. Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study. 3. Plan, analyze, design and implement a software project or gather knowledge over the field of research and design or plan about the proposed work. 4. Learn about and go through the software development cycle with emphasis on different processes - requirements, design, and implementation phases. 5. Learn to work as a team and to focus on getting a working project done on time with each student being held accountable for their part of the project. 6. Gain confidence at having conceptualized, designed, and implemented a working, medium sized project with their team.				
Course Outcomes	Students will be able to: 1. Demonstrate a sound technical knowledge of their selected project topic. 2. Undertake problem identification, formulation and solution. 3. Design engineering solutions to complex problems utilizing a systems approach. 4. Identify, analyze, and solve problems creatively through sustained critical investigation. 5. Communicate with engineers and the community at large in written an oral form. 6. Demonstrate the knowledge, skills and attitudes of a professional engineer.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	CHOOSING YOUR PROJECT TOPIC & LITERATURE SURVEY: Deciding upon the area of your research is the most difficult part of undertaking a final year project. Usually, most students are simply overwhelmed by the sheer amount of options available. At this initial stage, it is quite common to find students consulting each and every person possible, both friends and professors, for advice.	15	1, 2	CO1, CO2	PSO2

	Perform a thorough research before coming to the conclusion and never back away from taking advice. However, you must make sure that the final decision lies in your hands. Conducting a thorough literature review is more than essential here. It not only surveys the past instances of your project ideas, but also encapsulates, appraises, compares and contrast, and draws a relation between various research articles, scholarly books, and other relevant sources that may be related to your project topic. A rich literature review gives your paper an upper hand.				
2	DEVELOPING YOUR PROJECT PROPOSAL & PLANNING YOUR PROJECT: Your project proposal will determine whether it will get approved or not since it will outline what your project intends to accomplish and how feasible it is. It should include the specific objectives as well as the anticipated results of the proposed project. The proposal should be capable of convincing the reviewer that the project is worth the time, money and effort required to complete the work. Once your project has been approved, it's game time! You have to make extensive plans as to how to go about it since they are the foundation of a proposed research. The plan should cover every aspect of your research and what it requires. You must make an agenda to procure the required hardware/software. Take care of the minutest details and you're set to begin your project.	10	2, 3	CO3	PSO3
3	IMPLEMENTATION: This is the body of your project where your actual execution takes place. Make sure that you execute plans properly so that your project is well constructed. Pay attention to every detail and try to minimize the mistakes and be confident in what you do. There may be some trials and errors at first but make sure you do not give up.	25	3	CO4	PSO2, PSO6
4	TESTING & RE-ITERATING: After your project is built, it is time to test it. This can be done with the help of some test cases to check whether your project is yielding the results as per your expectations. If the results are not what you wanted, do not stress out. You can check the implementation of your project and try again. It is essential that you develop some test cases to implement on your project. Keep repeating the experiment with different inputs so that the results can be compared and a conclusion can be drawn. This helps in determining the efficiency of your research. Make corrections to your plans as necessary.	10	4,5	CO4	PSO2, PSO6
5	CONCLUSION & FINAL VERSION OF THE REPORT AND PRESENTATION: After implementation, testing and corrections, it is time to conclude the project. This stage is critical and determines whether the project is a success or a failure. Draw the necessary conclusions as per the results and obtain a summary of the executed procedure.	15	5, 6	CO5, CO6	PSO3, PSO5

	Final Version of the Report is the concluding step in case you are not planning to publish your paper. Go through your entire project from the beginning before preparing your final report. This is your final version and in the event of your paper being published, this is the one to be given. At the end of the semester, the student/group of students shall give a presentation of their work followed by a viva-voce examination.				
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Text books:

1. Prasanna Chandra; Projects- Planning, Analysis, Selection, Financing, Implementation and Review', VI Edition, Tata Mc Graw Hill, 8th Edition 2015.
2. Roger Pressman, "Software Engineering: A Practitioners Approach", 6th Edition, McGraw Hill, 2010.

Reference books:

1. Chaudhary S.; Project Management, Tata Mc Graw Hill
2. Ian Somerville, "Software Engineering", 9 th Edition, Addison Wesley, 2011
3. James F. Peters and Witold Pedrycz, "Software Engineering – An Engineering Approach", Wiley, 2000.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS415		Course Name	Big data analytics and Data Visualization	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019	Prerequisites (If any)	Data Structure
Qualifies for	BigData,DataAnalytics,DataScience				
Course Description	1. Understand big data for business intelligence. Learn business case studies for big data analytics. 2. Understand no SQL big data management. Perform map-reduce analytics using Hadoop and related tools.				
Course Objective	Student will be able to 1. Describe big data and use cases from selected business domains 2. Explain No Sql big data management 3. Install, configure, and run Hadoop and HDFS 4. Perform map-reduce analytics using Hadoop 5. Use Hadoop related tools such as HBase, Cassandra, Pig, and Hive for big data analytics				
Course Outcomes	Student will be able to 1. Demonstrate the functioning of a Compiler. 2. Design and implement the prototype compiler. 3. Apply various optimization techniques. 4. Use different compiler construction tools.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	What is big data, why big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics	5	1,2	CO1	PSO1
2	Introduction to NoSQL, aggregate data models, aggregates, key-value and document data models, relationships, graph databases, schemaless databases, materialized views, distribution models, sharding, master-slave replication, peerpeer replication, sharding and replication, consistency, relaxing consistency, version stamps, map- reduce, partitioning and combining, composing map-reduce calculations	6	1,2	CO2	PSO1
3	Data format, analyzing data with Hadoop, scaling out, Hadoop streaming, Hadoop, pipes, design of Hadoop distributed file system	8	1,2	CO3	PSO7 ,PSO8

	(HDFS), HDFS concepts, Java, interface, data flow, Hadoop I/O, data integrity, compression, serialization, Avro, filebased data structures				
4	Map Reduce workflows, unit tests with MR Unit, test data and local tests, anatomy of Map Reduce job run, classic Map-reduce, YARN, failures in classic Map-reduce and YARN, job scheduling, shuffle and sort, task execution, Map Reduce types, input formats, output formats	8	1, 2	CO4	PSO1, PSO3
5	Hbase, data model and implementations, Hbase clients, Hbase examples, praxis. Cassandra, Cassandra data model, Cassandra examples, Cassandra clients, Hadoop integration. Pig, Grunt, pig data model, Pig Latin, developing and testing Pig Latin scripts.Hive, data types and file formats, HiveQL data definition, HiveQL data manipulation, HiveQL queries.	15	1	CO5	PSO1, PSO5

Text books:

1. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley, 2013.
2. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", Addison-Wesley Professional, 2012.
3. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley, 2012

Reference books:

1. Eric Sammer, "Hadoop Operations", O'Reilley, 2012.
2. E. Capriolo, D. Wampler, and J. Rutherglen, "Programming Hive", O'Reilley, 2012.
3. Lars George, "HBase: The Definitive Guide", O'Reilley, 2011.
4. Eben Hewitt, "Cassandra: The Definitive Guide", O'Reilley, 2010.
5. Alan Gates, "Programming Pig", O'Reilley, 2011.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	PS415		Course Name	KHOJ	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-2020	Prerequisites (If any)	
Qualifies for					
Course Description	Khoj is a hands-on course which offers an opportunity to work on complex real-life problems challenging the society. Students identify and analyse challenges confronting the local community through field visits involving interaction with the stakeholders and observation of the community site. Working in interdisciplinary and disciplinary groups, several ideas to engage with and overcome the challenge get discussed in an interdisciplinary platform. The final endeavour is to create a strategy, a solution and a possible social innovation aimed at betterment of the community.				
Course Objective	1. Learners will be able to explain a community / social problem identified 2. Learners will be able to draw multiple perspectives from different disciplines to explain the problem 3. Learners will be able to develop local field context based strategies to address the community / social problem identified 4. Learners will be able to work in interdisciplinary teams.				
Course Outcomes	1. Learners identify various aspects about the identified community / social problem 2. Learners develop tools to obtain information about the community / social problem faced by the local stakeholders with respect to the identified community / social problem. 3. Learners collate information from various stakeholders and contexts about the identified community / social problem 4. Learners analyse various aspects of the community / social problem faced by the local stakeholders. 5. Learners explain the community / social problem faced by the local stakeholders from different disciplinary perspectives 6. Learners identify multiple perspectives from different disciplines to address the identified community / social problem 7. Learners identify various strategies, case studies, researches on similar community /social problem. 8. Learners analyse the existing strategies, case studies, researches on similar community / social problem to identify points of adaptability with the identified community / social problem. 9. Learners develop strategy /solution based on an interdisciplinary understanding and inputs for the problem identified community / social problem. 10. Learners develop local context specific strategies / solutions to solve the community / social problem based on the field needs and stakeholders' expectations.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Problem analysis	15	1,2,3,4	CO1 CO2 CO3 CO4 CO5 CO6	PSO1 PSO2
2	Literature Review	10	2,3,4	CO7 CO8	PSO3 PSO4
3	Develop strategy / solution to the problem Identified	20	3,4,5,6	CO9 CO10	PSO5

Suggested readings

Mashelkar, R. A. (2015) New Panchsheel of the New Millennium: Presidential Address
<http://www.mashelkar.com/index.php/keynote-addresses/145-indian-science-congress-inaugural-address>

Radjou, N. Prabhu, J. & Ahuja, S. 2012. Jugaad innovation: Think frugal, be flexible, generate breakthrough growth. San Francisco, CA: Jossey Bass.

Radjou, N. & Prabhu, J. 2015. Frugal innovation: How to do more with less. New York, NY: Public Affairs.

<https://www.infosys.com/infosys-foundation/aarohan-social-innovation-awards/overview.html>

<https://nif.org.in/>

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code			Course Name	Data Science and Machine Learning	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2021-22	Prerequisites (If any)	Introduction to Computer Programming, Probability and Statistics, Data Warehouse and Data Mining.
Qualifies for	Major Project				
Course Description	This course is to impart necessary knowledge of the mathematical foundations needed for data science and develop programming skills required to build data science applications.				
Course Objective	1. To demonstrate understanding of the mathematical foundations needed for data science. 2. To collect, explore, clean, munge and manipulate data. 3. To implement models such as k-nearest Neighbors, Naive Bayes, linear and logistic regression, decision trees, neural networks and clustering. 4. To build data science applications using Python based toolkits. 5. To understand the basic concepts of machine learning				
Course Outcomes	Students will be able to: 1. Demonstrate understanding of the mathematical foundations needed for data science. 2. Collect, explore, clean, munge and manipulate data. 3. Implement models such as k-nearest Neighbors, Naive Bayes, linear and logistic regression, decision trees, neural networks and clustering. 4. Build data science applications using Python based toolkits. 5. Understand the concepts of machine learning.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to Data Science Concept of Data Science, Traits of Big data, Web Scraping, Analysis vs Reporting	8	1,2,3,6	CO1	PSO1
2	Introduction to Programming Tools for Data Science: Toolkits using Python: Matplotlib, NumPy, Scikit-learn, NLTK, Visualizing Data: Bar Charts, Line Charts, Scatterplots, Working with data: Reading Files, Scraping the Web, Using APIs (Example: Using the Twitter APIs), Cleaning and Munging, Manipulating Data, Rescaling, Dimensionality Reduction.	12	1,2,3,6	CO3	PSO1
3	Mathematical Foundations:	8	1,2,3,6	CO2,CO3	PSO3,PSO4

	Linear Algebra: Vectors, Matrices, Statistics: Describing a Single Set of Data, Correlation, Simpson's Paradox, Correlation and Causation, Probability: Dependence and Independence, Conditional Probability, Bayes's, Theorem, Random Variables, Continuous Distributions, The Normal Distribution, The Central Limit Theorem, Hypothesis and Inference: Statistical Hypothesis Testing, Confidence Intervals, P-hacking, Bayesian Inference				
4	Machine Learning: Overview of Machine learning concepts – Over fitting and train/test splits, Types of Machine learning – Supervised, Unsupervised, Reinforced learning, Introduction to Bayes Theorem, Linear Regression- model assumptions, regularization (lasso, ridge, elastic net), Classification and Regression algorithms- Naïve Bayes, K-Nearest Neighbors, logistic regression, support vector machines (SVM), decision trees, and random forest, Classification Errors, Analysis of Time Series- Linear Systems Analysis, Nonlinear Dynamics, Rule Induction, Neural Networks Learning And Generalization, Overview of Deep Learning.	12	1,2,3,6	CO4,CO5	PSO2,PSO3
5	Case Studies of Data Science Application: Weather forecasting, Stock market prediction, Object recognition, Real Time Sentiment Analysis.	5	1,2,3,6	CO4	PSO2,PSO3

Text books:

1. Joel Grus, "Data Science from Scratch: First Principles with Python", O'Reilly Media.
2. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn and Tensor Flow: Concepts, Tools, and Techniques to Build Intelligent Systems", 1st Edition, O'Reilly Media.

Reference books:

1. Jain V.K., "Data Sciences", Khanna Publishing House, Delhi.
2. Jeeva Jose, "Machine Learning", Khanna Publishing House, Delhi.
3. Chopra Rajiv, "Machine Learning", Khanna Publishing House, Delhi.
4. Jiawei Han and Jian Pei, "Data Mining Concepts and Techniques", Third Edition, Morgan Kaufmann Publishers

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code			Course Name	Soft Skills	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2018-2019	Prerequisites (If any)	NII
Qualifies for	1. To equip the students with the skills to effectively communicate in English 2. To teach the students practical, everyday communication in English through innovative teaching methods 3. To train the students in interview skills, group discussions and				
Course Description	The objective of the course is to inculcate potential skills in the learners to prepare them to deal with the external world in a collaborative manner, communicate effectively, take initiative, solve problems, and demonstrate a positive work ethic so as to hold a good impression and positive impact.				
Course Objective	Students should be able to 1. Develop effective communication skills (spoken and written). 2. Develop effective presentation skills. 3. Conduct effective business correspondence and prepare business reports which produce results.				
Course Outcomes	1. Students will be enabled to understand the basic objective of the course by being acquainted with specific dimensions of communication skills i.e. Reading, Writing, Listening, Thinking and Speaking. 2. Students would be able to create substantial base by the formation of strong professional vocabulary for its application at different platforms and through numerous modes as Comprehension, reading, writing and speaking etc. 3. Students will apply it at their work place for writing purposes such as Presentation/official drafting/administrative communication and use it for document/project/report/research paper writing.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Unit – 1 Interpersonal communication 1: Individuals, groups and cultures Building Relationships Understanding Group Dynamics- I Understanding Group Dynamics- II Groups, Conflicts and their Resolution Social Network, Media and Extending Our Identities	4	1,2	CO1,CO2	PSO2
2	Interpersonal communication 2: Emotional and social skills Developing key traits 1: Creativity, critical thinking and problem solving Developing key traits 2: Motivation, persuasion, negotiation and leadership	7	1,2	CO2	PSO2

	Motivating oneself				
3	The art of persuasion-I The art of persuasion-II From persuasion to negotiation Leadership and motivating others	6	1,2	CO2,CO3	PSO3,PSO4
4	Essential and vocational skills: survival strategies Managing time Managing stress	7	1,2	CO3	PSO3,PSO4
5	Resilience Work-life balance Applying soft-skills to workplace	6	1,2	CO4	PSO2,PSO3

Text books:

1. Technical Communication – Principles and Practices by Meenakshi Raman & Sangeeta Sharma, Oxford Univ. Press, 2016, New Delhi.
2. Improve Your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, New Delhi.

Reference books:

1. Word Power Made Easy by Norman Lewis, W.R.Goyal Pub. & Distributors, 2009, Delhi.
2. Manual of Practical Communication by L.U.B. Pandey; A.I.T.B.S. Publications India Ltd.; Krishan Nagar, 2013, Delhi.
3. English Grammar and Usage by R.P.Sinha, Oxford University Press, 2005, New Delhi.
4. English Grammar, Composition and Usage by N.K.Agrawal&F.T.Wood, Macmillan India Ltd., New Delhi.
5. Effective Communication Skill, Kulbhusan Kumar, RS Salaria, Khanna Publishing House
6. English Grammar & Composition by Wren & Martin, S.Chand& Co. Ltd., New Delhi.
7. Communication Skills for Engineers and Scientists, Sangeeta Sharma et.al. PHI Learning Pvt. Ltd, 2011, New Delhi.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS324		Course Name	Artificial Intelligence	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2018	Prerequisites (If any)	Mathematics, Design Analysis and of Algorithm (CS 312), Computer Organization and Architecture (CS 307)
Qualifies for	Major project				
Course Description	This course studies how to realize the intelligent human behaviours on a computer. The ultimate goal is to make a computer that can learn, plan, and solve problems autonomously. Also, methods for problem solving are useful both for reasoning and planning.				
Course Objective	1. Gain an understanding of entire computer System. 2. To give brief knowledge about searching methodologies. 3. To provide depth knowledge about probability and reasoning. 4. To aware students about utility theory and learning process. 5. To make student familiar about recent technologies.				
Course Outcomes	1. Gain an understanding of basics of Artificial Intelligence. 2. To give brief knowledge about searching methodologies. 3. To provide depth knowledge about probability and reasoning 4. To aware students about utility theory and learning process 5. To make student familiar about recent technologies				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Unit-1 Introduction: Introduction, History, Foundations and Mathematical treatments, Problem solving with AI, AI models, Learning aspects in AI, What is an intelligent Agents, Rational agent, Environments types, types of Agents, Problem Formulations, Review of tree and graph structures, State space representation, Search graph and Search tree	8	1,2	CO1	PSO1,PSO2
2	Unit-2 Search Algorithms: Random search, Search with closed and open list, Depth first and Breadth first search, Heuristic search, Best first search, A* algorithm, Game Search	12	1,2	CO1,CO2	PSO1
3	Unit-3 Probabilistic Reasoning:	8	2,3	CO3	PSO3,PSO5

	Probability, conditional probability, Bayes Rule, Bayesian Networks- representation, construction and inference, temporal model, hidden Markov model				
4	Unit-4 Markov Decision process & Reinforcement Learning: MDP formulation, utility theory, utility functions, value iteration, policy iteration and partially observable MDPs , Passive reinforcement learning, direct utility estimation, adaptive dynamic programming, temporal difference learning, active reinforcement learning- Q learning	12	1,2	CO3	PSO3PSO5
5	Unit-5 Communicating, Perceiving, and Acting: Language Models, Text Classification, Information Retrieval, Information Extraction, Syntactic Analysis (Parsing), Machine Translation, Speech Recognition, Image Formation and object recognition, Early Image-Processing Operations, Object Recognition by Appearance	5	2,3,5	CO5	PSO3PSO5

Text books:

1. Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach" , 3rd Edition, Prentice Hall
2. Elaine Rich and Kevin Knight, "Artificial Intelligence", Tata McGraw Hill

Reference books:

1. Trivedi, M.C., "A Classical Approach to Artificial Intelligence", Khanna Publishing House, Delhi.
2. Saroj Kaushik, "Artificial Intelligence", Cengage Learning India, 2011.
3. David Poole and Alan Mackworth, "Artificial Intelligence: Foundations for Computational Agents", Cambridge University Press 2010.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	MG314		Course Name	Entrepreneurship	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VII	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019	Prerequisites (If any)	Enterprise Resource Planning, Management Information System
Qualifies for	Major Project				
Course Description	Students will develop skills for evaluating, articulating, refining, and pitching a new product or service offering, either as a start-up business or a new initiative within an existing firm. This course is appropriate for all students interested in innovation and design as necessary components of new businesses today. The aims to acquaint the students with challenges of starting new ventures and enable them to investigate, understand and internalize the process of setting up a business.				
Course Objective	Students will try to learn: 1. The prime motive of the course is to expose, excite and elevate the motivation of the students to the field of entrepreneurship. 2. To create entrepreneurial managers for corporate to enhance business performance and create sustainable competitive advantage for the corporate. 3. To understand the traits, skills, attitudes and drive necessary to be a successful entrepreneur. 4. To provide conceptual theoretical frameworks and tools, to enable students to make good decisions and realistic action plans about entrepreneurship. 5. Consider the legal and financial conditions for starting a business venture. 6. From different successful growth stories and will understand the entrepreneurial practices in India.				
Course Outcomes	Students will be able to: 2. Understand the nature of entrepreneurship and the function of the entrepreneur in the successful, commercial application of innovations. 2. Understand Business Models and Planning for Business and understand operations and management in business. 3. Know the parameters to assess opportunities and constraints for new business ideas. 4. Understand the systematic process to select and screen a business idea and learn how to start an enterprise and design business plans those are suitable for funding by considering all dimensions of business. 5. Consider the legal and financial conditions for starting a business venture. 6. Describe the entrepreneurial practices in India and understand entrepreneurial process by way of studying different case studies.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	ENTREPRENEURSHIP Entrepreneurship: Concept, knowledge and skills requirement; characteristic of successful entrepreneurs; role of entrepreneurship in economic development; entrepreneurship	6	1	CO1	PSO1, PSO3

	process; factors impacting emergence of entrepreneurship; managerial vs. entrepreneurial approach and emergence of entrepreneurship.				
2	STARTING THE VENTURE Starting the venture: generating business idea –sources of new ideas, methods of generating ideas, creative problem solving, opportunity recognition; environmental scanning, competitor and industry analysis; feasibility study: market feasibility, technical/operational feasibility, financial feasibility; drawing business plan; preparing project report; presenting business plan to investors.	11	2, 3	CO2, CO3, CO4	PSO2
3	FUNCTIONAL PLANS Functional plans: marketing plan –marketing research for the new venture, steps in preparing marketing plan, contingency planning; organizational plan: form of ownership, designing organization structure, job design, manpower planning; Financial plan: cash budget, working capital, Performa income statement Performa cash flow, perform balance sheet, break even analysis.	13	1, 2	CO4	PSO2, PSO5
4	SOURCES OF FINANCE Sources of finance: debt or equity financing, commercial banks, venture capital; financial institutions supporting entrepreneurs; legal issues: intellectual property rights patents, trademarks, copy rights, trade secrets, licensing; franchising.	6	2	CO3, CO5	PSO3, PSO4
5	GOVERNMENT SUPPORT FOR ENTREPRENEURSHIP DEVELOPMENT Initiatives for Start-up India Stand up India and Skill India, Government of Gujarat schemes for Start-up, Start-up and ecosystem, Stand-up India: Women and Minority Entrepreneurship, Case Studies of various entrepreneurships.	9	2, 3	CO6	PSO4, PSO6

Text books:

1. Entrepreneurship, Hisrich, Robert D., Michael Peters and Dean Shepherded, , Tata McGraw Hill, ND.
2. Entrepreneurship, Brace R., and R., Duane Ireland, Pearson Prentice Hall, New Jersey (USA).

Reference books:

1. Entrepreneurship, Lall, Madhurima, and ShikhaSahai, Excel Book, New Delhi.
2. Entrepreneurship Development and Small Business Enterprises, Charantimath, Poornima, Pearson Education, New Delhi.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS419		Course Name	Cloud Computing	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VII	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-20	Prerequisites (If any)	Discrete Mathematics, Computer Networks
Qualifies for	Big Data Analytics				
Course Description	Cloud Computing has transformed the IT industry by opening the possibility for infinite or at least highly elastic scalability in the delivery of enterprise applications and software as a service (SaaS). Amazon Elastic Cloud, Microsoft's Azure, Google App Engine, and many other Cloud offerings give mature software vendors and new start-ups the option to deploy their applications to systems of infinite computational power with practically no initial capital investment and with modest operating costs proportional to the actual use.				
Course Objective	Student will try: 1. To understand the concept of virtualization and how this has enabled the development of Cloud Computing. 2. To understand the various issues in cloud computing. 3. To familiarize themselves with the lead players in cloud. 4. To acquire knowledge on Aneka Cloud Application Platform. 5. To learn Industry Cloud Platforms.				
Course Outcomes	Student will be able to: 1. Articulate the main concepts, key technologies, strength 2. Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing. 3. Apply the fundamental concepts in datacenters to understand the tradeoffs in power, efficiency and cost. 4. Identify resource management fundamentals, i.e. resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing. 5. Analyze various cloud programming models and apply them to solve problems on the cloud.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION TO CLOUD: Cloud Computing at a Glance, The Vision of Cloud Computing, Defining a Cloud, A Closer Look, Cloud Computing Reference Model. Characteristics and Benefits, Challenges Ahead, Historical Developments. Virtualization: Introduction, Characteristics of Virtualized Environment, Taxonomy of Virtualization Techniques, Virtualization and Cloud computing, Pros and Cons of Virtualization, Technology Examples- VMware and Microsoft Hyper-V. Before the Move into the Cloud: Know Your Software	8	1,2	CO1	PSO1

	Licenses, The Shift to a Cloud Cost Model, Service Levels for Cloud Applications.				
2	CLOUD COMPUTING ARCHITECTURE: Introduction, Cloud Reference Model, Architecture, Infrastructure / Hardware as a Service, Platform as a Service, Software as a Service, Types of Clouds, Public Clouds, Private Clouds, Hybrid Clouds, Community Clouds, Economics of the Cloud, Open Challenges, Cloud Interoperability and Standards, Scalability and Fault Tolerance. Ready for the Cloud: Web Application Design, Machine Image Design, Privacy Design, Database Management, Data Security, Network Security, Host Security, Compromise Response.	8	1,2	CO2	PSO2
3	DEFINING THE CLOUDS FOR ENTERPRISE: Storage as a service, Database as a service, Process as a service, Information as a service, Integration as a service and Testing as a service. Scaling a cloud infrastructure - Capacity Planning, Cloud Scale. Disaster Recovery: Disaster Recovery Planning, Disasters in the Cloud, Disaster Management	9	2, 4	CO3	PSO2
4	ANEKA: Cloud Application Platform Framework Overview, Anatomy of the Aneka Container, From the Ground Up: Platform Abstraction Layer, Fabric Services, Foundation Services, Application Services, Building Aneka Clouds, Infrastructure Organization, Logical Organization, Private Cloud Deployment Mode, Public Cloud Deployment Mode, Hybrid Cloud Deployment Mode, Cloud Programming and Management, Aneka SDK, Management Tools.	10	2, 3	CO4, CO5	PSO6
5	CLOUD APPLICATIONS & PLATFORMS: Scientific Applications – Health care, Geoscience and Biology. Business and Consumer Applications- CRM and ERP, Social Networking, Media Applications and Multiplayer Online Gaming. Cloud Platforms in Industry: Amazon Web Services- Compute Services, Storage Services, Communication Services and Additional Services. Google App Engine-Architecture and Core Concepts, Application Life-Cycle, cost model. Microsoft Azure- Azure Core Concepts, SQL Azure.	10	3, 4	CO5	PSO6

Text books:

1. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S.Thamarai Selvi from TMH 2013.
2. George Reese Cloud Application Architectures, First Edition, O'Reilly Media 2009.

Reference books:

1. Cloud Computing and SOA Convergence in Your Enterprise A Step-by-Step Guide by David S. Linthicum from Pearson 2010.
2. Cloud Computing 2nd Edition by Dr. Kumar Saurabh from Wiley India 2012.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS414		Course Name	Major Project	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	1
Semester	VII	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	4
		Year of Syllabus Revision:	2021	Total Credit:	3
	Practical	Year of Introduction	2019-20	Prerequisites (If any)	Minor Project
Qualifies for					
Course Description	Major Project is a significant component of higher education degrees which helps to consolidate students' learning by providing an opportunity for applying a broad range of knowledge to a real life problem while developing other useful skills. Students work in close proximity with their assigned guides who guide them on a learning path during their project work. Research claims that the quality of overall outcome is dependent on the supervisor/student relationship.				
Course Objective	Students will try to: 1. Apply fundamental and disciplinary concepts and methods in ways appropriate to their principal areas of study. 2. Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study. 3. Plan, analyze, design and implement a software project or gather knowledge over the field of research and design or plan about the proposed work. 4. Learn about and go through the software development cycle with emphasis on different processes - requirements, design, and implementation phases. 5. Learn to work as a team and to focus on getting a working project done on time with each student being held accountable for their part of the project. 6. Gain confidence at having conceptualized, designed, and implemented a working, medium sized project with their team.				
Course Outcomes	Students will be able to: 1. Demonstrate a sound technical knowledge of their selected project topic. 2. Undertake problem identification, formulation and solution. 3. Design engineering solutions to complex problems utilizing a systems approach. 4. Identify, analyze, and solve problems creatively through sustained critical investigation. 5. Communicate with engineers and the community at large in written and oral form. 6. Demonstrate the knowledge, skills and attitudes of a professional engineer.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	CHOOSING YOUR PROJECT TOPIC & LITERATURE SURVEY: Deciding upon the area of your research is the most difficult part of undertaking a final year project. Usually, most students are simply overwhelmed by the sheer amount of options available. At this initial stage, it is quite common to find students consulting each and every person possible, both friends and professors, for advice. Perform a thorough research before coming to the conclusion and	15	1, 2	CO1, CO2	PSO2

	never back away from taking advice. However, you must make sure that the final decision lies in your hands. Conducting a thorough literature review is more than essential here. It not only surveys the past instances of your project ideas, but also encapsulates, appraises, compares and contrast, and draws a relation between various research articles, scholarly books, and other relevant sources that may be related to your project topic. A rich literature review gives your paper an upper hand.				
2	DEVELOPING YOUR PROJECT PROPOSAL & PLANNING YOUR PROJECT: Your project proposal will determine whether it will get approved or not since it will outline what your project intends to accomplish and how feasible it is. It should include the specific objectives as well as the anticipated results of the proposed project. The proposal should be capable of convincing the reviewer that the project is worth the time, money and effort required to complete the work. Once your project has been approved, it's game time! You have to make extensive plans as to how to go about it since they are the foundation of a proposed research. The plan should cover every aspect of your research and what it requires. You must make an agenda to procure the required hardware/software. Take care of the minutest details and you're set to begin your project.	10	2, 3	CO3	PSO3
3	IMPLEMENTATION: This is the body of your project where your actual execution takes place. Make sure that you execute plans properly so that your project is well constructed. Pay attention to every detail and try to minimize the mistakes and be confident in what you do. There may be some trials and errors at first but make sure you do not give up.	25	3	CO4	PSO2, PSO6
4	TESTING & RE-ITERATING: After your project is built, it is time to test it. This can be done with the help of some test cases to check whether your project is yielding the results as per your expectations. If the results are not what you wanted, do not stress out. You can check the implementation of your project and try again. It is essential that you develop some test cases to implement on your project. Keep repeating the experiment with different inputs so that the results can be compared and a conclusion can be drawn. This helps in determining the efficiency of your research. Make corrections to your plans as necessary.	10	4,5	CO4	PSO2, PSO6
5	CONCLUSION & FINAL VERSION OF THE REPORT AND PRESENTATION: After implementation, testing and corrections, it is time to conclude the project. This stage is critical and determines whether the project is a success or a failure. Draw the necessary conclusions as per the results and obtain a summary of the executed procedure. Final Version of the Report is the concluding step in case you are not planning to publish your paper. Go through your entire project	15	5, 6	CO5, CO6	PSO3, PSO5

	from the beginning before preparing your final report. This is your final version and in the event of your paper being published, this is the one to be given. At the end of the semester, the student/group of students shall give a presentation of their work followed by a viva-voce examination.				
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Text books:

1. Prasanna Chandra; Projects- Planning, Analysis, Selection, Financing, Implementation and Review', VI Edition, Tata Mc Graw Hill, 8th Edition 2015.
2. Roger Pressman, "Software Engineering: A Practitioners Approach", 6th Edition, McGraw Hill, 2010.

Reference books:

1. Chaudhary S.; Project Management, Tata Mc Graw Hill
2. Ian Somerville, "Software Engineering", 9 th Edition, Addison Wesley, 2011
3. James F. Peters and Witold Pedrycz, "Software Engineering – An Engineering Approach", Wiley, 2000.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS401		Course Name	Research Methodology	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VII	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-20	Prerequisites (If any)	Basics of Science
Qualifies for	Quality Research Paper publications				
Course Description	This course will provide an opportunity for participants to establish or advance their understanding of research through critical exploration of research language, ethics, and approaches. The course introduces the language of research, ethical principles and challenges, and the elements of the research process within quantitative, qualitative, and mixed methods approaches. Participants will use these theoretical underpinnings to begin to critically review literature relevant to their field or interests and determine how research findings are useful in forming their understanding of their work, social, local and global environment.				
Course Objective	This course introduces and discusses approaches, strategies, and data collection methods relating to research in social sciences. Students will consider how to select the appropriate methodology for use in a study to be performed. Additionally, these students will learn how to collect data based on different data collection methods, construct these tools, and pilot them before they become ready for use. Finally, this course elucidates the requirements for an academic work, considering aspects related to language, writing style, and lay-out. To culminate this final stage, students will learn to write a comprehensive research proposal that may be conducted in the future.				
Course Outcomes	This course aims to guide Master One students at the Section of English in the Navrachana University, Vadodara, of towards achieving competence and proficiency in the theory of and practice to research. This fundamental objective can be realised through helping these students to develop the subject of their research, encourage the formation of higher level of trained intellectual ability, critical analysis, rigour, and independence of thought, foster individual judgement, and skill in the application of research theory and methods, and develop skills required in writing research proposals, reports, and dissertation; and In more specific practical aims, the present course aims to: 1.Enable Master One students understand what research is and what is not.Raise awareness of crucial aspect of the nature of Knowledge and the value of scientific method. 2.Introduce the concept at the heart of every research project –the research problem- and to discuss what a researchable problem is. 3.Evaluate literature, form a variety of sources, pertinent to the research objectives. 4.Identify and justify the basic components of the research framework, relevant to the tackled research problem. 5.Explain and justify how researchers will collect research data. Discuss how to cite sources, using the American Psychological Association (APA), and justify this choice.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION	8	1	CO1	PSO1

	A review of the Research Fundamentals-Meaning and definition of research, Objectives of Research, Motivation in Research, Research methods, Research Methodology-definition, mathematical tools for analysis, General Characteristics of Research, Criteria of Good Research, Types of Research- Descriptive,. Analytical, Applied, Fundamental, Quantitative, Qualitative, Conceptual, Empirical, exploratory research, conclusive research, modelling research, algorithmic research, Research process- steps. Defining and formulating the research problem, selecting the problem, necessity of defining the problem, Technique Involved in Defining a Problem Sources of research problem, Statement of a Problem, Delimiting a Problem, Evaluation of a Problem.				
2	LITERATURE SURVEY Meaning of Review of Literature, Need of Review of Literature, Objectives of Review of Literature, Sources of Literature, The Functions of Literature, How to Conduct the Review of Literature, Some Hints for the Review of Literature, importance of literature review in defining a problem, literature review-primary and secondary sources, monograph, patents, research databases, web as a source, searching the web, critical literature review, identifying gap areas from literature and research database, Errors In research, development of working hypothesis. Precautions in Library Use, Reporting the Review of Literature.	8	1,2	CO2	PSO2, PSO3
3	DATA COLLECTION AND ANALYSIS Data collection method - Primary data - observation method, personal interview, telephonic interview, mail survey, questionnaire design, Secondary Data - internal sources of data, external sources of data, Data Presentation, Accepts of method validation, Processing and Analysis of Data: Statistics in Research, Measures of Central Tendency,. Measures of Dispersion (variation), Measures of Asymmetry (Skewness), Measures of Relationship, Forecasting, Linear Regression and Time series. Data analysis with statically package (Sigma STAT, SPSS for student t-test, ANOVA, etc.), Introduction to Discriminant analysis, Factor analysis, cluster analysis, multidimensional scaling, conjoint analysis.	10	2, 4	CO3	PSO3
4	SCALING AND SAMPLING Scales – measurement, Types of scale – Thurstone’s Case V scale model, Osgood’s Semantic Differential scale, Likert scale, Q- sort scale. Sampling- Meaning and Definition of Sampling, Functions of Population and Sampling, Methods of Sampling: Probability sampling methods – simple random sampling with replacement, simple random sampling without replacement, stratified sampling, cluster sampling. Nonprobability sampling method – convenience sampling, judgment sampling, quota sampling. Sampling Distribution of mean, Sampling Distributions of Variance.	10	4, 5	CO4	PSO5, PSO6
5	INTERPRETATION AND REPORT WRITING	9	4, 5	CO6	PSO2, PSO6

	Meaning of Interpretation, Technique of Interpretation, Precaution in Interpretation, Significance of Report Writing, Different Steps in Writing Report, Layout of the Research Report, Types of Reports, Oral Presentation, Mechanics of Writing a Research Report, Precautions for Writing Research Reports, Conclusions. Use of tools / techniques for Research: methods to search required information effectively, Reference Management Software like Zotero/Mendeley, Software for paper formatting like LaTeX/MS Office, Software for detection of Plagiarism.				
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Text books:

1. C.R. Kothari, Research Methodology Methods and Techniques (Second Revised Edition), New Age. International Publication.
2. R.Panneerselvam, Research Methodology, PHI Ranjit Kumar, Research methodology: a step-by-step guide for beginners, SAGE Publication.
2. Montgomery, Douglas C;(2007), 5/e, Design and Analysis of Experiments, Wiley India)
3. Montgomery, Douglas C. & Runger; George C. (2007), 3/e, Applied Statistics & Probability for Engineers (Wiley Endia)
4. Krishimswamy, K.N.. Sivakumar, Appa Iyer and Mathirattian M. (2006), Management.

Reference books:

1. Research Methodology; Integration of Principles, Methods and Techniques (Pearson Education, New Delhi).
2. Stuart Melville and Wayne Goddard, "Research methodology: an introduction for science & engineering students"
3. Wayne Goddard and Stuart Melville, "Research Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd, 2007. Mayall, "Industrial Design", McGraw Hill, 1992.
4. Niebel, "Product Design", McGraw Hill, 1974. 6. Merges, Peter S. Menell, Mark A. Lemley, "Intellectual Property in New Technological Age", 2016. Property Rights Under WTO", S. Chand, 2008

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS410		Course Name	Cyber Security	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	3
Semester	VII	Elective Subject (Yes/No):		Tutorial	
Typology of Course	Lecture	Foundation Subject (Yes/No):		Practical:	
		Year of Syllabus Revision:		Total Credit:	3
		Year of Introduction	2019-20	Prerequisites (If any)	Basic understanding of Technology, Computer Network, Information Security
Qualifies for	Blockchain				
Course Description	Understand the foundation and history behind the cyber security Explain different types of threats and attacks in cyber world Understand how to recognize the threats and classify different methods of defence Process the risk management techniques in networks Identify different types of spywares, malwares or malicious codes in cyber security Understand the policies and regulations of cyber security networks Manage the detection and protection solution depends on the type of attack.				
Course Objective	Students will try to learn: 1. Understand types of cyber crimes and cyber laws and also how to protect them self and ultimately society from such attacks 2. Understand the core concepts related to malware, hardware and software vulnerabilities and their causes. 3. Fundamentals regarding cryptography. 4. Understand various system vulnerabilities scanning tools. 5. About Network Security. 6. Regarding Cyber Law and its policies.				
Course Outcomes	Students will be able: 1. To exploit the vulnerabilities related to computer system and network using state of the art tools and technologies. 2. To understand various Web application tools to rectify the loop-hauls of the websites. 3. To have the knowledge of fundamentals of Cyber Crime Investigation and its terminologies. 4. To understand the different cyber offences and its types. 5. Be able to use cyber laws and cyber/computer forensics software/tools.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	CYBER CRIME: Definition and Origin of the Word, Cyber Crime and Information Security, who are Cyber Criminals? Classification of Cyber-crimes, E-mail Spoofing, Spamming, Cyber Defamation, Internet Time Theft, Salami Attack, Salami technique Data Diddling, Forgery, Web Jacking, Newsgroup Spam, Industrial Spying, Hacking, Online Frauds, Pornographic Offenders, Software Piracy, Computer Sabotage Email Bombing, Computer Network Intrusion, Password Sniffing, Credit Card Frauds, Identity Theft.	6	1	CO1	PSO1
2	CYBER SECURITY:	5	2	CO2	PSO3

	Organizational Implications: Web Threats for Organization, Security and Privacy Implications, Social Media Marketing: Security risk for organizations, Incident handling: An Essential Component of Cyber Security, Intellectual Property in the Cyberspace, The Ethical Dimensions of Cybercrime, The Psychology, Mindset and skills of Hackers and the Other criminals				
3	CYBER CRIME ON MOBILE: Mobile and Wireless Devices, Proliferation of Mobile and Wireless devices, Trends in Mobility, Credit card Frauds in Mobile and wireless devices, Authentication Service Security, Attacks on Mobile/Cellphones, Mobile Devices: Security Implications for Organizations, Organization Security policies and Measures in Mobile Computing Era.	4	2, 3	CO3	PSO1, PSO2
4	CYBER OFFENCES: How Criminals plan them, Categories of Cyber Crimes, How Criminal Plans the Attack: Active Attacks, Passive Attacks, Social Engineering, Classification of Social Engineering, Cyber Stalking: types of Stalkers, Cyber Cafe and Cyber Crimes, Botnets, Attack Vectors, Cyber Crime and Cloud Computing.	10	2, 4	CO4	PO3, PSO4
5	UNDERSTANDING CYBER LAWS: The legal perspectives: Cybercrimes and the legal Landscape around the world, why do we need cyber laws: The Indian context, The Indian IT ACT: Admissibility of Electronic records, Amendments made in Indian ITA 2000, Positive Aspects and weak areas of ITA 2000, Challenges to Indian law and cybercrime scenario in India, Digital signatures and the Indian ITA act, Cybercrime and punishment, Cyber law Technology and students: Indian Scenario. Understanding Cyber Forensics: Digital forensic Science, Need for Computer Forensic, Cyber Forensic and digital Evidence and rules of Evidence, Forensics Analysis of E-Mail, Digital Forensic Life Cycle	20	2, 4	CO5	PO3, PSO4

Text books:

1. Cyber security - Attack and Defence Strategies: Infrastructure security with Red Team and Blue Team tactics by Yuri Diogenes, Erdal Ozkaya.
2. Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Nina Godbole, Sunit Belapur, Wiley
3. Cyber Law of Information Technology and Internet (Lexix Nexis) Anirudh Rastogi
4. Understanding Laws—Cyber Laws And Cyber Crimes(Lexix Nexis)
5. Cyber Crime Manual by Bibhas Chatterjee, Lawman Publication

Reference books:

1. Atul Kahate, Cryptography and Network Security, McGraw Hill
2. Kaufman, c., Perlman, R., and Speciner, M., Network Security, Private Communication in a public world, 2nd ed., Prentice Hall PTR., 2002

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code			Course Name	Industry Internship	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	
Semester	VIII	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	28
		Year of Syllabus Revision:		Total Credit:	14
	Practical	Year of Introduction	2021-22	Prerequisites (If any)	Major Project
Qualifies for					
Course Description	Students undertake a significant experiential learning opportunity, typically with a company, governmental, or community-based organization. The internship represents an educational strategy that links classroom learning and student interest with the acquisition of knowledge in an applied work setting. Through direct observation, reflection, analysis, implementation and evaluation, students gain an understanding of the internship site's work, mission, and audience, how these potentially relate to their academic study, as well as the organization's position in the broader industry or field. Students will produce a critical reflection on their internship experience demonstrating how they have addressed specific learning goals.				
Course Objective	Students will try to: 1. Develop creative thinking. 2. Identify, analyze and articulate projects with a comprehensive and systematic approach. 3. Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study. 4. Develop ability to solve industrial problems or real-life problems. 5. Learn to work as a team and to focus on getting a working project done on time with each student being held accountable for their part of the project.				
Course Outcomes	Students will be able to: 1. Learn the technology of choice and apply that knowledge in solving real life problems. 2. Understand information about various existing and future technologies. 3. Design engineering solutions to complex problems utilizing a systems approach. 4. Create the documentation on the basis of their project. 5. Demonstrate the knowledge, skills and attitudes of a professional engineer.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	SELECTION OF PROJECT TITLE & LITERATURE REVIEW: After selecting the Project and proposed title, get approval from the Expert Panel. Study in detail about the project definition chosen.	50	1,2	CO1	PSO1
2	SYSTEM DESIGN: Prepare the required documents in consultation with Mentor. Introduction, Purpose, System Overview, System Design Constraints, Roles and Responsibilities, System Architecture, Database Design, System Security Controls, Project Reference.	80	2, 3	CO2	PSO2

3	IMPLEMENTATION: Implementation of the project in any of the programming languages.	200	3, 4, 6	CO3	PSO2, PSO6
4	REPORT WRITING: The report must be prepared as per suggested guidelines consisting of Preamble, Objectives, Scope, Introduction, Conclusions, Recommendations and Annexure.	60	6	CO4	PSO3, PSO6
5	PRESENTATION PREPARATION: At the end of the semester, the student/group of students shall give a presentation of their work followed by a viva-voce examination.	30	6	CO5	PSO3, PSO5, PSO6

Text books:

Reference books:

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CMP606		Course Name	Natural Language Processing	
Year	4	Core Subject (Yes/No):		Lecture:	2
Semester	VII	Elective Subject (Yes/No):	Yes	Tutorial	0
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	0
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2021-2022	Prerequisites (If any)	Basic knowledge Probability and statistics, Programming and data structures
Qualifies for	Nil				
Course Description	Natural language processing deals with written text. Students will learn how to process written text from basic of fundamental knowledge starts with Finite automata, Regular expression and probabilistic model with n-grams. Recognizing Speech and parsing with grammar. This course also covers basis of semantic analysis and discourse analysis and drives it to machine translation. This NLP course will boost student knowledge to research level where they can conduct new level of research. It really helpful for undergraduate students.				
Course Objective	Students will be able to: 1. Teach students the leading trends and systems in natural language processing. 2. Make them understand the concepts of morphology, syntax, semantics and pragmatics of the language and that they are able to give the appropriate examples that will illustrate the above mentioned concepts. 3. Teach them to recognize the significance of pragmatics for natural language understanding. 4. Enable students to be capable to describe the application based on natural language processing and to show the points of syntactic, semantic and pragmatic processing. 5. Interacting with humans in dialogue systems or cooperative tasks. Particular emphasis on Indian languages.				
Course Outcomes	Students will be able to: 1. Understand Natural Language Processing (Understanding). 2. Probabilistic model of defining language and techniques. (Application) 3. Applying Hidden Markov model and Speech Recognition. (Application) 4. Application of context free grammar and language parsing. (Application) 5. Implement probabilistic and language parsing.(Application) and Differentiation of semantic and discourse in terms of NLP.(Analyze)				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction: Ambiguity and uncertainty in language, processing paradigms, phases in natural language processing. Text representation in computers: encoding schemes	6	1,2	CO1	PSO1
2	Linguistics resources:Introduction to corpus, elements in balanced corpus, WordNet, VerbNet. Part of Speech tagging: Stochastic POS tagging, HMM, Transformation based tagging (TBL), handling of unknown words, named entities, multi word expressions.	5	3,4	CO2	PSO2

3	Natural language grammars:lexeme, phonemes, phrases and idioms, word order, agreement, tense, aspect and mood and agreement, context free grammar, spoken language syntax. Parsing- unification, probabilistic parsing, tree-bank.	6	3,4,5	CO2,CO3	PSO2, PSO3
4	Semantics: meaning representation, semantic analysis, lexical semantics. Word Sense Disambiguation: selection restriction, machine learning approaches, dictionary based approaches.	5	4,5,6	CO4,CO5	PSO3, PSO4
5	Discourse:Reference resolution, constraints on co-reference, algorithm for pronoun resolution, text coherence, discourse structure. Real time Applications of NLP: text to speech, text summarization, information retrieval, sentiment analysis, machine translation.	8	5,6	CO5	PSO5, PSO6

Text books:

1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition Jurafsky, David, and James H. Martin, PEARSON
2. Foundations of Statistical Natural Language Processing, Manning, Christopher D., and Hinrich Schütze, Cambridge, MA: MIT Press
3. Natural Language Understanding, James Allen. The Benjamin/Cummings Publishing Company Inc..
4. Natural Language Processing with Python – Analyzing Text with the Natural Language ToolkitSteven Bird, Ewan Klein, and Edward Loper.

Reference books:

1. Daniel Jurafsky and James H.Martin Speech and Language Processing(2nd Edition),Prentice Hall:2 edition,2008.
2. Machine Learning for Text by Charu C.Aggarwal,Springer,2018 edition
3. Foundations of Statistical Natural Language Processing by Christopher D.Manning and Hinrich Schuetze,MIT press, 1999
4. Steven Bird,Ewan Klein and Edward Loper Natural Language Processing with Python,O'Reilly Media;1 edition,2009
5. Roland R.Hausser, Foundations of Computational Linguistics:HumanComputer Communication in Natural Language,Paperback,MIT press,2011
6. D. Jurafsky, J. H. Martin, Speech and Language Processing, (2e), Pearson Education, 2009.
7. T. Siddiqui, U. S. Tiwary, Natural language processing and Information retrieval, Oxford University Press, 2008.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CMP502		Course Name	Fundamentals of Deep Learning	
Year	4	Core Subject (Yes/No):		Lecture:	2
Semester	VII	Elective Subject (Yes/No):	Yes	Tutorial	0
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	0
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2021-2022	Prerequisites (If any)	Basic knowledge of Artificial Intelligence, Machine Learning.
Qualifies for	Nil				
Course Description	Deep Learning is one of the most exciting and promising segments of Artificial Intelligence and machine learning technologies. This course with Tensor Flow is designed to help you master deep learning techniques and build using Tensor Flow, an open-source software library developed by Google for the purpose of conducting machine learning and deep neural networks. It is one of the most popular software platforms used for deep learning and contains powerful tools to help you build and implement artificial neural networks.				
Course Objective	The student will be able to understand 1. Understand complexity of Deep Learning algorithms and their limitations 2. Understand modern notions in data analysis oriented computing; 3. Student capable of confidently applying common Deep Learning algorithms in practice and implementing their own; 4. They can be performing distributed computations 5. They can be performing experiments in Deep Learning using real-world data.				
Course Outcomes	Students will be able to: 1. Describe the feedforward and deep networks. 2. Design single and multi-layer feed-forward deep networks and tune various hyper-parameters. 3. Analyze performance of deep networks. 4. Build own deep learning project 5. Differentiate between machine learning, deep learning and artificial intelligence				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction: Historical context and motivation for deep learning; basic supervised classification task, optimizing logistic classifier using gradient descent, stochastic gradient descent, momentum, and adaptive sub-gradient method.	6	1,2	CO1	PSO1
2	Neural Networks and Convolution Neural Networks Feedforward neural networks, deep networks, regularizing a deep network, model exploration, and hyperparameter tuning. Introduction to convolution neural networks: stacking, striding and pooling, applications like image, and text classification.	5	2,3,4	CO2, CO3	POS1 PSO2
3	Sequence Modeling: Recurrent Nets: unfolding computational graphs, recurrent neural networks (RNNs), bidirectional RNNs, encoder-decoder sequence to sequence architectures, deep recurrent networks	6	3,4,5	CO3 CO4	PSO2, PSO3

4	Autoencoders: Undercomplete autoencoders, regularized autoencoders, sparse autoencoders, denoising autoencoders, representational power, layer, size, and depth of autoencoders, stochastic encoders and decoders.	5	2,3	C04	PSO4 PSO5
5	Structuring Machine Learning Projects: Orthogonalization, evaluation metrics, train/dev/test distributions, size of the dev and test sets, cleaning up incorrectly labeled data, bias and variance with mismatched data distributions, transfer learning, multi-task learning.	8	4,5	C05	PSO5 PSO6

Text books:

1. Ian Goodfellow, Deep Learning, MIT Press, 2016.
2. Jeff Heaton, Deep Learning and Neural Networks, Heaton Research Inc, 2015.
3. Mindy L Hall, Deep Learning, VDM Verlag, 2011.

Reference books:

1. Li Deng (Author), Dong Yu, Deep Learning: Methods and Applications (Foundations and Trends in Signal Processing), Now Publishers Inc, 2009
2. Golub, G., H., and Van Loan, C., F., Matrix Computations, JHU Press, 2013.
3. Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CMP701		Course Name	Robotics	
Year	4	Core Subject (Yes/No):		Lecture:	2
Semester	VII	Elective Subject (Yes/No):	Yes	Tutorial	0
Typology of Course	Lecture	Foundation Subject (Yes/No):	Yes	Practical:	0
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2021-2022	Prerequisites (If any)	Basic knowledge of Digital Electronics And Microprocessors
Qualifies for	Nil				
Course Description	Robotics as an application draws from many different fields and allows automation of products as diverse as cars, vacuum cleaners, and factories. This course is a challenging introduction to basic computational concepts used broadly in robotics. Topics include simulation, kinematics, control, optimization, and probabilistic inference. The mathematical basis of each area is emphasized, and concepts are motivated using common robotics applications and programming exercises. Students will participate in a series of projects over the course of the semester, in which they will implement algorithms that apply each of the topics discussed in class to real robotics problems.				
Course Objective	The student will be able to understand 1. To understand the functions of the basic components of a Robot. 2. To study the use of various types of End of Effectors and Sensors 3. To impart knowledge in Robot Kinematics and Programming 4. To learn Robot safety issues and economics. 5. To understand the concepts of motion control of a mobile robot navigation				
Course Outcomes	Students will be able to: 1. learn the evolution of robotics and modern happenings in robotics technology 2. Understand fundamental concepts in robotics 3. Student become familiar with terminology of robotics 4. Deepen the knowledge of the robots and their components 5. Learn the principles of operation of a robotic system and analyze the problems and challenges on the fundamentals of the methods in robotics research				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction: Brief history, types, classification and usage, science and technology of robots, Artificial Intelligence in Robotics, some useful websites, textbooks and research journals	6	1,2	CO1	PSO1
2	Elements of Robots-Joints, Links, Actuators, and Sensors :Representation of joints, link representation using D-H parameters, Examples of D-H parameters and link transforms, different kind of actuators, stepper-DC-servo-and brushless motors- model of a DC servo motor-types of transmissions-purpose of sensor-internal and external sensor-common sensors-encoders-tachometers-strain gauge based force torque sensor-proximity and distance measuring sensors-and vision	5	2,3,4	CO2	PSO2, PSO3

3	End Effectors and Planning, Navigation: Classification of end effectors-tools as end effectors-drive system for grippers-mechanical adhesive- vacuum magnetic-grippers-hooks and scoops-gripper force analysis-and gripper design- active and passive grippers Introduction, path planning-overview-road map path planning-cell decomposition path planning- potential field path planning-obstacle avoidance-case studies	6	3,4	CO2, CO3	PSO3, PSO4
4	Vision system and Robot Programming: Robotic vision systems-image representation-object recognition-and categorization-depth measurement- image data compression-visual inspection-software considerations Introduction to robot languages VAL-RAPID-language-basic commands-motion instructions- pick and place operation using industrial robot manual mode-automatic mode-subroutine command based programming-move master command language-introduction-syntax-simple problems	5	4,5	CO4	PSO3, PSO4, PSO5
5	Field and service robots / Industrial Robots: Ariel robots-collision avoidance robots for agriculture-mining-exploration-underwater-civilian- and military applications-nuclear applications-space applications-Industrial robots-artificial intelligence in robots-application of robots in material handling-continuous arc welding-spot welding-spray painting-assembly operation-cleaning-etc	8	5,6	CO4, CO5	PSO5, PSO6

Text books:

1. Klafter R.D., Chmielewski T.A and Negin M., Robotic Engineering – An Integrated Approach, Prentice Hall, 2003.
2. Groover M.P., Industrial Robotics -Technology Programming and Applications, McGraw Hill, 2001.
3. Industrial robotic technology-programming and application by M.P.Groover et al, McGrawhill 2008
4. Saeed B.Nikku, Introduction to Robotics, analysis, control and applications Wiley-India
2nd edition-2011

Reference books:

1. Craig J.J., Introduction to Robotics Mechanics and Control, Pearson Education, 2008.
2. Deb S.R., Robotics Technology and Flexible Automation Tata McGraw Hill Book Co., 1994.
3. Koren Y., Robotics for Engineers”, Mc Graw Hill Book Co., 1992.
4. Fu.K.S.,Gonzalz R.C. and Lee C.S.G., Robotics Control, Sensing, Vision and Intelligence, McGraw Hill Book Co., 1987.
5. Janakiraman P.A., Robotics and Image Processing, Tata McGraw Hill, 1995.
6. Rajput R.K., Robotics and Industrial Automation, S.Chand and Company, 2008.
7. Surender Kumar, Industrial Robots and Computer Integrated Manufacturing, Oxford and IBH Publishing Co. Pvt. Ltd., 1991.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	EE119		Course Name	Basic Electrical and Electronics Laboratory	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	3
		Year of Syllabus Revision:		Total Credit:	1.5
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course develops a basic understanding of the fundamentals and principles of digital circuits and electronic.				
Course Objective	1) To identify polarities voltage and direction of current in loops and nodes. 2) The capability to use abstractions to analyse and design circuits. 3) To be able to design different wave shaping circuits and mathematically analyzes the circuits. 4) To be able Apply concepts of Junction Transistor to design and analyze BJT circuits. 5) To understand different machine behaviour against dynamics produced while in operation. 6) Provides an exhaustive practical exposure to digital logic gates and digital circuits				
Course Outcomes	CO1: Alternating voltages and currents and their vector and time domain presentations. CO2: R -L, R-C and R-L-C series and parallel circuits, impedance and admittance, series and parallel resonance. CO3: star and delta connections and transformations CO4: Construction, Working and Principles of Different electrical machines. CO5: Forward and reverse biased characteristics of Diode CO6: Half-wave and Full wave diode rectifier characteristics CO7: Transistor Characteristic-NPN and PNP Configurations (CE, CB, CC). CO8: Introduction to Logic gates, Boolean Algebra, SOP & POS canonical form				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Power measurement in Single-phase with R-L Loads.	3	3,4,5	CO2	PSO1,PSO2
2	Study of input output characteristics of BJT CE configuration.	3	3,4,5	CO7	PSO1,PSO4
3	Study of input output characteristics of BJT CB configuration.	3	2,3,5	CO7	PSO1,PSO2
4	Study of Diode Characteristics.	3	2,4,5	CO5	PSO1,PSO2,PSO4
5	Half wave and Full Wave diode rectifier.	3	3,4,5	CO6	PSO1,PSO2
6	Study of load test on single phase transformer.	3	3,4,5	CO6	PSO1,PSO2
7	Study of load test on single phase transformer.	3	2,4,5	CO4	PSO1,PSO4
8	Study of load test on Single phase Induction motor	3	2,3,4	CO4	PSO1,PSO2,PSO4
9	Study of load test on Three phase Induction motor	3	2,3,4	CO4	PSO1,PSO2, PSO4
10	Study of logic gates and its implementation.	3	2,3,4	CO8	PSO1,PSO2,PSO4
11	Implementation of simple SOP & POS logic circuits	3	1,2,3	CO1	PSO1,PSO2

Text books:

Nil

Reference books:

1. Hayt W. H., Kemmerly J. E, Durbin S. M., "Engineering Circuit Analysis", Tata McGraw Hill, 6th Edition, 2006.
2. Boylestad & Nashlesky, "Electronic Devices & Circuit Theory", PHI Publication, 2nd edition, 2000.
3. Edminister Joseph A., "Electrical circuits", Schaum's outline series, McGraw hill, 2nd edition, 1983.
4. G. K. Mithal , " Electronic Devices And Circuits" KHANNA PUBLISHERS. ISBN 9788174091777.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	ME111		Course Name	Workshop Practices	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	3
		Year of Syllabus Revision:		Total Credit:	1.5
	Practical	Year of Introduction	2014-2015	Prerequisites (If any)	Nil
Qualifies for	Industrial Job				
Course Description	This course intends to impart basic know-how of various hand tools and their use in different sections of manufacturing. Irrespective of branch, the use of workshop practices in day to day industrial as well domestic life helps to dissolve the problems.				
Course Objective	The workshop experiences would help to build the understanding 1. Complexity of the industrial job 2. Time and skills requirements of the job 3. Hands on experiences				
Course Outcomes	This subject includes practicing of the students in following trades by making various jobs. • Carpentry • Plumbing • Electric Wiring • Welding • Facing & turning on Lathe • Milling • Fitting • Tin smithy operations • Drilling • Slotting • Assembling of various parts				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	INTRODUCTION TO WORKSHOP - Sketch general workshop layout and Follow preliminary safety rules in workshop		1,2	CO1	PSO1,PSO2
2	FITTING - Select appropriate fitting tools for the required application.		3,4,5,6	CO2	PSO4,PSO5
3	PREPARE - Prepare the simple job as per specification using tin smithy tools		4,5,6	CO2	PSO4,PSO5
4	PIPE FITTING - Types, specification, material and applications of pipes.		3,4,5,6	CO3	PSO5,PSO6
5	METAL JOINING - Types, specification, material and applications of arc welding accessories and consumables.		4,5,6	CO3	PSO5,PSO6

Text books:

1 H. S. Bawa: Workshop Practice, Tata McGraw Hill, 2004

Reference books:

1 AchuthanS. K. Hajra Choudhury et. al.: Elements of Workshop Technology, Vol. I and 2, 14th Ed. Media Promoters and rPublishes 2009

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS1007		Course Name	Introduction to Web Design Laboratory	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	2
		Year of Syllabus Revision:	Nil	Total Credit:	1
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course is an introduction to web design and development of website in the information fields. It is designed to provide understanding of basic web development concepts and process, web design best practices, web page creation.				
Course Objective	This course is an introduction to web design and development of website in the information fields. It is designed to provide understanding of basic web development concepts and process, web design best practices, web page creation.				
Course Outcomes	1. Learn the basic concepts of Internet and Web. 2. Use fundamental skills to maintain web server services required to host a website. 3. Select and apply mark-up languages for processing, identifying, and presenting of information in web pages. 4. Implement a responsive web site using Bootstrap. 5. Build Website using online website builder tools Wix, Weebly etc				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Program for using html tags, linking a webpage and inserting pictures in the document.	2	1,2,3	CO1	PSO1, PSO2
2	Program using core attributes, formatting tags.	2	1,2,3	CO1	PSO1, PSO2
3	Program for time table creation.	2	1,2,3,4,5,6	CO1, CO2	PSO1, PSO2, PSO3
4	Program for different types of CSS.	2	1,2,3,4	CO1, CO2, CO3	PSO2, PSO3, PSO4
5	Program for creating frames and list.	2	2,3,4,5	CO1, CO2 CO3	PSO2, PSO3, PSO4
6	Program for 2D and 3D transforms	2	1,2,3,4,5,6	CO3	PSO2, PSO3, PSO4, PSO5
7	Program for XML tree.	2	1,2,3,4,5,6	CO3	PSO1, PSO2, PSO3, PSO4, PSO5
8	Program for creating form validation.	2	1,2,3,4,5,6	CO3	PSO1, PSO2, PSO3, PSO4, PSO5
9	Program for bootstrap custom components.	2	1,2,3,4,5,6	CO4	PSO1, PSO2, PSO3, PSO4, PSO6
10	Program for bootstrap tables, images and list.	2	1,2,3,4,5,7	CO4	PSO1, PSO2, PSO3, PSO4, PSO6
11	Figma tool for website building.	2	1,2,3,4,5,8	CO5	PSO1, PSO2, PSO3, PSO4, PSO6
12	Weebly and Wix for website building.	2	1,2,3,4,5,9	CO5	PSO1, PSO2, PSO3, PSO4, PSO6

Text books:

Nil

Reference books:

- 1.HTML 5, Black Book, Dreamtech Press.
- 2.Web Design, Joel Sklar, Cengage Learning

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	LC137		Course Name	Communication Skills	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Successful completion of elementary grammar course.
Qualifies for	Communication Skills II				
Course Description	This Communication course is conceptualized as a course aimed at enhancing English language as a tool of learning and English language as a tool of communication in a professional context. It offers a framework for English language proficiency required of a student in academic work in general as well as in discipline specific areas. As a tool of learning, the four skills of language and learning: listening, speaking, reading and writing are practiced. The focus is on building language skills required in discipline specific areas.				
Course Objective	This course will enable students to - 1. Develop their overall listening, speaking, reading and writing skills 2. Develop knowledge of vocabulary and grammar 3. Read and comprehend texts of varying length at basic/low intermediate level. 4. Understand and use effective writing skills to express ideas/give information. 5. Develop students' general capacity to a level that enables them to use English for their academic and professional requirements				
Course Outcomes	Upon completion of the course, the following outcomes are expected- - A better understanding of the prerequisites of effective communication. - An understanding of other people's communication styles and needs - An understanding of techniques to facilitate effective solo and group communications - Emotional Intelligence to deal with barriers to communication. Upon completion of the course, the following outcomes are expected- 1. A better understanding of the prerequisites of effective communication. 2. An understanding of other people's communication styles and needs 3. An understanding of techniques to facilitate effective solo and group communications 4. Emotional Intelligence to deal with barriers to communication. Upon completion of the course, the following outcomes are expected- - A better understanding of the prerequisites of effective communication. - An understanding of other people's communication styles and needs - An understanding of techniques to facilitate effective solo and group communications - Emotional Intelligence to deal with barriers to communication.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	COMMUNICATION SKILLS - Comprehension (including all grammar components) - Essay writing		1,2,3	CO1	PSO1

	- Paraphrasing and notetaking - Precis writing / Abstract writing - Report Writing (Compiling the report & Report structure) - Importance of Digital Literacy ; Use of social media - Capstone Paper and presentation				
2	PROFESSIONAL SKILLS A - Letter Writing - Resume writing - Interview, Group Discussion skills.		2,3	CO2	PSO2
3	PROFESSIONAL SKILLS B - Designing the KWL Chart - Formulating the central message - Arranging the ideas, facts & supportive arguments - Making a positive impact (appearance, gestures, eye contact, body language, style of speaking) - Effective use of visual aids (types of visual aid equipment, using the equipment correctly) - Maximizing delivery (fielding questions, managing answers, handling difficult situations, short talk guidelines, impromptu sessions) - Practical Session: Delivery of a two minute presentation (each student delivers a presentation on the KWL Chart)		2,3,4,5	CO3	PSO5
4	SOCIAL SKILLS - Leadership and Management skills		4,5,6	CO4	PSO5
5	CRITICAL THINKING SKILLS - Significance of Critical thinking skills - Concept map designing - Fishbone Diagrams		4,5,6	CO5	PSO5,PSO6

Text books:

Nil

Reference books:

1. John Seely; Oxford Guide to Effective Writing and Speaking; Oxford University Press; 2009 Ed.
2. L. Gartside; Modern Business Correspondence; The English Language Book Society and Macdonald and Evans Ltd.
3. Lester and Beason; The McGraw Hill Handbook of English Grammar and Usage; Tata McGraw Hill Education Private Limited; 2010 Ed.
4. Herrmann Robert Ned; The Whole Business Brain; McGraw-Hill, 1998

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS1009		Course Name	Introduction to Computer programming laboratory	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	4
		Year of Syllabus Revision:	Nil	Total Credit:	2
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course covers the basics of programming in the “C” programming language and demonstrates fundamental programming techniques including the most common library functions and the usage of the pre-processor. It also introduces to object-based programming using C++. The problem solving begins with writing algorithm and drawing flowcharts which will be mapped on to a computer program. This course is designed for students having no or very little programming knowledge and experience				
Course Objective	1. Understand the fundamentals of C programming. 2. Learn how to manage the Input and output Operations in C with the various control structures and looping statements and functions for transforming a problem solution into programs involving programming constructs. 3. Study the concept of array and OOP. 4. To get familiar with Class, Object and pointers using C++. 5. Understand how the files can be managed and dynamically how the memory can be allocated.				
Course Outcomes	1. Learn the basic concepts of algorithms, flowcharts, constants, variables, data types, operators, expressions and be able to design and solve complex problems. 2. To identify the loops, decision making statements and function to solve the problem. 3. Implement different Operations on arrays and understand basic concepts of OOP. 4. Understand the concept of Class, Object and pointers using C++. 5. Implement file Operations in C programming for a given application.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Program to implement use of datatypes, variables, formatting characters and mathematical operators.	4	1,2,3	CO1	PSO1,PSO2
2	Program to implement use of Decision-making statements.	4	1,2,3	CO1	PSO2
3	Program to implement use of Looping structures using while and do-while.	4	2,3,4	CO2	PSO2,PSO3
4	Program to implement use of Looping structures with for loop and Loop nesting.	4	3,4,5	CO2	PSO3,PSO4
5	Program to implement use of switch case control and implementing menu driven structure.	4	1,2,3,4	CO2	PSO4,PSO5

6	Program to implement use of array. (Storing, Retrieving, finding maximum and minimum, etc.	4	3,4,5	CO3	PSO5
7	Program to implement sorting in an array using different methods.	4	4,5,6	CO3	PSO4,PSO5
8	Program to access arrays using pointers.	4	4,5	CO4	PSO5,PSO6
9	Program to implement OOP concepts. (Use of class, Object, Constructors, Destructors)	4	4,5,6	CO4	PSO4,PSO5
10	Program to implement dynamic creation of object and its member variables.	4	2,3,4,5	CO4	PSO4,PSO5,PSO6
11	Program to implement inheritance and function polymorphism.	4	4,5,6	CO4	PSO4,PSO5
12	Program to implement File/Stream handling. Writing and reading primitive types, objects to a file.	4	4,5,6	CO5	PSO5,PSO6

Text books:

Nil

Reference books:

1. B .W. Kernighan, Dennis M. Ritchie: The C Programming Language, Phi/Pearson.
2. R. Lafore, "The Waite Group's Object Oriented Programming In C++", 3rd Edition, Galgotia Publications, 2001.
3. B. Stroustrup, "C++ Programming Language", 3rd Edition, Pearson Education, 199

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	PH121		Course Name	Engineering Physics Laboratory	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	I	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	2
		Year of Syllabus Revision:	Nil	Total Credit:	1
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course intends students to achieve the following: · conceptualize physics principles effectively through experiments. · correlate experiment and theory. · understand how laws in physics got formulated through experimental data.				
Course Objective	Physics is a science to fundamentally understand how things work in nature and around us. The importance of the subject is thus unparalleled for an engineering student of any branch. Some of the immediate and direct objectives of this course are listed below. 1. To develop clear understanding about concepts in physics which form basis of present day engineering practices and cutting edge technology. 2. To develop problem solving skills. 3. To have sufficient knowledge base for research and development work in the field of semiconductors, nanotechnology, materials, electromagnetic radiation, waves, optics and communication.				
Course Outcomes	CO1 To get awareness of various measuring/calibrating instruments. CO2 To get an expertise in handling and using routine and some special instruments. CO3 To learn to correlate the manual instructions and video demonstrations to the actual lab instruments. CO4 To strengthen their theoretical understanding by doing experiments. CO5 To learn to do basic error analysis and get to learn how well their performed experiment was! CO6 To learn to work in a small group to complete a task (team building). CO7 To learn experiments through simulations using virtual labs				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Error Analysis	2	1,2,3,4	CO1 CO5	PSO1 ,PSO2, PSO3 , PSO4, PSO5 ,PSO6
2	Young's modulus by Koenig's method	2	1,2,3,4	CO1 CO5	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6

3	Kundt's Tube	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
4	Helmholtz coil	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 , PSO4, PSO5 ,PSO6
5	Thermal conductivity by Lee's disk	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
6	LCR circuit	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
7	Compound pendulum	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
8	Frsenel's Biprism	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
9	Single slit diffraction	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
10	Grating Spetrometer	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
11	Hydrogen Spectrum	2	1,2,3,4	CO1 CO2 CO3 CO4 CO6	PSO1 ,PSO2, PSO3 ,PSO4, PSO5 ,PSO6
12	Virtual lab	2	1,2,3,5	CO7	PSO1 ,PSO2,

					PSO3 ,PSO4, PSO5 ,PSO6
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Text books:

Nil

Reference books:

1. B. L. Worsnop and H. T. Flint: Advanced Practical Physics for students, Asia Publishing House, 1971
2. G. L. Squires: Practical Physics, Cambridge University Press, 2001
3. M. Nelkon and P. Parker: Advanced Level Physics, 7th Edition, Cbs, 2006
4. B.K. Mathur: Introduction to Geometrical and Physical Optics, 7th Edition, Gopal Printing, 1967

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS194		Course Name	Problem Solving Using Python Laboratory	
Year	1	Core Subject (Yes/No):		Lecture:	
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	4
		Year of Syllabus Revision:	Nil	Total Credit:	2
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	Introduction to Python Programming course is intended for students with little or no programming experience. It aims to provide students with an understanding of the role computation can play in solving problems and, regardless of their major, feel justifiably confident of their ability to write small programs that allow them to accomplish useful goals for an IOT application				
Course Objective	1. To understand the nature of programming as human activity. 2. To learn and experience main components of programming process. 3. To understand main control structures of procedural programming languages. 4. To learn and being able to use major programming patterns.				
Course Outcomes	Student will be able to: 1. Explain basic principles of Python programming language. 2. Make Simple application using python. 3. Understand any python code's structure and fundamental aspects. 4. Apply basic python for real time problem solutions.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Program for basic operations in python	4	1,2	CO1	PSO1,PSO2
2	Program for using data type and operators	4	1,2,3	CO1	PSO1,PSO2
3	Program for if and else statement	4	1,2,3,4	CO2	PSO1,PSO2,PSO3
4	Program for for and while statement	4	1,2,3,4	CO2	PSO1,PSO2,PSO3
5	Program for break , continue and pass statement	4	1,2,3	CO2	PSO2,PSO3,PSO4
6	Program for arrays and slicing in python	4	1,2,3,4,5	CO2	PSO2,PSO3,PSO4,PSO5
7	Program for list and tuples	4	1,2,3	CO3	PSO3,PSO4
8	Program for string manipulation	4	1,2,3	CO3	PSO4,PSO5
9	Program for dictionaries	4	1,2,3	CO3	PSO4,PSO5
10	Program for date and time classes in python	4	1,2,3	CO4	PSO4,PSO5,PSO6
11	Program for defining & calling function	4	1,2,3	CO4	PSO4,PSO5, PSO6

Text books:

Nil

Reference books:

1. Wesley J. Chun. "Core Python Programming - Second Edition", Prentice Hall.
2. Kenneth A. Lambert, "Fundamentals of Python – First Programs", CENGAGE Publication.
3. Luke Sneeringer, "Professional Python", Wrox.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CH141		Course Name	Engineering Chemistry	
Year	2	Core Subject (Yes/No):		Lecture:	
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	2
		Year of Syllabus Revision:	Nil	Total Credit:	1
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	A laboratory course illustrating the concepts of 1) Water analysis, i.e., determination of water hardness, chemical oxygen demand and dissolved oxygen of a given sample 2) Redox titrations 3) Quantitative and Qualitative analysis using UV-Visible Spectrophotometer 4) Potentiometry 5) Adsorption 6) Catalysis (heterogeneous and homogeneous) 7) Chemical Kinetics 8) Organic Synthesis/Preparation 9) Chromatography 10) Solvent Extraction.				
Course Objective	To impart the students the knowledge on phase rule, its applications and alloys, their importance, composition and applications. · To demonstrate the application of spectroscopic methods in mechanistic studies of photochemical reactions · To introduce basic analytical techniques and practical aspects of classical chemical analysis. · To solve problems related to chemical analysis and interpret analytical results.				
Course Outcomes	CO1 Understanding basic principles of analytical chemistry, fundamentals of sophisticated analytical instrumentation techniques (Chromatography and Spectroscopy) as well as water analysis and water treatment. CO2 Thorough knowledge on catalysis and their industrial applications. CO3 Develop understanding and significance of Green Chemistry. CO4 Develop understanding of phase transitions, construction of phase diagram, its interpretation, and its applications (alloy formation). Gain insights on liquid crystals, their classification and industrial applications. CO5 Thorough understanding of subject areas like corrosion, corrosion prevention and protection, fuel cells, CO6 Describe and classify various types of polymers, polymer matrix composites, cement, ceramics, and refractory materials with respect to various engineering applications.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Organic preparation: p – Bromoacetanilid	2	1, 2, 3, 4, 5, 6	CO5	PSO1,PSO2
2	Organic extraction: Caffeine	2	1, 2, 3, 4, 5, 6	CO5	PSO1,PSO2, PSO5
3	Determination of strength of a weak acid by potentiometric titration	2	1,2,3,4,5	CO4	PSO1,PSO2
4	Determination of strength of a weak acid conductometric titration	2	1,2,3,4,5	CO4	PSO1,PSO2
5	Determination of rate constant of acid catalyzed hydrolysis of sucrose (inversion of sucrose).	2	1,2,3,4,5	CO3,CO4	PSO1,PSO2
6	Catalytic decomposition of hydrogen peroxide using metal ions.	2	1,2,3,4,5	CO3,CO4	PSO1,PSO2,PSO5
7	Preparations of TLC plates, and use it to separate an unknown mixture and also to calculate the R _f values	2	1,2,3,4,5,6	CO4	PSO1,PSO2,PSO5
8	Determination of λ_{max} and concentration of given potassium permanganate solution using visible spectrometry	2	1,2,3,4,5	CO4	PSO1,PSO2
9	Determination of hardness of a given water sample by complexometric method	2	1,2,3,4,5	CO1	PSO1,PSO2
10	Determination of dissolved oxygen (DO) in a given water sample by Winkler's iodometric method	2	1,2,3,4,5	CO1	PSO1,PSO2,PSO3
11	Determination of chemical oxygen demand (COD) for a given polluted water sample.	2	1,2,3,4,5	CO1	PSO1,PSO2,PSO3
12	Determination of critical micelle concentration (CMC) of a given surfactant (Sodium Dodecylsulfate, SDS) using conductivity method	2	1,2,3,4,5	CO1,CO4	PSO1,PSO2

Text books:

Nil

Reference books:

1. D.P. Shoemaker, C.W. Garland and J.W. Nibler: Experiments in Physical Chemistry, McGraw Hill International Edition, 1996
2. V.D. Athawale and P. Mathur: Experimental Physical Chemistry, 1st Edition, New Age International Publication, New Delhi, 2001.
3. J.B. Yadav: Advanced Practical Physical Chemistry, Goel Pub., Meerut, 2003
- 4 S. M. Khopkar: Basic Concepts of Analytical Chemistry, 3rd Edition, New Age International Publication, New Delhi, 2008
- 5 P. Samnani: Experiments in Chemistry, Anmol Publication Pvt. Ltd. New Delhi, 2007.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CE111		Course Name	Applied Mechanics Laboratory	
Year	3	Core Subject (Yes/No):		Lecture:	
Semester	II	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):	Yes	Practical:	2
		Year of Syllabus Revision:	Nil	Total Credit:	1
	Practical	Year of Introduction	2020-2021	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course provides a sound working knowledge of the fundamentals of applied engineering mechanics including kinematics, dynamics, statics and hydraulics.				
Course Objective	1 At the end of of the course students will able to solve simple problem of work and energy. 2 At the end of of the course students will able to understand the importance and application of various laws of mechanics.				
Course Outcomes	.Examine a mechanical system and derive all forces, couples and moments about it 2.Solve linear velocity and friction problems 3.Evaluate hydrostatic problems related to ship's systems 4.Describes the different energies stored in a liquid				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Law of polygon	2	1,2,3	CO1	PSO1
2	Equilibrium of concurrent forces in space	2	1,2,3	CO1	PSO1,PSO2
3	Forces in simple plane roof truss	2	1,2,3	CO1	PSO1,PSO2
4	Equilibrium of coplanar parallel forces in simple/overhang beam	2	1,2,3,4	CO1	PSO1,PSO2,PSO3
5	Modulus of elasticity of steel wire	2	1,2,3,4,5	CO1	PSO1,PSO2,PSO3
6	Coefficient of friction between belt and pulley	2	1,2,3,4,5	CO2	PSO2,PSO3,PSO4,PSO5
7	Coefficient of static friction between different surface in contact	2	1,2,3,4,5	CO2	PSO2,PSO3,PSO4, PSO5
8	Acceleration of earth 'g' by Fletcher's trolley	2	1,2,3,4,5,6	CO3	PSO3,PSO4,PSO5
9	Mass moment of inertia of a flywheel	2	1,2,3,4,5,6	CO3	PSO3,PSO4, PSO5
10	Forces through jib crane apparatus	2	1,2,3,4,5,6	CO3	PSO3,PSO4,PSO5
11	Forces in space using shear legs apparatus	2	1,2,3,4,5,6	CO4	PSO3,PSO4, PSO5
12	Performance of simple lifting machine	2	1,2,3,4,5,6	CO4	PSO3,PSO4,PSO5, PSO6

Text books:

Nil

Reference books:

Reference Books

1. R.S. Khurmi, Engineering Mechanics
2. H.J.Shah and Junarkar, Applied Mechanics
3. Bear and Jonstan, Engineering Mechanics 7th edition

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS158		Course Name	Web Technology Laboratory	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:	2017-2018	Total Credit:	1
	Practical	Year of Introduction	2017-2018	Prerequisites (If any)	Introduction to Programming, Web Technology
Qualifies for	Advanced Web Technology				
Course Description	This course is an introduction to web design and development of website in the information fields. It is designed to provide understanding of basic web development concepts and process, web design best practices, web page creation with X/HTML and CSS, web client-side scripting using JavaScript and server-side. The acquired knowledge will be used in and reinforced by developing a website that aligns to a set of user specified requirements.				
Course Objective	1.To get familiar with the fundamentals of Web and Internet technologies. 2.To design web pages using HTML, CSS, XML along with the scripting languages. 3.To understand the and implement server-side programming. 4.To learn and implement mysql database for website development. 5. To able to make use of JQuery and Ajax.				
Course Outcomes	Student will be able to 1.To be able to learn basic concepts of Internet and Web Technologies. 2.To be able to design Website using HTML, CSS, XML and scripting languages. 3.To be able to expose to server-side programming languages. 4. Make web applications in the given scripting languages 5.To be able to understand and design mysql database using PHP.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Write a html code for performing basic html tags.	1	3	CO1 CO2	PSO1
2	Write a html code for explaining inline, internal, external CSS	1	3,6	CO1 CO2	PSO2
3	Create CD Catalogue Table in XML and display it using XSL Style Sheet	1	3,6	CO1 CO2	PSO1
4	To study Javascript function.	1	3,6	CO2 CO4	PSO2
5	To study Javascript events.	1	3,6	CO2 CO4	PSO2
6	a) WAP to perform form validation (contact number) using javascript b) WAP to extract information from Jason and displaying it in html table.	1	3,6	CO2 CO4	PSO6
7	Create a Registration form when user submit the form, show the information user has inserted for the confirmation using php.	1	3,6	CO3 CO5	PSO6
8	Create table using Mysql and perform queries on the data stored in table. Introduction to Ajax, setting up and implementing Ajax, PHP and AJAX, AJAX Database, AJAX XML, AJAX Live Search, AJAX RSS Reader, AJAX Poll. Introduction to jQuery, jQuery effects	1	3,6	CO3 CO5	PSO6

9	a) Create registration form (include username and password field) and store information to the database when user submit data. B) Create login page and perform authentication using Mysql and Php c)Fetch the User information from database and show it in ta	2	3,6	CO3 CO5	PSO6
10	a) Write a script that shows user information stored in JSON file when the user clicks on the button without reloading the webpage. b) Write script that show the updated value of number of likes, if user clicks on Like button without refreshing/reloading	2	3,6	CO3 CO5	PSO6
11	a)WAP that demonstrate different events and effects using jquery. b) Perform form validation using jquery.	1	3,6	CO3 CO5	PSO6
12	Developing website on WordPress.	2	3,6	CO1	PSO6

Text books:

1. B.A. Forouzan And R.F. Gilberg: Computer Science,A Structured Programming Approac Using C, Third Edition, Thomson.
2. E. Balaguruswamy, "Object Oriented Programming with C++", Fourth Edition, Tata Mcgraw Hill Publishing Pvt. Ltd.

Reference books:

1. B.W. Kernighan, Dennis M. Ritchie: The C Programming Language, Phi/Pearson.
2. R. Lafore, "The Waite Group's Object Oriented Programming In C++", 3rd Edition, Galgotia Publications, 2001.
3. B. Stroustrup, "C++ Programming Language", 3rd Edition, Pearson Education, 1997

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS232		Course Name	Object Oriented Programming using Python Laboratory	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	3
		Year of Syllabus Revision:	2019-2020	Total Credit:	1.5
	Practical	Year of Introduction	2019-2020	Prerequisites (If any)	Programming in Python
Qualifies for	Advance Python Programming				
Course Description	This course in computer science is designed to introduce students to top-down structured problem-solving, algorithm development, procedural and data abstraction. Topics include selection, repetition, and sequence control structures; program design, development, test, and documentation; coding style, readability and maintainability; array, record and file manipulation. Students will examine elements of a basic python programming language.				
Course Objective	1. To teach basic object oriented programming concepts 2. To study Classes & Objects 3. To Study Input-Output, Modules & Packages 4. To Study OOP Building Blocks 5. Implement small projects using OOP				
Course Outcomes	Student will be able to 1. To implement basic object oriented programming concepts 2. To make programmes using Classes & Objects 3. To use Input-Output, Modules & Packages of python 4. Learn OOP Building Blocks 5. Implement projects using OOP paradigm				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Program to implement use of Python basics, complex types, basic class structures and objects.	2	2,3	CO2	PSO2
2	Program to implement OOP using Class, class-attributes, member functions, member variables	1	2,3	CO1	PSO1
3	Program to implement use of constructors, data hiding, data abstractions.	2	1,3	CO1	PSO1
4	Program to implement use of Inheritance and Mixins.	1	1,3	CO2	PSO2
5	Program to implement use of Function polymorphism with overloading and overriding.	1	2,6	CO3	PSO6
6	Program to implement use of Abstract class and methods.	1	1,3	CO4	PSO2
7	Program to implement Inheritance with Single Level, Multilevel, Multiple and Diamond shape inheritance.	2	2,3	CO1	PSO3
8	Program to implement complete class with getter-setter, static methods and decorators.	1	1,6	CO2	PSO3
9	Program to implement File Input output with Text and Binary files. Use Text, JSON and Pickle format to handle data.	1	1,6	CO5	PSO2
10	Program to implement module, packages and garbage collection mechanism.	1	2,3	CO4	PSO1

11	Program to implement use of Exception handling. Also create user define exception.	1	1,3	CO3 CO5	PSO2
12	Program to implement invocation of C program from python for various datatypes and vice-versa.	1	1,3	CO4	PSO2

Text books:

2. E. Balaguruswamy, "Object Oriented Programming with C++", Fourth Edition, Tata Mcgraw Hill Publishing Pvt. Ltd.

Reference books:

1. Dusty Phillips , "Python 3 Object Oriented Programming" ,PACLT publications.
2. R. Nageswara Rao, "Core Python Programming",dreamtech

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS230		Course Name	Data Structures Laboratory	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	
Semester	III	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	3
		Year of Syllabus Revision:	2017-2018	Total Credit:	1.5
	Practical	Year of Introduction	2017-2018	Prerequisites (If any)	Introduction to Programming
Qualifies for	Design Analysis and Algorithm				
Course Description	The course introduces about how to design, write, and analyze the performance of C/C++ programs that handle structured data and perform more complex tasks, typical of similar to module of a larger software projects. The course aims at developing the skills in using generic principles for data representation & manipulation with a view for efficiency, maintainability and code reusability.				
Course Objective	1. The objective of the course is to teach 'c' programming (with an emphasis on problem solving) and introduce elementary data structures. 2. To introduce the fundamental concept of data structures and to emphasize the importance of data structures in developing and implementing efficient algorithms. 3. Topics include elementary data structures, (including arrays, stacks, queues, and lists), advanced data structures (including trees and graphs), the algorithms used to manipulate these structures, and their application to solving practical engineering problems. 4. Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data. 5. To explain the importance of the subject for various engineering fields and their uses over there.				
Course Outcomes	Students will be able to : 1. Formulate and apply 'C' programming, as a modern tool to solve engineering problems. 2. Demonstrate an understanding of basic data structures (such as an array-based list, linked-list, stack, queue, binary search tree) and algorithms. 3. Demonstrate the ability to analyze, design, apply and use data structures and algorithms to solve engineering problems and evaluate their solutions. 4. Describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms. 5. Discuss the efficiency of the principal algorithms for sorting, searching.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Programs for Basic C Programming	1	1,2	CO1	PSO1
2	Programs based on Functions	1	1,2	CO1	PSO3
3	a)Programs based on Pointers b)Programs based on Functions and Pointers c)Programs based on Recursion	1	1,2	CO1	PSO2
4	a)Programs based on Strings b)Programs based on String Handling Functions & Pointers c)Programs based on Structures & Union	1	1,2	CO1	PSO1
5	a)Programs based on File Handling in C b)Menu Driven Program to implement various operations of One-Dimensional Array	1	1,2	CO1	PSO1

	c)Menu Driven Program to implement various operations of Multi-Dimensional Array				
6	a)Menu Driven Program to implement various operations of Singly Linked List b)Menu Driven Program to implement various operations of Doubly Linked List c)Menu Driven Program to implement various operations of Circular Linked Lis	2	2,3	CO2 CO4	PSO2
7	Menu Driven Program to implement various operations of Stack using Array and Linked Lis	1	3,6	CO2 CO4	PSO1
8	a)Menu Driven Program to implement various operations of Queue using Array and Linked List b)Menu Driven Program to implement various operations of Circular Queue using Array and Linked List	1	3,6	CO2 CO4	PSO2
9	a)Program to implement Bubble Sort b)Program to implement Selection Sort c)Program to implement Insertion Sort	1	3,6	CO5	PSO3
10	a) Program to implement Quick Sort b) Program to implement Merge Sort	2	3,6	CO5	PSO3
11	a) Program to implement Linear Search b)Program to implement Binary Search	1	3,6	CO5	PSO2
12	a) Program to implement Binary Search Tree and its Traversals (Prefix and Postfix) b) Program to implement Shortest Path Algorithm: Dijkstra's Algorithm c) Program to implement Minimum Spanning Trees: Kruskal's and Prim's Algorithm	2	3,6	CO3	PSO1

Text books:

1. Data Structures & Program Design In C, 2/E by Robert Kruse, C.L. Tondo, Bruce Leung, Shashi Mogalla.

Reference books:

1. Data Structures and Algorithms: Concepts, Techniques and Applications by Dr. G. A. Vijayalakshmi Pai.
2. Introduction to Algorithms, 3E Paperback by Thomas H. Cormen (Author), Charles E. Leiserson (Author), Ronald L. Rivest (Author), Clifford Stein (Author).

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS238		Course Name	Database Management System Laboratory	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:	2017-2018	Total Credit:	1
	Practical	Year of Introduction	2017-2018	Prerequisites (If any)	Introduction to Computers, Introduction to programming, Data Structure
Qualifies for	Software Engineering				
Course Description	This course introduces students to basic database design and the use of databases in applications. Throughout the course students are introduced to Database and Database Management System, advantages and disadvantages of the different database models, comparison of relational model with the Structured Query Language (SQL); Understanding constraints and controversies associated with relational database model as well as the rules guiding transaction ACID. It includes the design using various diagrams like ER, Sequence, and Activity. The designed system will produce the 'reports' for specific queries.				
Course Objective	1.To introduce fundamental concepts of DBMS. 2.To design and create the Database model. 3.To understand the Normalization techniques to reduce the data Redundancy. 4.To Learn the query processing system, recovery techniques and management of transactions in database. 5. To empower and expose the students with database implementation using SQL and PL/SQL languages.				
Course Outcomes	Student will be able to 1.To be able to apply analytical skills to map out the conceptual design for a real-life problem and explain the fundamental elements of Database management System. 2.To be able to draw the model of Relational Database using Entity Relationship (ER) model 3.To be able to Improve the database design by normalization. 4.To be able to evaluate the logical design to translate into a specific data model and physical design to meet system storage requirements. 5.To be able to evaluate Oracle features and Oracle related products for maintaining the integrity and performance of databases.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	To study DDL-create and DML-insert commands.	2	1	CO1	PSO2 PSO6
2	To study various options of LIKE predicate	1	1	CO1	PSO1
3	To Perform various data manipulation commands, aggregate functions and sorting concept on all created tables.	1	2,3	CO3	PSO1
4	To study Single-row functions.	1	1	CO1	PSO2
5	Displaying data from Multiple Tables (join)	2	2,3	CO4	PSO6
6	To apply the concept of Aggregating Data using Group functions.	1	1,3	CO4	PSO2
7	To solve queries using the concept of sub query	2	1,3	CO4	PSO2

8	Write SQL query using group by function.	1	1,3	CO1	PSO2
9	To apply the concept of security and privileges	1	1	CO1	PSO1
10	Write SQL queries to create views.	1	1	CO1	PSO2
11	To study Transaction control command	1	1	CO4	PSO1
12	Write programme by the use of PL/SQL.	1	1	CO5	PSO6

Text books:

1. An introduction to Database Systemsby C J Date -Wesley.
2. Understanding SQLby Martin Gruber, -BPB

Reference books:

1. Database System Concepts by Abraham Silberschatz, Henry F. Korth & S. Sudarshan - McGraw Hill.
2. ORACLE PL/SQL by example. Benjamin Rosenzweig, Elena Silvestrova, Pearson Education 3rd Edition.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS313		Course Name	Design analysis and algorithm Laboratory	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:	2018 – 2019	Total Credit:	1
	Practical	Year of Introduction	2018 – 2019	Prerequisites (If any)	Introduction to computers and programming (CS 139), Data Structures
Qualifies for	Major Project				
Course Description	This course enables to develop problem solving by analysing algorithms and their running time. It also gives insight into analysing space complexity in order to write efficient programs.				
Course Objective	1. To teach theoretical concepts underlying computer algorithm development. 2. Teach problem solving by different methods. 3. Synthesize efficient algorithms in common engineering design situations. 4. To teach analysing algorithms for time and space complexity through different methods such as statistical methods and dynamic programming. 5. Ability to understand how the choice of data structures and the algorithm design methods impact the performance of programs.				
Course Outcomes	Student will be able to: 1. Choose methods for problem solving. 2. Analyze the performance of algorithms and analyse algorithms to find mean running time. 3. Understand different algorithmic design strategies. 4. To understand how the choice of data structures and the algorithm design methods impact the performance of programs. 5. Apply the algorithms and design techniques to solve problems;				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Write a html code for performing basic html tags.	2	3,6	CO2 CO3	PSO3
2	Implementation and Time analysis of sorting algorithms for best case, average case & worst case.	1	2,4	CO2 CO3	PSO1
3	Implementation and Time analysis of Factorial program using iterative and recursive method.	1	2,4	CO2 CO3	PSO2
4	Implementation and Time analysis of Fibonacci program using recursive method.	1	4	CO2 CO3	PSO3
5	Implementation and Time analysis of program to calculate power of number using recursive method. Test it for big values.	1	3	CO2 CO3	PSO2
6	Optimize Experiment-3,4,5 using divide and conquer	2	2,4	CO4 CO5	PSO1
7	Write a program to find out whether there exist cycle in given graph or not	1	6	CO1	PSO2
8	Given an integer x, find square root of it. If x is not a perfect square, then return floor($\sqrt{x}$) (expected time complexity $\log n$)	1	6	CO1	PSO6
9	Write a program to solve 0-1 and fractional knapsack problem	1	6	CO6	PSO2

10	Write a program to implement longest common subsequence and longest increasing subsequence.	1	6	CO4	PSO3
11	Write a program to check if a given directed graph is strongly connected. Given a directed graph, A directed graph is strongly connected if there is a path between any two pairs of vertices.	2	6	CO1	PSO1
12	Write a program to implement KMP algorithm. Take a long string and compare output with naive approach.	1	3,6	CO1	PSO1

Text books:

- 1.Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles, Narasimha Karumanchi (Textbook)
- 2.Thomas H. Coreman, Charles E. Leiserson and Ronald L. Rivest, "Introduction to Algorithms", Printice Hall of India. (Textbook)

Reference books:

- 3.Fundamental of Algorithms by Gills Brassard, Paul Bratley, PH

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code			Course Name	Critical thinking/Technical Reading	
Year	2	Core Subject (Yes/No):	Yes	Lecture:	
Semester	IV	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:	2019-2020	Total Credit:	1
	Practical	Year of Introduction	2019-2020	Prerequisites (If any)	Nil
Qualifies for	Nil				
Course Description	This course will introduce you to critical thinking, technical reading and a small amount of formal logic. Its purpose is to provide you with the basic tools of analytical reasoning, which will give you a distinctive edge in a wide variety of careers and courses of study				
Course Objective	1. Use critical thinking and argumentation in university contexts to improve academic results 2. Understand the importance and function of critical thinking in academic culture				
Course Outcomes	Student will be able to 1. Apply critical thinking and technical reading to real world problems and issues 2. Use evidence to support claims in arguments 3. Use a variety of thinking tools to improve critical thinking 4. Identify types of argument, and bias within arguments, in order to better evaluate the strength of arguments 5. Acquire the use of grammar effectively (vocabulary and so on) through extensive coursework on writing reports and reading comprehensions, articles, essays, general discussion etc.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to Critical Thinking in Academic Culture	2	1,2	CO1	PSO3 PSO5
2	Critical Thinking in Academic Contexts	1	1,2	CO2 CO3	PSO3 PSO5
3	Logic & Reasoning	1	1,2	CO1	PSO3 PSO5
4	Argumentation	1	1,2	CO2	PSO3
5	Demonstrating Critical Thinking	1	1,2	CO3	PSO5
6	Reading for the Main idea, finding specific information, reading for detail, Reading and transferring information, Understanding the attitudes	2	1,2	CO1	PSO5
7	Reading for Opinion and writer's purpose, Reading for interpreting the visual reading for gist	1	1,2	CO5	PSO5
8	Reading for inference and Global meaning, Understanding Vocabulary and grammar in a short text	1	1,2	CO5	PSO5
9	Reading for understanding short, real world messages etc,	1	1,2	CO5	PSO5
10	Reading for detailed comprehension of detailed material; Skimming and Scanning.	1	1,2	CO5	PSO5
11	Practice Advanced Pre Viewing Techniques with Paragraphs	1	4	CO5	PSO1
12	Summative Assignment	2	6	CO1	PSO1

Text books:

Nil

Reference books:

1. Critical Thinking By Jonathan Haber · 2020 Publisher: MIT Press
2. Thinking from A to Z By Freelance Philosopher Podcaster and Writer Nigel Warburton, Nigel Warburton
Sevda Çalışkan, Nigel Warburton · 2000 Publisher: Routledge

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS309		Course Name	Computer Network Labarotary	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	0
Semester	V	Elective Subject (Yes/No):		Tutorial	0
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2018-2019	Prerequisites (If any)	Introduction to computers and programming (CS 139), Web Technology (CS 227)
Qualifies for	Distributed Operating System, Sensors and Cloud computing				
Course Description	This course delves into the details of internet and communication between computer systems. It helps students understand networking concepts using the OSI reference model. OSI, abbreviation for Open Systems Interconnection is a conceptual model for studying networks, its components and protocols. It partitions the communication system into 7 abstract layers and studies importance of each layer in detail				
Course Objective	1. To teach basic organization of network and internet through OSI model. 2. To introduce different hardware components used for networking computers. 3. Study of protocols used in networking. 4. Study of networking architectures 5. Studying physical aspects of networking such as bandwidth allocation, hardware and tools				
Course Outcomes	Student will be able to 1. Choose organization of networking elements 2. Understand working of different protocols for layers of OSI models 3. Apply concepts learned in domains like cloud computing and distributed systems. 4. Apply the networking concepts 5. Undertsand the security aspects				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Study of different types of Network cables and Practically implement the cross-wired cable and straight through cable using clamping tool.	2	1,2,3	CO2 CO1	PSO2,PSO6
2	Study of Network Devices in Detail.	2	1,2,3	CO2 CO1	PSO2,PSO6
3	Study of network IP.	2	1,2,3	CO3 CO1	PSO2,PSO6
4	Connect the computers in Local Area Network.	2	1,2,3,6	CO4	PSO2,PSO6
5	Study of basic network command and Network configuration commands	2	1,2,3,6	CO4	PSO2,PSO6
6	Configure a Network topology using packet tracer software	2	1,2,3,6	CO5	PSO2,PSO6
7	Configure a Network topology using packet tracer software.	2	1,2,3,6	CO5	PSO2,PSO6
8	Configure a Network using Distance Vector Routing protocol.	2	1,2,3,6	CO5	PSO2,PSO6
9	Configure Network using Link State Vector Routing protocol.	2	1,2,3,6	CO5	PSO2,PSO6
10	Performing an Initial Switch Configuration	2	1,2,3,6	CO2	PSO2,PSO6
11	Performing an Initial Router Configuration	2	1,2,3,6	CO2	PSO2,PSO6

12	Examining Network Address Translation (NAT)	2	1,2,3,6	CO5	PSO2,PSO6
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Text books:

1.Computer Networks (4th edition), Andrew Tanenbaum, Prentice Hall Data Computer Networking- A Top-Down approach, 5th edition, Kurose and Ross, Pearson 2.Computer Networking and the Internet (5th edition),Fred Halsall, Addison Wesley

Reference books:

Data Communications and Networking (5th edition), Behrouz Forouzan, McGraw Hill

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS347		Course Name	Mobile Application Development Lab	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	0
Semester	V	Elective Subject (Yes/No):		Tutorial	0
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2018-2019	Prerequisites (If any)	Introduction to Programming, Web Technology
Qualifies for	Mini Project, Minor Project				
Course Description	This course was designed to Mobile computing devices have become ubiquitous in our communities. In this course, we focus on the creation of mobile solutions for various modern platforms, including major mobile operating systems. Topics include mobile device architecture, programming languages, software engineering, user interface design, and app distribution.				
Course Objective	1. Design mobile based applications. 2. Able hybrid application development platform. 3. Develop mobile based applications. 4. Analyze and troubleshoot the problems while developing application. 5. Deploy application on available online app stores.				
Course Outcomes	1. Learn the basic concepts of mobile application development. 2. To understand the android development. 3. Design the Graphic user interface for android application. 4. Implement the android mobile application and make use of database. 5. Understand the iOS application development and Xcode IDE				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction to android operating system and study of basic widgets.	2	1,2,3	CO1	PSO1 PSO2 PSO6
2	Creating a simple Android Application	2	1,2,3	CO1	PSO1 PSO2 PSO6
3	Study of android lifecycle and demonstration of it.	2	1,2,3	CO2	PSO1 PSO2 PSO6
4	Study of intents and types of intents	2	1,2,3,6	CO2 CO1	PSO1 PSO2 PSO6
5	Study of list views and adapters	2	1,2,3,6	CO3	PSO1

					PSO2 PSO6
6	Study of dialog interfaces in android	2	1,2,3,6	CO3	PSO1 PSO2 PSO6
7	Study of Sensors in android	2	1,2,3,6	CO4	PSO1 PSO2 PSO6
8	Study of Services in android	2	1,2,3,6	CO4	PSO1 PSO2 PSO6
9	Study of touch in android	2	1,2,3,6	CO4	PSO1 PSO2 PSO6
10	Creation of Application over the Operation of Database using Android	2	1,2,3,6	CO5	PSO1 PSO2 PSO6
11	Creation of Application to locate the map using geocoding	2	1,2,3,6	CO5	PSO1 PSO2 PSO6
12	Implement the application for Bluetooth scanning of device	2	1,2,3,6	CO5	PSO1 PSO2 PSO6

Text books:

1. J2ME: The Complete Reference, James Keogh, Tata McGrawHill
2. Android application development for java programmers. By James C. Sheusi. Publisher: Cengage Learning, 2013.

Reference books:

1. Lauren Darcey and Shane Conder, —Android Wireless Application Development||, Pearson Education, 2nd ed. (2011)
2. Learning iOS Programming: From Xcode to App Store by Alasdair Allan published by O'Reilly Media; 3rd edition, 2013.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS241		Course Name	Computer Graphics Laboratory	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	0
Semester	V	Elective Subject (Yes/No):		Tutorial	0
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2016-2017	Prerequisites (If any)	Basic programming C,C++
Qualifies for	Augmented Reality -Virtual Reality				
Course Description	The purpose of this course is to study the process of creating graphics using computer programs				
Course Objective	1. To teach creation of graphics using computers 2. Teach different algorithms used in the process 3. To design basic animations using latest graphics package software. 4.Students will be able to apply the knowledge, techniques, skills and modern tools to become successful professionals in graphics industries				
Course Outcomes	Draw basic lines and 2 dimensional figures using computer programs				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	BASIC GRAPHICS FUNCTION	2	1,2,3	CO1	PSO2,PSO6
2	program to draw a line using Bresenham's Line Algorithm (BLA	2	1,2,3	CO2 CO1	PSO2,PSO6
3	program to draw a line using Digital Differential Analyzer (DDA) Algorithm	2	1,2,3	CO2 CO1	PSO2,PSO6
4	program to draw a circle using Bresenham's Circle Algorithm	2	1,2,3,6	CO2 CO1	PSO2,PSO6
5	program to draw a circle using MidPoint Circle Algorithm.	2	1,2,3,6	CO2 CO3	PSO2,PSO6
6	program to fill different types of geometric shapes using Boundary	2	1,2,3,6	CO4 CO5	PSO2,PSO6
7	PROGRAM TO ROTATE A TRIANGLE ABOUT ORIGIN.	2	1,2,3,6	CO4 CO5	PSO2,PSO6
8	PROGRAM TO SCALE THE TRIANGLE	2	1,2,3,6	CO4	PSO2,PSO6
9	PROGRAM TO TRANSLATE A TRIANGLE	2	1,2,3,6	CO3	PSO2,PSO6
10	To create a wine glass using EP curve tool	2	1,2,3,6	CO4	PSO2,PSO6
11	To create a building with balcony and a door at the base of the building using polygon and NURBS primitives	2	1,2,3,6	CO4 CO5	PSO2,PSO6
12	To bounce a ball using Animation.	2	1,2,3,6	CO5	PSO2,PSO6

Text books:

1. K. Venugopal, Engineering Drawing and Graphics + AutoCAD, New Age International, 2011 2. N.D. Bhat and V. M. Panchal, Engineering Drawing , Charotar Publishing House, 2011

Reference books:

1. B. Agrawal and C. M. Agrawal, Engineering Drawing, Tata McGraw-Hill Publishing Company Limited, 2008. 2. K. R. Gopalakrishna, Engineering Drawings, Subhas Stores, Bangalore, 2001. 3. N. Sidheswar, P. Kanniah and V. V. S. Sastry, Machine Drawing, Tata-McGraw Hill, New Delhi, 1980. 4. T. E. French, C. J. Vireck and R. J. Foster, Engineering Drawing and Graphic Technology, 14th Ed, McGraw Hill, New York, 1993. 5. W. J. Luzadder and J. M. Duff, Fundamentals of Engineering Drawing, Prentice-Hall India, New Delhi, 2001.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS311		Course Name	Operating Systems Laboratory	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	0
Semester	V	Elective Subject (Yes/No):		Tutorial	0
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2018-2019	Prerequisites (If any)	Introduction to computers and programming
Qualifies for	Distributed Operating System, Sensors and Cloud computing				
Course Description	This course studies computer from its system software i.e. the operating system perspective. The operating system (OS) allows users to perform the basic functions of a computer. Operating system acts as intermediary between hardware and the user programs. The OS manages all software and peripheral hardware, and accesses the central processing unit (CPU) for memory or storage purposes. The purpose of this course is to provide an overview of computer operating systems.				
Course Objective	1. To learn the fundamentals of Operating Systems. 2. To learn the mechanisms of OS to handle processes and threads and their communication. 3. To learn the mechanisms involved in memory management in contemporary OS. 4. Provide comprehensive introduction to understand the underlying principles, techniques and approaches which constitute a coherent body of knowledge in operating systems. 5. To understand the basic components of a computer operating system including Scheduling, Memory Management, I/O Management, and the interactions among the various components.				
Course Outcomes	1. To learn the fundamentals of Operating Systems. 2. To learn the mechanisms of OS to handle processes and threads and their communication. 3. To learn the mechanisms involved in memory management in contemporary OS. 4. Provide comprehensive introduction to understand the underlying principles, techniques and approaches which constitute a coherent body of knowledge in operating systems. 5. To understand the basic components of a computer operating system including Scheduling, Memory Management, I/O Management, and the interactions among the various components.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Implement FCFS Scheduling	2	1,2,3	CO1	PSO1 PSO2 PSO6
2	Implement SJF Scheduling	2	1,2,3	CO1	PSO1 PSO2 PSO6
3	Implement RRT Scheduling	2	1,2,3	CO1	PSO1 PSO2 PSO6
4	Implement Priority Scheduling	2	1,2,3,6	CO1	PSO1 PSO2

					PSO6
5	Implement Page Replacement Algo	2	1,2,3,6	CO1	PSO1 PSO2 PSO6
6	Implement Disk Scheduling Algorithm-FIFO	2	1,2,3,6	CO2	PSO1 PSO2 PSO6
7	Implement Disk Scheduling Algorithm-SJF	2	1,2,3,6	CO3	PSO1 PSO2 PSO6
8	Implement Disk Scheduling Algorithm-SCAN Algo	2	1,2,3,6	CO3	PSO1 PSO2 PSO6
9	Implement Disk Scheduling Algorithm-Optimal Page	2	1,2,3,6	CO4	PSO1 PSO2 PSO6
10	Implement File level operation	2	1,2,3,6	CO4	PSO1 PSO2 PSO6
11	Installation of Virtual Box and Ubuntu Operating System –FileLevel Command	2	1,2,3,6	CO5	PSO1 PSO2 PSO6
12	Implement System Level Command	2		CO5	PSO1 PSO2 PSO6

Text books:

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts. Sixth edition. Addison-Wesley (2003).

Reference books:

- 1 Andrew Tanenbaum, Modern Operating Systems, Prentice Hall.
- 2 William Stallings, Operating Systems, Prentice Hall.
- 3 Harvey M. Deitel, An introduction to operating systems. Addison-Wesley.
- 4 Andrew Tanenbaum & Albert Woodhull, Operating Systems: Design and Implementation. Prentice-Hall.
- 5 Douglas Comer, Operating System Design - The XINU Approach. Prentice-Hall.
- 6 A.M. Lister, Fundamentals of Operating Systems. Macmillan (1979).

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code			Course Name	Big data analytics and Data Visualization Laboratory	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2019-2020	Prerequisites (If any)	Data Structure
Qualifies for	BigData,DataAnalytics,DataScience				
Course Description	1. Understand big data for business intelligence. Learn business case studies for big data analytics. 2. Understand no SQL big data management. Perform map-reduce analytics using Hadoop and related tools.				
Course Objective	Student will be able to 1. Describe big data and use cases from selected business domains 2. Explain No Sql big data management 3. Install, configure, and run Hadoop and HDFS 4. Perform map-reduce analytics using Hadoop 5. Use Hadoop related tools such as HBase, Cassandra, Pig, and Hive for big data analytics				
Course Outcomes	Student will be able to 1. Demonstrate the functioning of a Compiler. 2. Design and implement the prototype compiler. 3. Apply various optimization techniques. 4. Use different compiler construction tools.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Basic of R programming Write a R code for creating Vectors, List and matrix. Write a R code for creating Arrays, factors and data frames.	2	1,2,3	CO1	PSO2,PSO6
2	Write a program to display casting and melting using datasets in R.	2	1,2,3	CO2 CO1	PSO2,PSO6
3	Basic of R interfaces Write a R code to load the CSV file. Write a R code to load the JSON and Excel file.	2	1,2,3	CO2 CO1	PSO2,PSO6
4	R charts and graphs R code for Pie chart, Bar graph and Box plot. R code for Histogram, line graph and scatter plot	2	1,2,3,6	CO2 CO1	PSO2,PSO6
5	Statistical analysis in R Finding out the mean, median and mode and regressions. Normal distribution and Binomial distribution Poisson Regression	2	1,2,3,6	CO2 CO3	PSO2,PSO6
6	Statistical test in R Analysis of covariance Nonlinear least square Chi square test	2	1,2,3,6	CO4 CO5	PSO2,PSO6

7	Data analysis in R. Time series analysis in R Survival analysis in R Creating decision Tree in R using dataset.	2	1,2,3,6	CO4 CO5	PSO2,PSO6
8	R charts and graphs Performing map and reduce operation using datasets. Word count program using map reduce concept.	2	1,2,3,6	CO4	PSO2,PSO6
9	Clustering in R K means clustering in R	2	1,2,3,6	CO3	PSO2,PSO6
10	Implement the following file management tasks in Hadoop System (HDFS): Adding files and directories, Retrieving files, Deleting files	2	1,2,3,6	CO4	PSO2,PSO6
11	Build WordCloud, a text mining method using R for understanding and visualization of data.	2	1,2,3,6	CO4 CO5	PSO2,PSO6
12	Term Frequency and Inverse Document Frequency (tf-idf) Matrix for Recommendation Systems using R.	2	1,2,3,6	CO5	PSO2,PSO6

Text books:

Reference books:

1. Big Data: A Revolution That Will Transform How We Live, Work, and Think Book by Kenneth Cukier and Viktor Mayer-Schönberger
2. Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking Book by Tom Fawcett
3. Too Big to Ignore: The Business Case for Big Data Book by Phil Simon

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code			Course Name	Data Science and Machine Learning Lab	
Year	3	Core Subject (Yes/No):	Yes	Lecture:	
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
	Practical	Year of Introduction	2021-2022	Prerequisites (If any)	Introduction to Computer Programming, Probability and Statistics, Data Warehouse and Data Mining.
Qualifies for	Major Project				
Course Description	This course is to impart necessary knowledge of the mathematical foundations needed for data science and develop programming skills required to build data science applications.				
Course Objective	1. To demonstrate understanding of the mathematical foundations needed for data science. 2. To collect, explore, clean, munge and manipulate data. 3. To implement models such as k-nearest Neighbors, Naive Bayes, linear and logistic regression, decision trees, neural networks and clustering. 4. To build data science applications using Python based toolkits. 5. To understand the basic concepts of machine learning				
Course Outcomes	Students will be able to: 1. Demonstrate understanding of the mathematical foundations needed for data science. 2. Collect, explore, clean, munge and manipulate data. 3. Implement models such as k-nearest Neighbors, Naive Bayes, linear and logistic regression, decision trees, neural networks and clustering. 4. Build data science applications using Python based toolkits. 5. Understand the concepts of machine learning.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	To implement Map Reduce programs for processing big data	2	1,2,3	CO1	PSO2,PSO6
2	To realize storage of big data using H base, Mongo DB	2	1,2,3	CO2 CO1	PSO2,PSO6
3	To analyse big data using linear models	2	1,2,3	CO2 CO1	PSO2,PSO6
4	The probability that it is Friday and that a student is absent is 3 %. Since there are 5 school days in a week, the probability that it is Friday is 20 %. What is the probability that a student is absent given that today is Friday?	2	1,2,3,6	CO2 CO1	PSO2,PSO6
5	Apply Baye's rule in python to get the result. (Ans: 15%)	2	1,2,3,6	CO2 CO3	PSO2,PSO6
6	Extract the data from database using python	2	1,2,3,6	CO4 CO5	PSO2,PSO6

7	Implement k-nearest neighbours classification using python	2	1,2,3,6	CO4 CO5	PSO2,PSO6
8	Implement linear regression using python.	2	1,2,3,6	CO4	PSO2,PSO6
9	Implement Naïve Bayes theorem to classify the English text	2	1,2,3,6	CO3	PSO2,PSO6
10	Implement an algorithm to demonstrate the significance of genetic algorithm	2	1,2,3,6	CO4	PSO2,PSO6
11	Visualize data using any plotting framework	2	1,2,3,6	CO4 CO5	PSO2,PSO6
12	Implement an application that stores big data in Hbase / MongoDB / Pig using Hadoop / R.	2	1,2,3,6	CO5	PSO2,PSO6

Text books:

Reference books:

1. Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python by Andrew Bruce, Peter C. Bruce, and Peter Gedeck, O'Reilly Media
2. Python Data Science Handbook: Essential Tools for Working with Data by Jake VanderPlas, O'Reilly Media
3. R for Data Science by Garrett Golemund and Hadley Wickham, O'Reilly Media

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS325		Course Name	Artificial Intelligence Laboratory	
Year	3	Core Subject (Yes/No):		Lecture:	
Semester	VI	Elective Subject (Yes/No):		Tutorial	
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:		Total Credit:	1
		Year of Introduction	2018-2019	Prerequisites (If any)	Mathematics, Design Analysis and of Algorithm (CS 312), Computer Organization and Architecture (CS 307)
Qualifies for	Major project				
Course Description	This course studies how to realize the intelligent human behaviours on a computer. The ultimate goal is to make a computer that can learn, plan, and solve problems autonomously. Also, methods for problem solving are useful both for reasoning and planning.				
Course Objective	1. Gain an understanding of entire computer System. 2. To give brief knowledge about searching methodologies. 3. To provide depth knowledge about probability and reasoning. 4. To aware students about utility theory and learning process. 5. To make student familiar about recent technologies.				
Course Outcomes	1. Gain an understanding of basics of Artificial Intelligence. 2. To give brief knowledge about searching methodologies. 3. To provide depth knowledge about probability and reasoning 4. To aware students about utility theory and learning process 5. To make student familiar about recent technologies				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Introduction of Artificial Intelligence and its application.	2	1,2,3	CO1	PSO2,PSO6
2	Implementation of Depth-First Search(DFS).	2	1,2,3	CO2 CO1	PSO2,PSO6
3	Write a program to implement towers of Hanoi	2	1,2,3	CO2 CO1	PSO2,PSO6
4	Write a program to implement water jug problem.	2	1,2,3,6	CO2 CO1	PSO2,PSO6
5	Write a program to implement tic tac toe game for 0 and X.	2	1,2,3,6	CO2 CO3	PSO2,PSO6
6	Write a program to implement production system.	2	1,2,3,6	CO4 CO5	PSO2,PSO6
7	Write a program to implement heuristic search procedure.	2	1,2,3,6	CO4 CO5	PSO2,PSO6
8	Write a program to implement expert system.	2	1,2,3,6	CO4	PSO2,PSO6

9	Write a program to implement Single Player Game (Using Heuristic Function)	2	1,2,3,6	CO3	PSO2,PSO6
10	WAP to implement search problem of 3 x 3 puzzle.	2	1,2,3,6	CO4	PSO2,PSO6
11	Program to implement a* algorithm.	2	1,2,3,6	CO4 CO5	PSO2,PSO6
12	Write a program to solve travelling salesman problem.	2	1,2,3,6	CO5	PSO2,PSO6

Text books:

1. "Artificial Intelligence" -By Elaine Rich and Kevin Knight (2nd Edition) Tata Mcgraw-Hill
2. "Artificial Intelligence: A Modern Approach" -By Stuart Russel, Peter Norvig, PHI

Reference books:

1. "PROLOG Programming for Artificial Intelligence" -By Ivan Bratko (Addison-Wesley)
2. "Programming with PROLOG" -By Klocks in and Mellish.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CS420		Course Name	Cloud Computing Laboratory	
Year	4	Core Subject (Yes/No):	Yes	Lecture:	0
Semester	VII	Elective Subject (Yes/No):		Tutorial	0
Typology of Course		Foundation Subject (Yes/No):		Practical:	2
		Year of Syllabus Revision:	2021-22	Total Credit:	1
	Practical	Year of Introduction	2019-2020	Prerequisites (If any)	Discrete Mathematics, Computer Networks
Qualifies for	Big Data Analytics				
Course Description	Cloud Computing has transformed the IT industry by opening the possibility for infinite or at least highly elastic scalability in the delivery of enterprise applications and software as a service (SaaS). Amazon Elastic Cloud, Microsoft's Azure, Google App Engine, and many other Cloud offerings give mature software vendors and new start-ups the option to deploy their applications to systems of infinite computational power with practically no initial capital investment and with modest operating costs proportional to the actual use.				
Course Objective	Student will try: 1. To understand the concept of virtualization and how this has enabled the development of Cloud Computing. 2. To understand the various issues in cloud computing. 3. To familiarize themselves with the lead players in cloud. 4. To acquire knowledge on Aneka Cloud Application Platform. 5. To learn Industry Cloud Platforms.				
Course Outcomes	Student will be able to: 1. Articulate the main concepts, key technologies, strength 2. Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing. 3. Apply the fundamental concepts in datacenters to understand the tradeoffs in power, efficiency and cost. 4. Identify resource management fundamentals, i.e. resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing. 5. Analyze various cloud programming models and apply them to solve problems on the cloud.				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Sketch out and analyze architecture of CloudSim Simulator. Write a program in cloudsim NetBeans IDE/Eclipse to create a datacenter with one host and one virtual machine on it.	2	1,2,3,6	CO1 CO2 CO3	PSO2 PSO6
2	Write a program in cloudsim using NetBeans IDE/Eclipse to create a datacenter with one host and run four cloudlet on it.	2	1,2,3,6	CO1 CO2 CO3	PSO2 PSO6
3	Write a program in cloudsim using NetBeans IDE Eclipse to create a datacenter with three hosts and run three cloudlets on it.	2	1,2,3,6	CO1 CO2 CO3	PSO2 PSO6

4	Write program in cloudsim using NetBeans IDE/ Eclipse to create one datacenter with four host, one databroker , 40 cloudlets, 10 virtual machines.	2	1,2,3,6	CO1 CO2 CO3	PSO2 PSO6
5	Sketch out and analyze architecture of Aneka / Eucalyptus / KVM identify different entities to understand the structure of it.	2	1,2,3,6	CO4	PSO2 PSO6
6	Create a scenario in Aneka / Eucalyptus to create a datacenter and host. Also create virtual machines with static configuration to run cloudlets on them.	2	1,2,3,6	CO4	PSO2 PSO6
7	Make and perform scenario to pause and resume the simulation in Aneka / Eucalyptus entity, and create simulation entities dynamically.	2	1,2,3,6	CO4	PSO2 PSO6
8	Organize a case in Aneka / Eucalyptus for simulation entities in run-time using a its toolkit support and manage virtual cloud.	2	1,2,3,6	CO4	PSO2 PSO6
9	Sketch out and analyze architecture of Microsoft Azure.	2	1,2,3,6	CO3 CO5	PSO2 PSO6
10	Sketch out and analyze architecture of Amazon Web Service (AWS).	2	1,2,3,6	CO3 CO5	PSO2 PSO6
11	Categorize Amazon Web Service (AWS) and implement its various cloud entities using its Cloud Toolbox support.	2	1,2,3,6	CO3 CO5	PSO2 PSO6
12	Implement and use sample cloud services with the help of Microsoft Azure.	2	1,2,3,6	CO3 CO5	PSO2 PSO6

Text books:

1. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S.Thamarai Selvi from TMH 2013.
2. George Reese Cloud Application Architectures, First Edition, O'Reilly Media 2009.

Reference books:

1. Cloud Computing and SOA Convergence in Your Enterprise A Step-by-Step Guide by David S. Linthicum from Pearson 2010.
2. Cloud Computing 2nd Edition by Dr. Kumar Saurabh from Wiley India 2012.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CMP607		Course Name	Natural Language Processing Laboratory	
Year	4	Core Subject (Yes/No):		Lecture:	0
Semester	VII	Elective Subject (Yes/No):	Yes	Tutorial	0
Typology of Course	Laboratory	Foundation Subject (Yes/No):	Yes	Practical:	4
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2021-2022	Prerequisites (If any)	Basic knowledge Probability and statistics, Programming and data structures
Qualifies for	Nil				
Course Description	Natural language processing deals with written text. Students will learn how to process written text from basic of fundamental knowledge starts with Finite automata, Regular expression and probabilistic model with n-grams. Recognizing Speech and parsing with grammar. This course also covers basis of semantic analysis and discourse analysis and drives it to machine translation. This NLP course will boost student knowledge to research level where they can conduct new level of research. It really helpful for undergraduate students.				
Course Objective	Students will be able to: 1. Teach students the leading trends and systems in natural language processing. 2. Make them understand the concepts of morphology, syntax, semantics and pragmatics of the language and that they are able to give the appropriate examples that will illustrate the above mentioned concepts. 3. Teach them to recognize the significance of pragmatics for natural language understanding. 4. Enable students to be capable to describe the application based on natural language processing and to show the points of syntactic, semantic and pragmatic processing. 5. Interacting with humans in dialogue systems or cooperative tasks. Particular emphasis on Indian languages.				
Course Outcomes	Students will be able to: 1. Understand Natural Language Processing (Understanding). 2. Probabilistic model of defining language and techniques. (Application) 3. Applying Hidden Markov model and Speech Recognition. (Application) 4. Application of context free grammar and language parsing. (Application) 5. Implement probabilistic and language parsing.(Application) and Differentiation of semantic and discourse in terms of NLP.(Analyze)				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Sentiment Analysis.	2	1,2	CO1	PSO1
2	Article Recommendation	2	1,2	CO1	PSO1
3	Distinguishing Opinion	2	2,3	CO2	PSO1,PSO2
4	Predicting Sentiment from any Movie Reviews.	2	2,3	CO2	PSO2,PSO3
5	Information Extraction from Collection of Resumes.	2	3	CO2	PSO3
6	Text Summarization.	2	3	CO2,CO3	PSO3,PSO4

7	Question Answering System.	2	3,4	CO3,CO4	PSO4
8	Part of Speech Tagging	2	3,4	CO4	PSO5
9	Machine Translation	2	4,5	CO4	PSO5
10	Spelling Correction System	2	4,5	CO4,CO5	PSO5,PSO6
11	Morphological Analyzer	2	4,5,6	CO5	PSO5
12	Building Transliterate System	2	5,6	CO5	PSO5,PSO6

Text books:

1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition Jurafsky, David, and James H. Martin, PEARSON
2. Foundations of Statistical Natural Language Processing, Manning, Christopher D., and Hinrich Schütze, Cambridge, MA: MIT Press
3. Natural Language Understanding, James Allen. The Benjamin/Cummings Publishing Company Inc..
4. Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit Steven Bird, Ewan Klein, and Edward Loper.

Reference books:

1. Daniel Jurafsky and James H.Martin Speech and Language Processing(2nd Edition),Prentice Hall:2 edition,2008.
2. Machine Learning for Text by Charu C.Aggarwal,Springer,2018 edition
3. Foundations of Statistical Natural Language Processing by Christopher D.Manning and Hinrich Schuetze,MIT press, 1999
4. Steven Bird,Ewan Klein and Edward Loper Natural Language Processing with Python,O'Reilly Media;1 edition,2009
5. Roland R.Hausser, Foundations of Computational Linguistics:HumanComputer Communication in Natural Language,Paperback,MIT press,2011
6. D. Jurafsky, J. H. Martin, Speech and Language Processing, (2e), Pearson Education, 2009.
7. T. Siddiqui, U. S. Tiwary, Natural language processing and Information retrieval, Oxford University Press, 2008.

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CMP503		Course Name	Fundamentals of Deep Learning Laboratory	
Year	4	Core Subject (Yes/No):		Lecture:	0
Semester	VII	Elective Subject (Yes/No):	Yes	Tutorial	0
Typology of Course	Laboratory	Foundation Subject (Yes/No):	Yes	Practical:	4
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2021-2022	Prerequisites (If any)	Basic knowledge of Artificial Intelligence, Machine Learning.
Qualifies for	Nil				
Course Description	Deep Learning is one of the most exciting and promising segments of Artificial Intelligence and machine learning technologies. This course with Tensor Flow is designed to help you master deep learning techniques and build using Tensor Flow, an open-source software library developed by Google for the purpose of conducting machine learning and deep neural networks. It is one of the most popular software platforms used for deep learning and contains powerful tools to help you build and implement artificial neural networks.				
Course Objective	The student will be able to understand 1. Understand complexity of Deep Learning algorithms and their limitations 2. Understand modern notions in data analysis oriented computing; 3. Student capable of confidently applying common Deep Learning algorithms in practice and implementing their own; 4. They can be performing distributed computations 5. They can be performing experiments in Deep Learning using real-world data.				
Course Outcomes	Students will be able to: 1. Describe the feedforward and deep networks. 2. Design single and multi-layer feed-forward deep networks and tune various hyper-parameters. 3. Analyze performance of deep networks. 4. Build own deep learning project 5. Differentiate between machine learning, deep learning and artificial intelligence				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Implement Simple Programs like vector addition in Tensor Flow.	2	1,2	CO1	PSO1
2	Implement a simple problem like regression model in Keras.	2	1,2	CO1	PSO1,PSO2
3	Implement a perceptron in Tensor Flow/Keras Environment.	2	2,3,4	CO1,CO2	PSO2,PSO3
4	Implement a Feed-Forward Network in Tensor Flow/Keras.	2	2,3,4	CO2,CO3	PSO2,PSO3
5	Implement an Image Classifier using CNN in Tensor Flow/Keras.	2	3,4	CO3	PSO3
6	Implement a Transfer Learning concept in Image Classification.	2	3,4	CO3	PSO4
7	Implement an Auto encoder in Tensor Flow/Keras.	2	3,4,5	CO3,CO4	PSO4
8	Implement a Simple LSTM using Tensor Flow/Keras.	2	3,4,5	CO4	PSO4,PSO5
9	Implement an Opinion Mining in Recurrent Neural network.	2	3,4,5	CO4	PSO5
10	Implement an Object Detection using CNN.	2	4,5	CO4,CO5	PSO5
11	Implement Statistical Machine Translation system.	2	4,5	CO5	PSO5,PSO6

12	Implement Machine Translation using Deep Learning.	2	4,5,6	CO5	PSO5,PSO6
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Text books:

1. Ian Goodfellow, Deep Learning, MIT Press, 2016.
2. Jeff Heaton, Deep Learning and Neural Networks, Heaton Research Inc, 2015.
3. Mindy L Hall, Deep Learning, VDM Verlag, 2011.

Reference books:

1. Li Deng (Author), Dong Yu, Deep Learning: Methods and Applications (Foundations and Trends in Signal Processing), Now Publishers Inc, 2009
2. Golub, G.,H., and Van Loan,C.,F., Matrix Computations, JHU Press,2013.
3. Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004

School: School of Engineering and Technology			Program: B. Tech. (Computer Science and Engineering)		
Course code	CMP702		Course Name	Robotics Laboratory	
Year	4	Core Subject (Yes/No):		Lecture:	0
Semester	VII	Elective Subject (Yes/No):	Yes	Tutorial	0
Typology of Course	Laboratory	Foundation Subject (Yes/No):	Yes	Practical:	4
		Year of Syllabus Revision:		Total Credit:	2
		Year of Introduction	2021-2022	Prerequisites (If any)	Basic knowledge of Digital Electronics And Microprocessors
Qualifies for	Nil				
Course Description	Robotics as an application draws from many different fields and allows automation of products as diverse as cars, vacuum cleaners, and factories. This course is a challenging introduction to basic computational concepts used broadly in robotics. Topics include simulation, kinematics, control, optimization, and probabilistic inference. The mathematical basis of each area is emphasized, and concepts are motivated using common robotics applications and programming exercises. Students will participate in a series of projects over the course of the semester, in which they will implement algorithms that apply each of the topics discussed in class to real robotics problems.				
Course Objective	The student will be able to understand 1. To understand the functions of the basic components of a Robot. 2. To study the use of various types of End of Effectors and Sensors 3. To impart knowledge in Robot Kinematics and Programming 4. To learn Robot safety issues and economics. 5. To understand the concepts of motion control of a mobile robot navigation				
Course Outcomes	Students will be able to: 1. learn the evolution of robotics and modern happenings in robotics technology 2. Understand fundamental concepts in robotics 3. Student become familiar with terminology of robotics 4. Deepen the knowledge of the robots and their components 5. Learn the principles of operation of a robotic system and analyze the problems and challenges on the fundamentals of the methods in robotics research				

Unit No.	Topic/Unit	Contact Hours	BT Level	CO	PSO
1	Determination of maximum and minimum position of links.	2	1,2	CO1	PSO1
2	Verification of transformation (Position and orientation) with respect to gripper and world coordinate system	2	1,2	CO1	PSO1
3	Estimation of accuracy, repeatability and resolution.	2	2,3	CO1, CO2	PSO1, PSO2
4	Robot programming and simulation for pick and place	2	2,3	CO1 CO2	PSO2
5	Robot programming and simulation for Color identification	2	3,4	CO2, CO3	PSO2, PSO3
6	Robot programming and simulation for Shape identification	2	3,4	CO3	PSO3

7	Robot programming and simulation for machining (cutting, welding)	2	4,5	CO4	PSO3, PSO4
8	Robot programming and simulation for writing practice	2	5	CO4	PSO4
9	Robot programming and simulation for any industrial process (Packaging, Assembly)	3	5,6	CO4,CO5	PSO4, PSO5
10	Robot programming and simulation for multi process	3	5,6	CO5	PSO5, PSO6

Text books:

1. Klafter R.D., Chmielewski T.A and Negin M., Robotic Engineering – An Integrated Approach, Prentice Hall, 2003.
2. Groover M.P., Industrial Robotics -Technology Programming and Applications, McGraw Hill, 2001.
3. Industrial robotic technology-programming and application by M.P.Groover et al, McGrawhill 2008
4. Saeed B.Nikku, Introduction to Robotics, analysis, control and applications Wiley-India
2nd edition-2011

Reference books:

1. Craig J.J., Introduction to Robotics Mechanics and Control, Pearson Education, 2008.
2. Deb S.R., Robotics Technology and Flexible Automation Tata McGraw Hill Book Co., 1994.
3. Koren Y., Robotics for Engineers”, Mc Graw Hill Book Co., 1992.
4. Fu.K.S.,Gonzalz R.C. and Lee C.S.G., Robotics Control, Sensing, Vision and Intelligence, McGraw Hill Book Co., 1987.
5. Janakiraman P.A., Robotics and Image Processing, Tata McGraw Hill, 1995.
6. Rajput R.K., Robotics and Industrial Automation, S.Chand and Company, 2008.
7. Surender Kumar, Industrial Robots and Computer Integrated Manufacturing, Oxford and IBH Publishing Co. Pvt. Ltd., 1991.

